



2024-2026

SERC REGIONAL RISK REPORT

Reliability Risk Working Group (RRWG)

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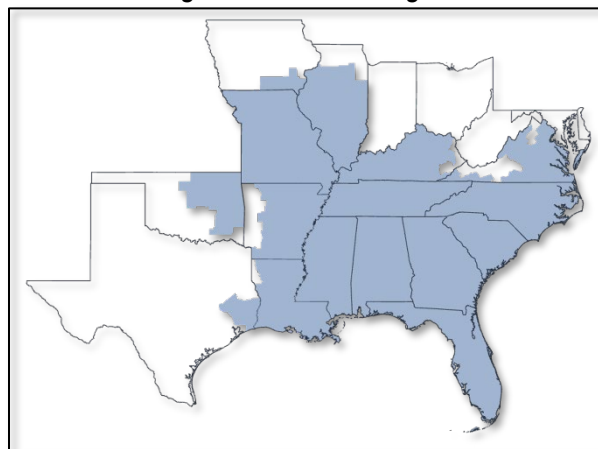
1.0 Executive Summary

Introduction

The electric grid is essential to modern life, supporting economic activity, societal functions, and technological advancements. The reliability and security of the electric grid is crucial for national security, public health and safety, and community well-being. As society's dependence on electricity grows due to technological advancements in communication, transportation, and overall moving toward more electrification, ensuring the grid's resilience is more critical than ever. This overall growing demand for electricity increases society's dependence on new generation technologies in the pursuit of reduced emissions.

SERC's mission is to assure the effective and efficient reduction of risks to the reliability and security of the bulk power system (BPS). SERC is one of only six companies across North America that is responsible for this important work pursuant to Federal Energy Regulatory Commission (FERC) approved delegation agreements with the North American Electric Reliability Corporation (NERC). SERC is responsible for the reliability and security of the electric grid across the southeastern and central regions of the United States. This area covers approximately 630,000 square miles, serves a population of more than 97 million, and includes all or portions of Florida, Georgia, Alabama, Mississippi, Louisiana, Texas, Oklahoma, Arkansas, Missouri, Iowa, Illinois, Kentucky, Tennessee, Virginia, North Carolina, and South Carolina.

Figure1-A: SERC Region



The SERC Regional Risk Report is SERC's independent assessment of the risks to the BES across SERC's footprint. The SERC Risk Report builds upon and serves as a regional supplement to the NERC 2023 ERO Reliability Risk Priorities Report, which provides a continent-wide view.¹ In developing the report, SERC staff also collaborated with SERC's technical committees, including the Reliability Risk Working Group (RRWG), to gather invaluable industry expertise, insights, and feedback.

This report highlights the essential themes and emerging risk trends in the SERC region.

¹ 2023 ERO Reliability Risk Priorities Report (nerc.com)

Identified Risk Themes and Takeaways

1. **Supply Chain Constraints:** The SERC footprint faces risks due to supply chain dependencies on suppliers; and the associated impact that geopolitical tensions, natural disasters, and cybersecurity threats pose to those suppliers, their products, and the corresponding supply chain. Diversifying suppliers, enhancing cybersecurity, and improving inventory management are crucial mitigating actions necessary to manage grid reliability and security. Additionally, as supply chains grow more complex, collaboration with key stakeholders and regulators will be crucial for managing these risks proactively.
2. **Extreme Weather Impacts:** The increasing occurrence of severe and extreme weather events such as hurricanes, extreme cold weather, and heatwaves have the potential to disrupt the BPS and the supply of energy across SERC footprint. The growing frequency of these events, driven by climate change, underscores the need for resilience in operational planning and infrastructure investments. Preparedness measures, including extreme weather readiness and collaborative restoration efforts, are essential for maintaining grid stability and operational readiness.
3. **Resource Uncertainty:** The shift toward renewable energy introduces challenges in managing resource adequacy and system stability. A diverse energy portfolio, advanced modeling techniques, implementing different resource adequacy metrics to accurately gauge risks, and flexible planning are necessary to successfully integrate these new generation resources. The transition from coal to natural gas, renewables, and battery energy storage increases operational risks, requiring updated tools and strategies to ensure reliability in an evolving energy landscape.
4. **Shortage of Required Skillsets:** Stakeholders across SERC's footprint are facing a shortage of skilled professionals in operations, planning, cybersecurity, and advanced technologies. The shift in workforce dynamics, alongside the growing need for expertise in emerging technologies, cybersecurity, and renewable integration, creates a potential talent gap. Expanded training programs, educational partnerships, and enhanced recruitment are vital to address this gap.
5. **Exploitation of Vulnerabilities:** Increased digitalization has heightened cyber security risks across SERC's footprint. Implementing robust protocols, network segmentation, encryption, leveraging advanced technologies for better security, and continuous risk management practices are key to protecting critical infrastructure from potential cyber-attacks. As the reliance on digital systems grows, entities must adopt advanced cybersecurity solutions and collaborative risk management practices to safeguard grid security from increasingly sophisticated threats.
6. **Rapid Nontraditional Load Growth:** The ongoing rise in demand from sources like data centers, crypto mining, artificial intelligence, electric vehicles, and semiconductor manufacturing puts stress on the transmission and distribution systems across SERC's footprint. This rapid load growth calls for infrastructure modernization, strategic investments, and the integration of advanced technologies like artificial intelligence (AI) and grid monitoring tools to maintain grid reliability. Upgrading infrastructure and balancing generation capacity with demand are crucial to maintaining grid reliability.

- 7. Lack of Fuel Diversity and Availability:** Heavy reliance on natural gas poses risks related to fuel supply disruptions and volatility in availability. Mitigation includes monitoring fuel availability, enhancing operational tools, enhancing infrastructure, and diversifying the fuel mix to ensure reliable power generation. The transition to alternative fuels and increased integration of storage technologies will be essential for long-term fuel security and grid reliability.
- 8. Legacy Architecture Compatibility:** Legacy systems often struggle with compatibility issues when integrated with modern technologies, thus affecting performance, reliability, and security. The rapid evolution of emerging technologies, including grid monitoring tools, AI, and advanced cybersecurity solutions, demands ongoing investment in infrastructure upgrades to ensure these systems can handle future challenges. Collaboration with industry experts and strategic prioritization of upgrades can help address these challenges.
- 9. Extreme Physical Events: Sabotage and Attacks:** The SERC footprint faces the risk of physical attacks from malicious actors, including sabotage and extreme vandalism, which can lead to major disruptions and equipment losses. Comprehensive security measures, including access controls, barriers, surveillance, and incident response plans are essential. Collaborative efforts with law enforcement and continuous vulnerability assessments will enhance regional security.
- 10. Integrating Variable Energy Resources:** The growth of renewable energy sources introduces challenges in grid management across SERC's footprint. As renewable integration increases, advancements in inverter technologies and updated modeling techniques are necessary to manage the variability of resources like wind and solar. Adhering to industry guidelines and investing in infrastructure are necessary to manage the variability of renewable energy resources and maintain grid reliability.

Figure 1 displays the top identified SERC Reliability Risks for 2024.

The 2024-2026 SERC Top 10 Regional Risks

Figure 1: Identified Risk Themes

Supply Chain Constraints

Increased risks from supplier dependencies, cyber threats, and natural disasters, require diversification, cybersecurity upgrades, and improved inventory management.

01

Extreme Weather

Impacts: Extreme weather in the SERC region can damage infrastructure and disrupt fuel supply.

02

Resource Uncertainty:

Accelerating changes in generating resources and fuel complicate planning and operations.

03

Shortage of Required

Skillsets: This risk arises from a critical shortage of skilled staff in electrical operations, planning, and cybersecurity.

04

Exploitation of

Vulnerabilities: Advanced tools and processes exploit bulk power system vulnerabilities, including ransomware.

05



Rapid Nontraditional Load

Growth: Rapid load growth from new data centers and AI, is challenging the planning and operation of the BPS.

06

Lack of Fuel Diversity and

Availability: Transition to natural gas and variable generation challenges energy adequacy.

07

Legacy Architecture

Compatibility: Aging infrastructure struggles with compatibility and support from vendors.

08

Extreme Physical Events:

Sabotage & Attacks: Deliberate disruptions to Bulk Electric System facilities and equipment.

09

Integrating Variable Energy

Resources: Renewable energy integration requires backup planning due to weather dependencies.

10

Risk Profiles

The SERC Regional Risk Report builds upon and serves as regional supplement to the 2023 ERO Reliability Risk Priorities Report.³ SERC's top regional risks fall within the following risk profiles identified in the 2023 ERO Reliability Risk Priorities report:

- **Energy Policy Changes:** Governmental decisions can profoundly influence BPS reliability. Key regional risks within this category include supply chain constraints, resource uncertainty, and challenges related to integrating variable energy resources.
- **Grid Transformation:** The shift toward non-synchronous and digital technologies introduces a new set of nontraditional risks. These include skill shortages, rapid nontraditional load growth, challenges with fuel diversity, legacy infrastructure compatibility, and the potential for sabotage.
- **Resilience to Extreme Events:** This category emphasizes the BPS's ability to recover from severe conditions such as hurricanes and heatwaves. Extreme weather events are identified as a significant concern affecting system reliability.
- **Security Risks:** As the BPS infrastructure rapidly expands, its vulnerability to cyber-physical threats increases. This includes risks from supply chain issues, skill shortages, and attacks on legacy systems.
- **Critical Infrastructure Interdependency:** The increasing interconnectedness between the energy sector and other essential sectors—such as transportation and communications—introduces cascading risks. Notable risks include lack of fuel diversity and sabotage.

Figure 2 shows how the SERC Regional Risks map to the 2023 ERO Reliability Risk Priorities Report³ high-level risk profiles.

³ 2023 ERO Reliability Risk Priorities Report (nerc.com)

Figure 2: SERC Risks by ERO Reliability Risk Profiles



Risk Management and Status

To further address these challenges, the 2024–2026 SERC Regional Risk Report categorizes risks into two groups: “Manage” and “Monitor.”

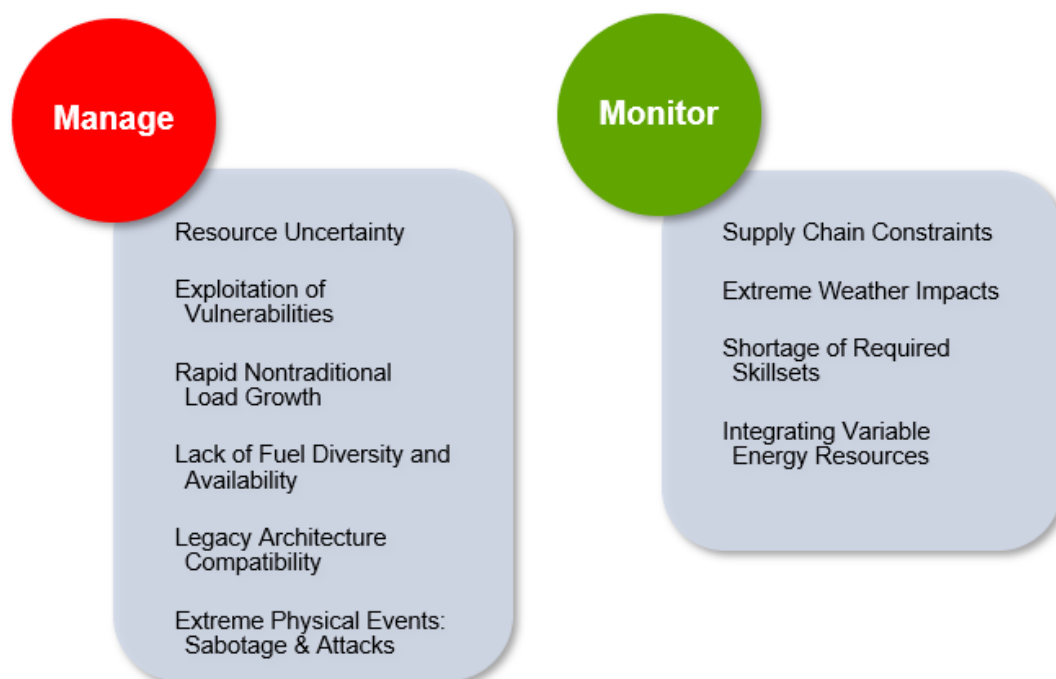
The **Manage** status group includes emerging risks where mitigation plans need to be developed and implemented—through SERC, NERC, or other Industry engagements—or additional mitigation plans need to be considered.

The **Monitor** status group includes risks where mitigation plans and guidance are already underway but additional time is required to fully implement these actions and assess their effectiveness in reducing risk.

In this report, the risks related to **Supply Chain Constraints** and **Shortage of Required Skillsets** have shifted from “manage” status to “monitor” status. This change reflects the proactive efforts of SERC stakeholders in developing and implementing mitigation strategies to deal with the supply chain for key parts and equipment (see section 4.1, **Supply Chain Constraints**). In addition, stakeholders have formed partnerships and strategies to improve workforce training and recruitment (see section 4.4, **Shortage of Required Skillsets**). As a result, these risks have been reclassified to “monitor” status.

Figure 3 displays the top identified SERC Reliability Risks and their risk status for the SERC footprint. The status assigned to each risk is based on the Manage and Monitor definitions.

Figure 3: 2024 Top Risks by Risk Status



Conclusion

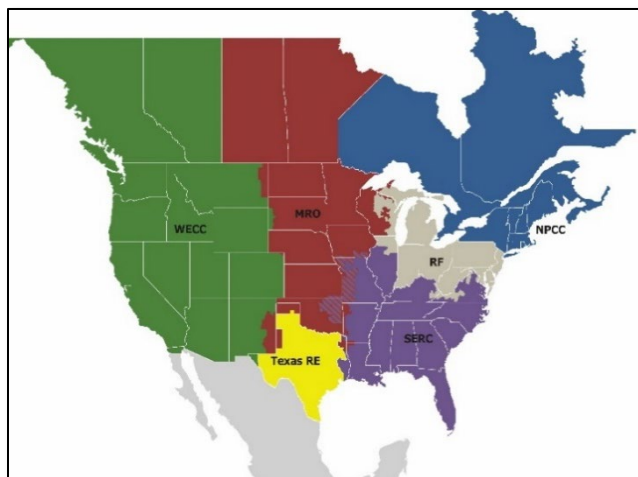
The SERC Regional Risk Report identifies key challenges and emerging trends that could impact the reliability and security of the BPS across SERC's footprint. SERC's top regional risks fall within the following risk profiles identified in the 2023 ERO Reliability Risk Priorities report: Energy Policy, Grid Transformation, Resilience to Extreme Events, Security Risks, and Critical Infrastructure Interdependencies.

The SERC Regional Risk Report highlights the evolving risks facing SERC's footprint as it adapts to increasing demand, technological advancements, and environmental changes. Protecting the grid's reliability and security requires a comprehensive approach, as many of these risks are interrelated and/or interconnected. Through proactive and strategic planning, and continued collaboration, the sector can sustain the reliability and resilience of the BPS in the SERC footprint. SERC remains committed to driving innovation and preparedness to meet the future demands of a secure and reliable electric grid.

2.0 Background

The mission of SERC Reliability Corporation (SERC) is to assure effective and efficient reduction of risks to the reliability and security of the Bulk Power System (BPS). To achieve this mission, SERC maintains a diverse team of experts across numerous disciplines to address the complex, evolving, and dynamic challenges facing the grid. Our team also partners with the best and brightest individuals from the power industry and the federal and state governments to understand and address the challenges facing the grid. These key partnerships make our work more informed, pragmatic, responsive, and impactful.

Figure 4: Six Regional Entities of North American BPS



The 2024–2026 SERC Regional Risk Report is an independent assessment of the risks affecting the SERC Region, developed by SERC. The Reliability Risk Working Group (RRWG), a committee of SERC stakeholders and industry experts, provided their valuable feedback and insights into the development of the risk registry and report.

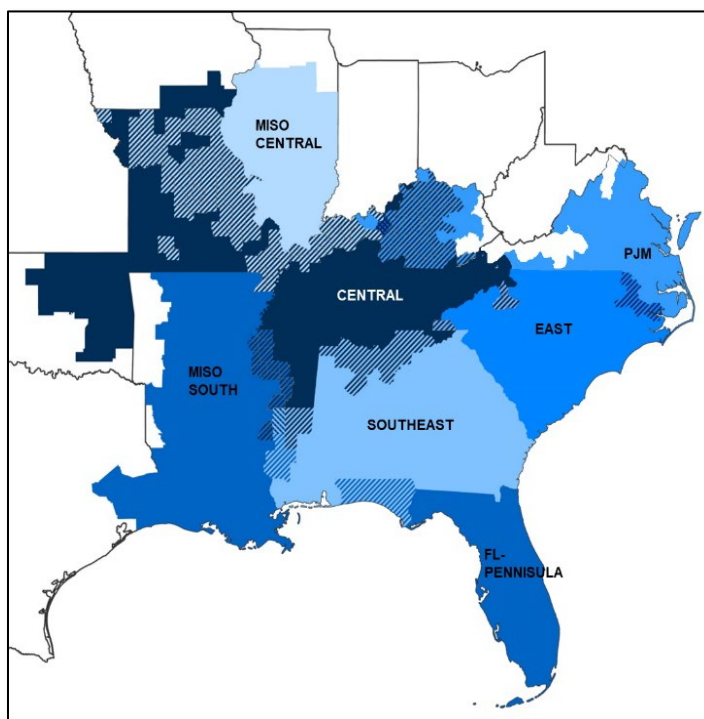
The SERC Regional Risk Report serves as a supplement to the NERC ERO Reliability Risk Priorities Report by providing a more granular focus on the SERC Region to complement the ERO Reliability Risk Priorities Report's continent-wide view.

ABOUT SERC

SERC is one of six Regional Entities across North America responsible for this important work under the Federal Energy Regulatory Commission (FERC) approved delegation agreements with the North American Electric Reliability Corporation (NERC). SERC is specifically responsible for the reliability and security of the electric grid across the southeastern and central regions of the United States. The SERC Region covers approximately 630,000 square miles and serves a population of more than 97 million. It includes all or portions of 16 states: Florida, Georgia, Alabama, Mississippi, Louisiana, Texas, Oklahoma, Arkansas, Missouri, Iowa, Illinois, Kentucky, Tennessee, Virginia, North Carolina, and South Carolina.

Geographically, the SERC Region is divided into seven subregions. These subregions are SERC Central, SERC East, SERC FL-Peninsula, SERC MISO-Central, SERC MISO-South, SERC PJM, and SERC Southeast. SERC's seven subregions encompass these six Reliability Coordinator (RC) Areas:

Figure 5: SERC Footprint and Subregions



- Florida Reliability Coordinating Council, Inc (FRCC) is the Reliability Coordinator for the Florida Peninsula subregion (FL-Peninsula).
- Midcontinent Independent System Operator (MISO) is the Reliability Coordinator for two of the SERC sub regions, MISO-Central, and MISO-South.
- Tennessee Valley Authority (TVA) is the Reliability Coordinator for the Central subregion.
- Southern Company Services (SCS) is the Reliability Coordinator for the Southeast subregion (SeRC).
- VACAR South (VACS) is the Reliability Coordinator for the East subregion.
- PJM Interconnection, LLC (PJM) is the Reliability Coordinator for the PJM subregion.

SERC currently has 86 members,⁴ which include Investor-Owned Utilities (IOU), Municipals, Cooperatives, Federal/State, Regional Transmission Organizations (RTO)/Independent System Operators (ISO)/Reliability Coordinators (RC), Merchant Electricity Generators, and Marketers. In the SERC Region, there are approximately 309 NERC registered entities.⁵

Acknowledgement

The 2024–2026 SERC Regional Risk Report is an independent assessment of the key risks affecting the SERC Region. Developed by SERC, this report was enriched by the valuable feedback and insights provided by the Reliability Risk Working Group (RRWG), comprised of dedicated SERC stakeholders and industry experts. The RRWG plays a critical role in identifying, analyzing, and prioritizing regional risks to the BPS, working collaboratively to ensure the security and reliability of the grid. Their expertise and ongoing efforts support the development of the risk registry and have been instrumental in aligning regional focus with broader, continent-wide risk considerations.

SERC would like to extend a sincere acknowledgment and thank you to the RRWG for their outstanding contributions to this report and their commitment to enhancing the region's reliability and security. The RRWG's ongoing collaboration with the SERC technical committees continues to strengthen the effectiveness of risk management and provides an input into the Compliance Monitoring and Enforcement Program (CMEP) Implementation Plan. Their work ensures that regional risks are clearly identified and prioritized, guiding both current mitigation strategies and future planning efforts.

⁴ SERC members as of September 24, 2024

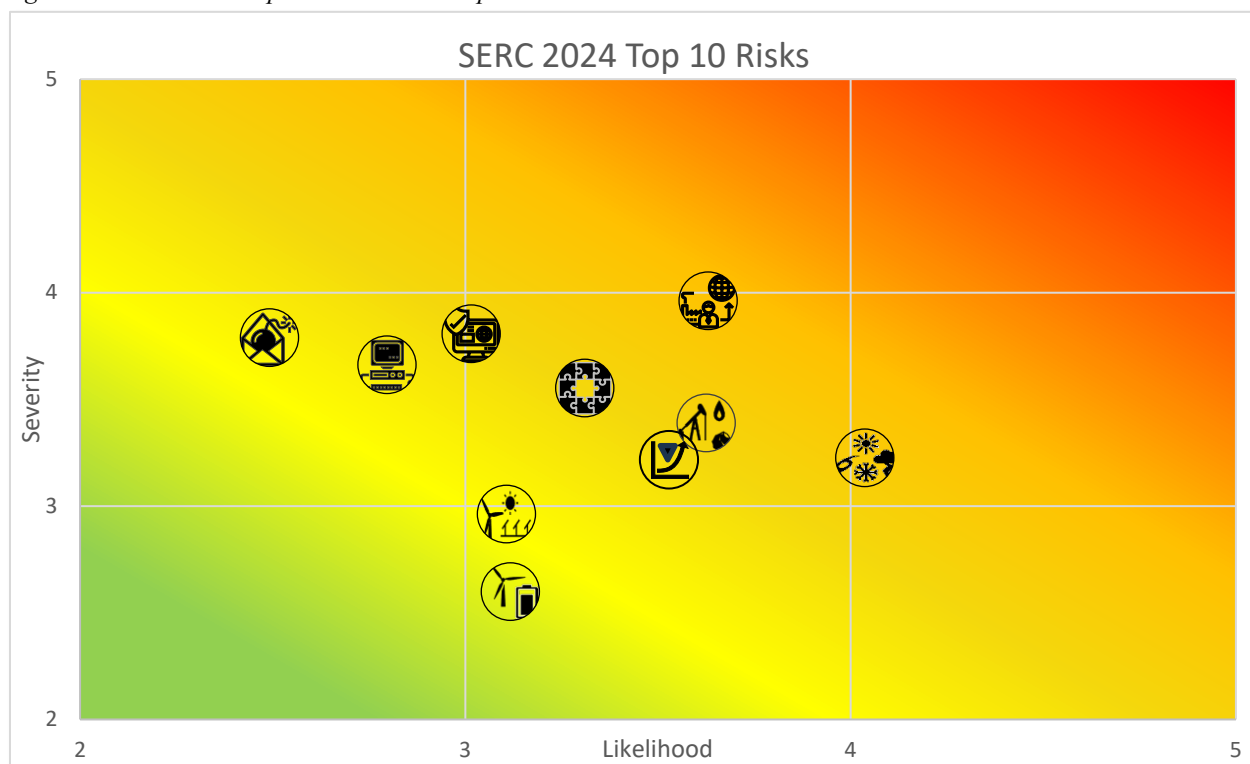
⁵ SERC registered entities as of September 24, 2024

3.0 Overview

The SERC Regional Risk Report focuses on identifying and prioritizing reliability and security risks within the SERC footprint of the BPS. The primary goal of the report is to raise awareness of evolving reliability and security challenges and to actively engage SERC members and registered entities in strategies to mitigate these risks.

Figure 6 is a heat map representation of the SERC 2024 top Major Reliability Risks identified for the SERC footprint.

Figure 6: SERC 2024 Top 10 Risks Heat Map



Regional Risk Heat Map Legend

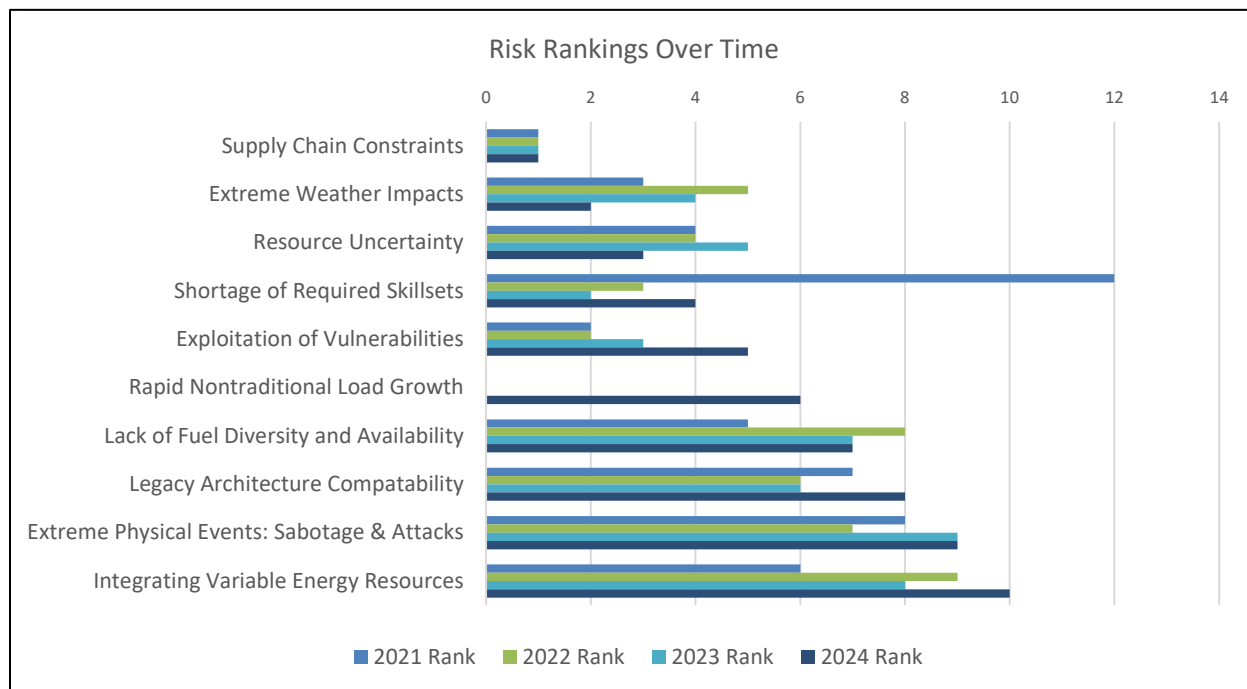
	SC – Supply Chain Constraints (Monitor)		RL - Rapid Nontraditional Load Growth (Manage)
	EW – Extreme Weather Impacts (Monitor)		FD – Lack of Fuel Diversity and Availability (Manage)
	RU – Resource Uncertainty (Manage)		LA – Legacy Architecture Compatibility (Manage)
	SS – Shortage of Required Skillsets (Monitor)		SA – Extreme Physical Events: Sabotage & Attacks (Manage)
	EV – Exploitation of Vulnerabilities (Manage)		VE – Integrating Variable Energy Resources (Monitor)

Figure 7 shows a comparison of composite rankings for SERC's top 10 risks in 2024 and their corresponding risk ranking in the 2021, 2022, 2023, and 2024 risk registries.

In 2021, SERC implemented a scoring change along with the development of a standard set of 15 impact areas to evaluate each risk. In 2023, four different scenarios were reviewed and analyzed (scoring 6, 8, 10, or 12 impact areas). It was found that scoring 10 or 12 impact areas

did not significantly alter risk rankings. As a result, the approach of scoring the top 10 impact areas for each risk was adopted for the 2024 Risk Register, resulting in a slight increase in the total risk score for each risk.

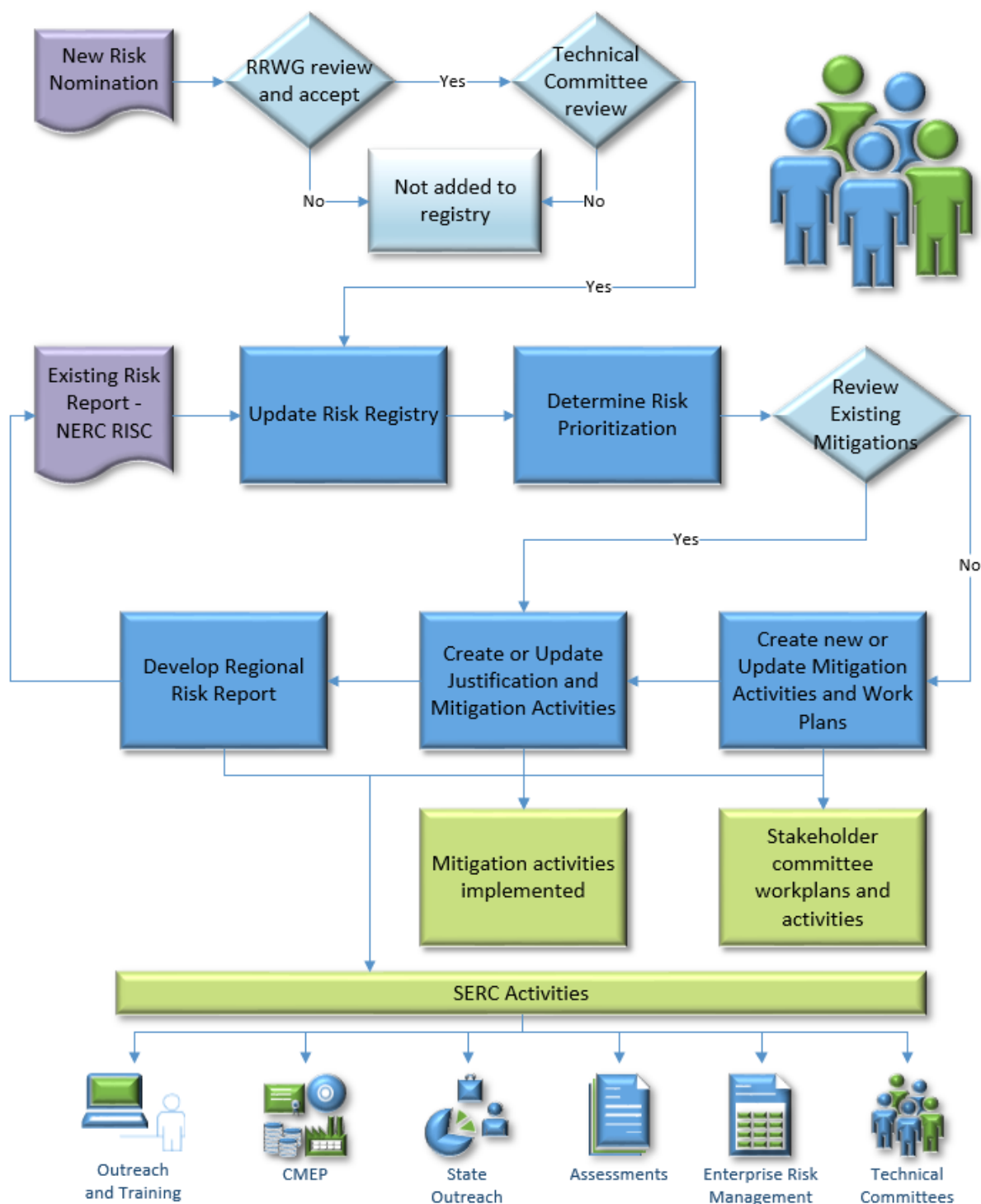
Figure 7: Risk Rankings for the 2024 Top Ten Risks for 2021, 2022, 2023, and 2024



3.1 SERC Risk Process Flow Diagram

Figure 8 provides a pictorial flow chart of the Risk Nomination and the SERC Regional Risk Management Process.

Figure 8: SERC Risk Process Flow Diagram

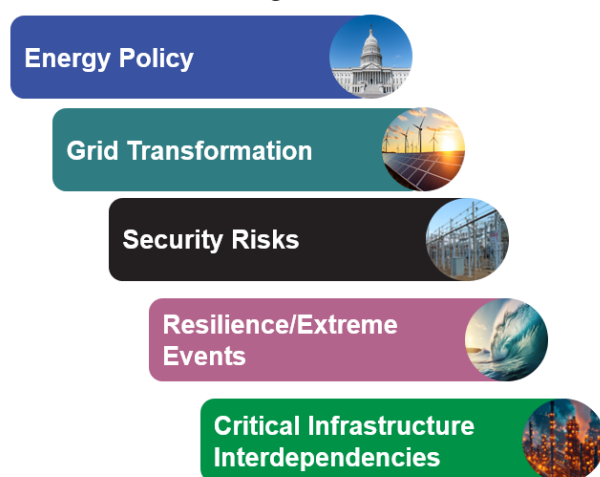


3.2 Tie to the ERO Reliability Risk Priorities Report

The 2024 report builds on the 2023 ERO Reliability Risk Priorities Report⁶, recognizing progress but also underscoring that new risks continue to emerge, and that the potential severity of risks can change from year to year as mitigating plans are put in place and implemented and as conditions change.

For the SERC Regional Risk Report, the process begins with the 2023 ERO Reliability Risk Priorities Report, which outlines five high-level risk profiles in [Figure 9](#): Energy Policy, Grid Transformation, Security Risks, Resilience/Extreme Events, and Critical Infrastructure Interdependencies. These risk categories are then applied to the SERC region and further prioritized based on a more detailed analysis, considering both the likelihood of occurrence and the potential severity of impact on reliability and security.

Figure 9: 2023 ERO Reliability Risk Priorities Report - Five High Level Risks



Energy Policy – Energy Policy can drive changes in the planning and operation of the BPS. Accordingly, policy can affect BPS reliability and resilience and could present risks to its reliable operation. Ensuring reliability during and after policy-driven transitions should be a key consideration in setting Energy Policy. The implementation of policy decisions can significantly affect the reliability and resilience of the BPS.⁶ Whether it drives changes in generating resources, accelerates retirements, or incentivizes the addition/location of certain kinds of new loads, energy policy at the state and federal level can have direct and subtle impacts on reliability and security. The following risks have been identified as SERC Regional Risks that may

be affected by Energy Policy: **Supply Chain Constraints, Resource Uncertainty, Lack of Fuel Diversity, and Integrating Variable Energy Resources.**

Grid Transformation – Transformation of the grid continues to accelerate toward generation and load resources that are increasingly non-synchronous, diverse, digital, and dynamic, while dependence on uninterrupted electric energy increases. At the same time, electricity demand for homes, transportation, commerce, and industry shows signs of growth at rates much more rapid than previously anticipated. This rapid transformation results in an array of nontraditional operating conditions and risks.⁶ The following risks have been identified as SERC Regional Risks related to Grid Transformation: **Supply Chain Constraints, Resource Uncertainty, Shortage of Required Skillsets, Rapid Nontraditional Load Growth, Lack of Fuel Diversity and Availability, Legacy Architecture Compatibility, Extreme Physical Events: Sabotage & Attacks, and Integrating Variable Energy Resources.**

Resilience to Extreme Events – BPS resilience to extreme events is an area of increasing focus. Reliability and resilience are related but distinct concepts:⁷ BPS reliability involves performance consistency under various reasonably expected known or historical operating conditions. Resilience, on the other hand, involves the ability of the BPS to withstand and

⁶ 2023 ERO Reliability Risk Priorities Report RISC Approved July 2023, Board Approved August, 2023

⁷ Reference NATF/EPRI Resilience definition and NERC Resilience Framework.

recover quickly from significant abnormal conditions or extreme events.⁸ This is especially important for the SERC Region given its large geographic exposure to hurricanes, tornadoes, winter storms, and extreme heat, among other extreme events. The following risks have been identified as SERC Regional Risks in the category of resilience to extreme events: ***Extreme Weather Impacts, Lack of Fuel Diversity and Availability, and Extreme Physical Events: Sabotage & Attacks.***

Security Risks – The rapid growth of BPS infrastructure with new facilities, services, and technology integration poses significant challenges to operational security, crucial for a reliable and resilient BPS. This expansion increases the cyber-attack surface and the risk of coordinated attacks. Cyber and physical security are interconnected, as vulnerabilities in one can compromise the other, potentially leading to asset damage, functionality loss, or impaired situational awareness essential for BPS operations and restoration. These vulnerabilities are further aggravated by insider threats, poor cyber hygiene, equipment limitations, and supply chain issues.⁸ The following risks have been identified as SERC Regional Risks in the category of security risks: ***Supply Chain Constraints, Shortage of Required Skillsets, Exploitation of Vulnerabilities, Legacy Architecture Compatibility, and Extreme Physical Events: Sabotage & Attacks.***

Critical Infrastructure Interdependency – The interdependencies between critical infrastructure sectors (e.g., communications, water, wastewater, financial, manufacturing, transportation) and energy subsectors (e.g., natural gas, oil) must be fully understood, communicated, coordinated, and planned. Otherwise, disruptions in other critical infrastructures can affect BPS reliability. The interdependence between energy subsectors is increasing, particularly with natural gas. The growing reliance on natural gas as a fuel source for electric generation is potentially heightening the risk of common-mode failures that could lead to reliability issues. Additionally, the dependency of BPS reliability on natural-gas-fired generation does not always align with the service priorities of the natural gas delivery system, nor with the weatherization requirements for natural gas gathering and delivery infrastructure. Furthermore, the natural gas delivery system depends on reliable electric service to deliver natural gas at acceptable pressures; however, the foreseeable growth and dependence of BPS reliability on natural-gas-fired generation does not align with the expected pace of pipeline development.

As the grid evolves and electrification of critical sectors like transportation increases, these interdependencies will grow, causing potential ripple effects across multiple infrastructures.⁸ The following risks have been identified as SERC Regional Risks that are notably interdependent with BPS reliability and security: ***Lack of Fuel Diversity and Availability, and Extreme Physical Events: Sabotage & Attacks.***

3.3 Risk Categories by the Manage and Monitor Definitions

Risk Categories

SERC also categorizes Regional Risk according to the maturity of mitigation plans that are in place to reduce the impacts or probability of occurrence. The status assigned to each risk is based on the existence of mitigation plans or guidance.

The **Manage** status group includes emerging risks where mitigation plans need to be developed and implemented—through SERC, NERC, or other Industry engagements—or additional mitigation plans need to be considered: ***Resource Uncertainty, Exploitation of Vulnerabilities, Rapid Nontraditional Load Growth, Lack of Fuel Diversity and Availability, Legacy Architecture Compatibility, and Extreme Physical Events: Sabotage & Attacks.***

⁸ 2023 ERO Reliability Risk Priorities Report RISC Approved July 2023, Board Approved August, 2023

The **Monitor** status group includes risks where mitigation plans and guidance are already underway, but additional time is required to fully implement these actions and assess their effectiveness in reducing risk: ***Supply Chain Constraints, Extreme Weather Impacts, Shortage of Required Skillsets, and Integrating Variable Energy Resources.***

In the review of SERC Reliability Risks the Engineering Committee (EC), Operations Committee (OC), and Critical Infrastructure Protection Committee (CIPC) separately identified the top Reliability Risks to the SERC footprint from their perspectives. Then the groups consider those risks for which two or more of the technical committees identified a risk as significant and determine the overall ranking for the most significant risks for the SERC footprint.

The Regional Risk Profile section of this report provides the following for each risk: the committee that identified and evaluated the risk, a definition and justification as to why this risk made the top ten list, and an update on the mitigation activities for the risk identified.

3.4 Improvements for the 2024–2026 SERC Regional Risk Report

Scoring Changes for 2024

In 2023, SERC conducted a statistical review to assess the effects of not scoring or assigning a zero to certain impact areas (Appendix C). The objective was to determine whether reducing the number of impact areas would significantly affect the total risk score and ranking of major risks in the SERC footprint. By scoring fewer impact areas, the three disciplines gained more flexibility in evaluating and scoring each impact area for each risk. The review analyzed four different scenarios (scoring 6, 8, 10, or 12 impact areas). It found that scoring 10 or 12 impact areas did not significantly alter risk rankings. Consequently, SERC adopted the approach of scoring 10 impact areas for the 2024 Risk Register, resulting in a slight increase in the total risk score for each risk.

Naming Convention Changes in 2024

To enhance understanding and clarity of the risks identified in the risk registry and report, SERC introduced additional descriptors in the naming convention for each risk in 2024. This effort aims to provide more detailed and precise descriptions, thereby improving comprehension of each risk.

Additional Resources

To bolster the justification and mitigation strategies of risks, SERC incorporated a variety of industry reports, articles, and white papers into the SERC Regional Risk Report. This comprehensive approach provided robust support for the ranking of each risk, ensuring a thorough explanation of the rationale behind each risk's position.

New Sections of the Report

To address the rapidly evolving BPS and the influx of risk nominations, SERC emphasized new and emerging risks in this report (Section 5). SERC received five new risk nominations: Rapid Nontraditional Load Growth, Lack of Generative Artificial Intelligence Governance, Coordination of Critical Generation or Load Through a Single Substation, Disruption of Cloud Services, and Lack of Insider Threat Governance Programs. This new section aims to provide an overview, introduction, additional resources, and details of ongoing industry activities related to these risks.

Additionally, to enhance understanding of the risk registry, SERC included the entire 2024 Reliability Risk Working Group Consolidated Risk Registry (Appendix A). This comprehensive registry details and scores all thirty risks reviewed annually, offering a complete picture of the current risk landscape.

4.0 SERC Top 10 2024 Regional Risk Profiles

Risk 1: Supply Chain Constraints (Monitor)



Identified and evaluated by the Critical Infrastructure Protection and Engineering Committees

Definition/Justification of risk: As the demand for electricity increases in tandem with technological advancements and the shift toward renewable energy, it is essential to address primary risks within supply chains. These risks include both vulnerabilities (the risks vendors could potentially introduce within industry environments) and product delays (i.e., delays in projects created by shortages of needed materials to produce final products). The electric sector

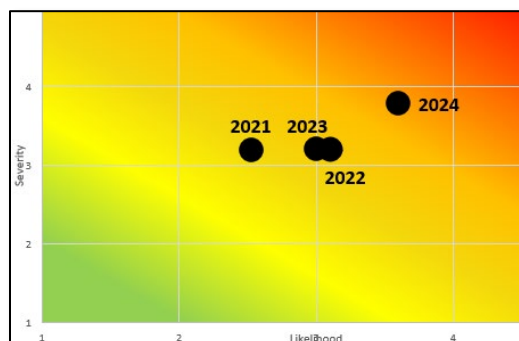


Figure 10: Supply Chain Attacks on the Risk



encompasses the generation, transmission, and distribution of electricity. Efficient supply chain management is crucial for ensuring reliable service delivery while minimizing costs and environmental impacts. However, several risks threaten the stability and efficiency of supply chains in this sector. See [Figure 10](#).⁹

Risks:

- **Supplier Dependency.** This is typified by over-reliance on a limited number of suppliers for critical components (e.g., transformers, circuit breakers, renewable technology, etc.). Potential disruptions can occur due to supplier bankruptcy, political instability, or natural disasters affecting production.
- **Geopolitical Risks.** Trade restrictions, tariffs, and geopolitical tensions can limit access to essential materials and technologies. This could result in increased costs and delays in project timelines, and potential shifts in market dynamics.
- **Natural Disasters and Climate Change.** The electric sector is particularly vulnerable to weather events such as hurricanes, floods, wildfires, and earthquakes, which can disrupt supply chains. Damage to infrastructure and facilities may lead to significant supply delays, increased repair costs, and regulatory challenges.
- **Cyber Security Threats.** The integration of smart technology and the Internet of Things (IoT) into supply chains can introduce vulnerabilities to cyber-attacks. A cyber breach can disrupt operations, lead to the theft of sensitive data, and compromise safety protocols.

⁹ European union's supply chain security guidelines for software suppliers

- **Regulatory Changes.** Changes in regulations related to environmental standards, labor laws, or import/export restrictions can impact supply-chain dynamics. Companies may face increased compliance costs, operational disruptions, and potential legal liabilities.
- **Operational Disruptions.** Supply-chain disruptions can lead to delayed project timelines, impacting electricity generation, transmission, and distribution.
- **Financial Loss.** Increased costs from sourcing alternative suppliers, expediting shipments, or dealing with inventory shortages can significantly impact business or entities.
- **Reputation Damage.** Failure to meet service expectations due to supply-chain disruptions can harm customer trust and brand reputation. Delays in complying with regulatory requirements can result in fines, sanctions, and further erosion of public trust. Mitigating supply chain risks might lead to unintended consequences affecting sustainability goals.
- **Vendor Risk Management.** Assessing and managing the security posture of third-party vendors is crucial. This includes evaluating their security policies, practices, and potential vulnerabilities. Vendors with weak security can become entry points for attacks, so ongoing risk assessments and monitoring are essential.
- **Incident Response and Management.** Developing and testing an incident response plan tailored specifically for supply chain disruptions is essential. This plan should outline how to quickly address and mitigate the impact of a cyber security incident that affects the supply chain, including communication strategies and coordination with affected parties.
- **Data Protection and Privacy.** Ensuring the protection of data shared across the supply chain is vital. This involves implementing strong encryption, access controls, and data loss prevention measures to safeguard sensitive information from unauthorized access or breaches.

Figure 11: Supply Chain Graphic

The Engineering Committee has identified significant supply chain risks that threaten the stability of the energy sector. Prolonged lead times for critical components like transformers and 230 kV breakers, disruptions in coal deliveries, and shortages of essential chemicals for thermal power plants could lead to operational shutdowns. [Figure 11.10](#) is a pictorial example of dependencies in the supply chain.

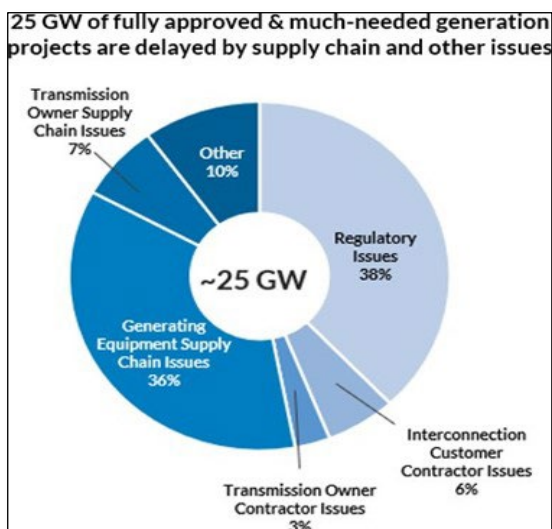
These challenges are delaying new generation projects, particularly in solar and storage, amidst the rapid retirement of fossil fuel plants and rising energy demand.



¹⁰ <https://www.liquidplanner.com/wp-content/uploads/2019/04/Supply-Chain-graphic-1.jpg>

As of late 2023, about 25 GW of fully approved generation projects in MISO's Generator Interconnection Queue had missed their in-service deadlines by an average of 650 days, with developers citing supply chain and permitting issues as the two biggest reasons for the delays. See Figure 12.¹¹

Figure 12: MISO Generation Queue Delays



Additionally, disruptions in oil refineries are raising concerns about the availability of liquid fuels during natural gas shortages. Political tensions in resource-rich regions are also threatening the supply of essential materials like titanium, magnesium, and precious metals needed for the BPS. While industry has limited control over these geopolitical factors, proactive planning and information sharing are vital to understanding and mitigating these risks. The COVID-19 pandemic and ongoing international tensions have underscored the vulnerability of the utility industry to supply chain disruptions, even during stable conditions.

Mitigation Status and Strategies:

- **Diversification of Suppliers.** Developing relationships with multiple suppliers across different regions will increase the diversification of suppliers and reduce dependency on single sources; therefore, improving the resilience to supplier disruptions.
- **Inventory Management Practices.** Implement safety stock policies to balance inventory levels, enhance flexibility, and reduce the risk of stockouts. By maintaining safety stock, organizations can adapt to market changes while minimizing excess inventory. This approach provides a buffer to prevent stockouts, resulting in an efficient inventory management system that supports operational continuity.
- **Enhanced Cyber Security Measures.** Develop robust cyber security protocols, including implementing regular security audits, employee training, and updated software systems. These protocols reduce vulnerability to cyber threats.
- **Scenario Planning and Risk Assessment.** Regularly conduct scenario analyses to evaluate the potential impact of various risks, including natural disasters and geopolitical events. This improves preparedness and enables quicker and more agile capabilities to respond to unforeseen events.
- **Stronger Regulatory Compliance Programs.** The minimization of non-compliance risks and avoidance of penalties are results of stronger regulatory compliance programs, achieved by establishing dedicated compliance teams to actively monitor and adapt to regulatory changes in real time.
- **Workforce Development and Engagement.** Invest in workforce training and development, while fostering strong relationships with labor unions and stakeholders. This could reduce the likelihood of labor-related disruptions and increase employee retention.

¹¹ Reliability Imperative (misoenergy.org)

The electric sector's supply chain management faces numerous risks that can potentially disrupt operations and hinder service delivery. By implementing proactive risk mitigation strategies such as diversifying suppliers, enhancing cyber security, and strengthening compliance programs, organizations can fortify their supply chain resilience.

During evaluation of this risk, the CIPC and the EC teams realized that to address the risk fully, the following areas must be included:

- Technologies and Services
 - Includes the technologies and cyber security risks associated with the fuel supply, communications, and other critical services
- Software
- Hardware
- Expanded planning to include long lead times for:
 - Materials and contracts in project planning to provide more realistic timelines for project completion
 - Limited vendors and suppliers
 - Fuel supply concerns

To address the supply chain risk fully we must look at not only the durable goods that our vendors provide but also the services provided and the interdependencies of vendors who support our business.

See the following documents for more information on supply chain risk:

- [*Security Guideline for the Electricity Sector – Supply Chain – Secure Equipment Delivery*](#)
- [*NERC Initiative - Supply Chain Risk Mitigation Program*](#)
- [*Supply Chain Industry Coordination \(natf.net\)*](#)

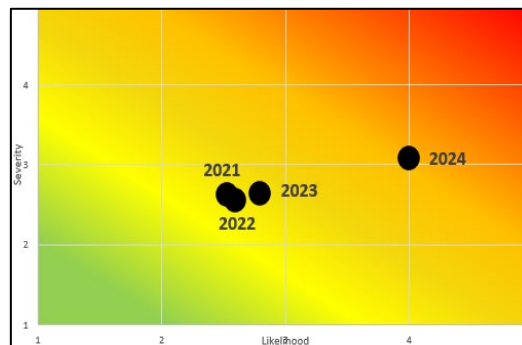
Risk 2: Extreme Weather (Monitor)



Identified and evaluated by the Operations Committee

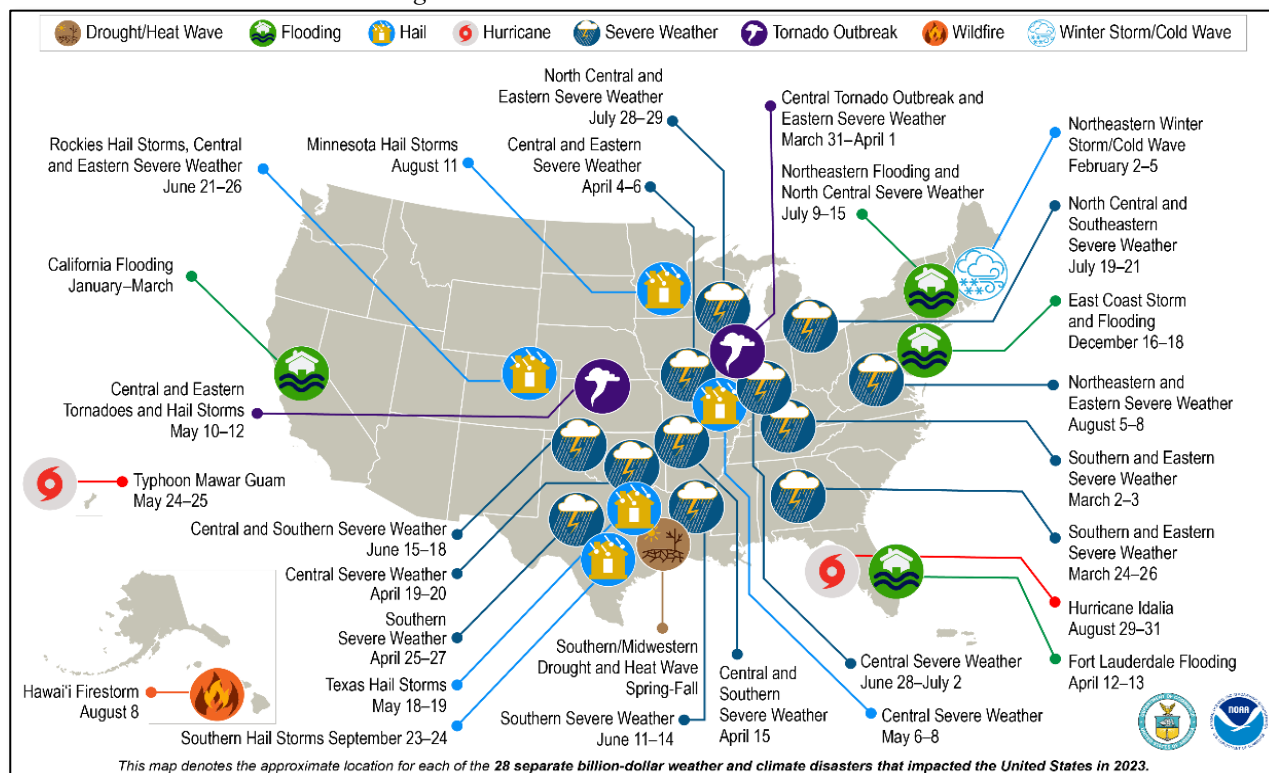
Definition/Justification of risk: The SERC Region frequently faces extreme weather events, such as cold weather with

ice and wind, and extreme heat, and typically endures tornadoes and tropical storms annually. These extreme weather conditions pose significant risks to BPS reliability by causing disruptions in fuel supply, loss of telecommunications, extended growing seasons that can lead to vegetation encroachment on transmission lines, reduced situational awareness, and forced outages of generation and transmission facilities.



Severe cold weather can disrupt natural gas pipeline supplies, cause freezing of critical piping or instrumentation at generation plants, and lead to frozen coal piles, all of which jeopardize the continuity of power generation. Conversely, extremely hot weather can impair generation capacity by limiting plant cooling options, reducing the availability of cooling water, and affecting hydroelectric generation due to drought or flooding. Additionally, in recent years, the United States and the SERC Region have experienced increasingly frequent and intense heat waves and heat domes. These extreme temperatures can reduce facility ratings and put additional strain on the system during peak demand periods. Weather-related challenges not only threaten grid reliability but also lead to substantial disruptions and resource strains. Figure 13 illustrates the impact of weather and climate disasters from 2023, as reported by the National Oceanic and Atmospheric Administration (NOAA).¹²

Figure 13: NOAA 2023 Weather and Climate Disasters

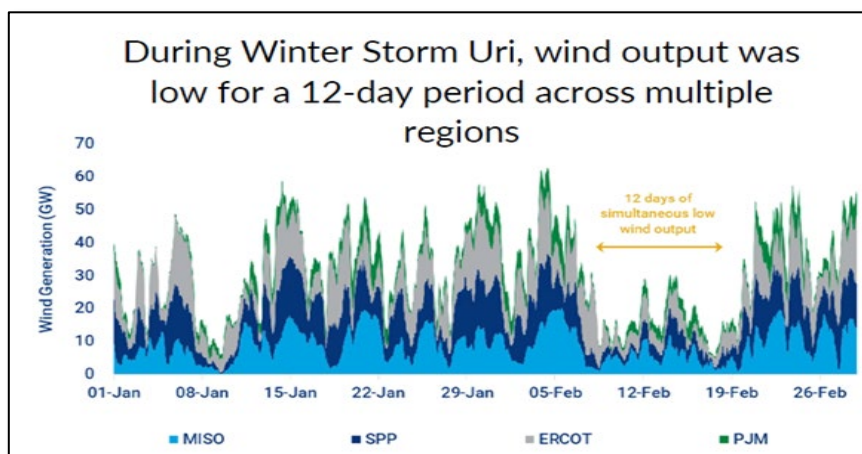


¹² 2023: A historic year of U.S. billion-dollar weather and climate disasters | NOAA Climate.gov

As the energy mix shifts with an increasing reliance on natural gas and inverter-based resources, reliability risks increase during winter, especially in the early morning or evening when temperatures are at their lowest, demand is high, and solar output is minimal. This situation becomes more critical if there is insufficient wind, a condition referred to as a wind drought.

Wind drought: MISO experienced a wind output of less than 1 percent of the wind generation nameplate aggregate for 39 consecutive hours on January 28-30, 2020. During this period, 20.2 GWs of wind decreased to almost 0. Another example of wind drought occurred during Winter Storm Uri (in February 2021), when wind output was low across multiple regions for 12 days (see [Figure 14](#)).

Figure 14: Wind Output during Winter Storm Uri



Solar in winter months: During the winter, a combination of shorter days with widespread cloud cover and fog greatly decreases solar output. Any calm night or calm cloudy day can create a potential capacity shortfall. This may become increasingly critical as we move away from conventional generation resources. In addition, uniform high-load factors leave little room for charging Battery Energy Storage Systems (BESS) or electric vehicle batteries.

“Imagine a night during a future wind drought where there is no solar, little wind and depleted batteries.”¹³

The potential of limited wind and/or solar output during winter and increased winter load during extremely cold weather limits the ability to restore reservoirs or charge batteries. Electrification exacerbates this issue.¹³

Mitigation Status and Strategies: The SERC Operating Committee (OC) actively mitigates the risks associated with extreme weather by utilizing the SERC OC [Conservative Operations Guideline](#), which was updated and approved in 2024. This guideline assists SERC members in developing conservative operations procedures tailored to the extreme weather challenges faced by the SERC footprint.

In addition, NERC’s [Reliability Guideline, Generating Unit Winter Weather Readiness](#) provides industry practices for preparing and operating through extreme cold weather. SERC has circulated this guideline among OC members to support their winter preparedness activities.

The OC updated a severe cold weather guideline document in 2024, drawing on member-submitted checklists. To maintain focus on preparedness, the OC established a standing agenda item at their fall meeting dedicated to cold weather operations.

¹³ Transition to renewables increases winter reliability risk (power-eng.com)

Guideline Documents:

- SERC OC [Conservative Operations Guideline](#)
- NERC's [Reliability Guideline, Generating Unit Winter Weather Readiness](#)
- SERC OC [Extreme Cold Weather Guidelines](#)

Sharing best practices is a cornerstone of SERC's approach. Members collaborate extensively during extreme weather events, utilizing mutual assistance for restoration efforts. SERC enhances regional resilience by covering Cold Weather Preparedness and Extreme Weather topics during annual outreach events, where entities exchange lessons learned and strategies for improving extreme weather resilience.

The OC has conducted a post-winter survey over the past four years, revealing ongoing improvements in generator preparedness, training, and knowledge regarding cold weather. The 2024 survey saw the highest participation yet, with around 80 percent of generator owners and operators contributing. Insights from this survey will guide the development of further mitigation efforts, including seminars, training sessions, and new guidelines.

The newly formed Generator Working Group is also addressing the implementation of new cold weather standards, such as EOP-011-3, and EOP-012-2. Additionally, the SERC Engineering Committee (EC) continues to evaluate the impacts of extreme weather in both seasonal and long-term planning horizons. The EC's working groups assess system constraints under higher-than-projected loads and widespread generation outages to identify potential vulnerabilities and improve regional support during extreme events.

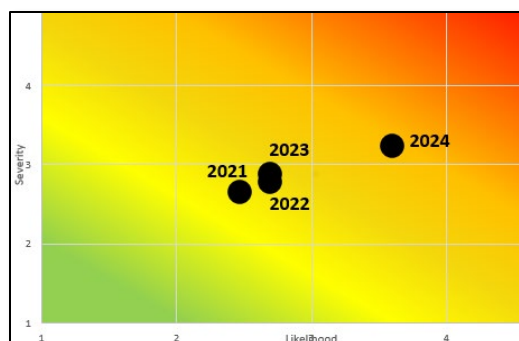
The EC is also incorporating the evaluation of extreme events into planning studies, including FERC Order 896, which initiated the forthcoming Transmission System Planning Performance Requirements for Extreme Temperature Events TPL-008 Reliability Standard. TPL-008 establishes transmission system planning performance requirements to ensure the BES operates reliably during extreme temperature events. While specific guidelines are still in development, these studies represent ongoing efforts to address the increasing frequency and severity of extreme weather events within the SERC footprint.

Risk 3: Resource Uncertainty (Manage)



Identified and evaluated by the Operations and Engineering Committees

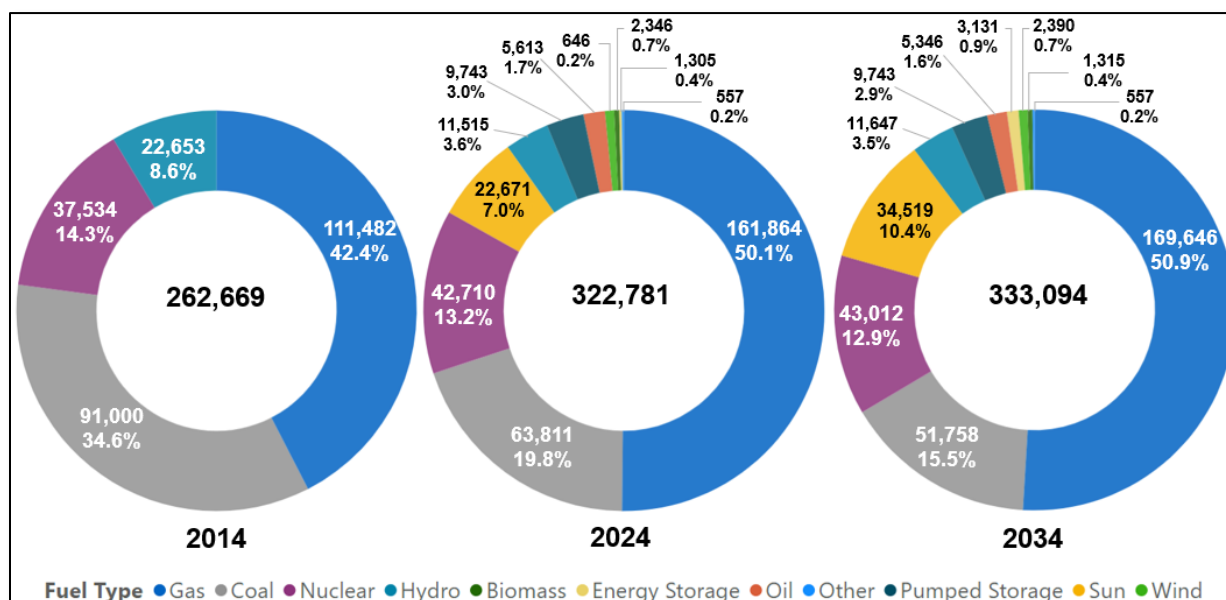
Definition/Justification of risk: The traditional methods of long-term planning, short-term planning, and real-time operations must adapt to the challenges presented by changes in generating resources. That are challenging uncertainties surrounding resource availability. While the energy mix is shifting toward renewable energy sources, the increasing reliance on natural gas complicates the narrative. Natural gas, while technically a fossil fuel, plays a crucial role in the current energy transition.



Natural gas remains a cornerstone of power generation, providing essential services that current renewables (like solar and wind) cannot fully supply, such as system inertia and ramping, and voltage regulation. Accelerating changes in these resources (see [Figure 15](#) and [Figure 16](#))—as well as a growing dependence on natural gas presents operational challenges, particularly as resource uncertainties—such as fuel deliverability and load variability—become more pronounced and complicates operations and planning.

The rapid integration of renewables, coupled with the ongoing retirements of coal plants, amplifies the need for reliable natural gas supply and infrastructure, especially since renewable energy sources currently lack the ability to provide essential ancillary services at the same level as conventional generation.

Figure 15: SERC Capacity Resources for 2014, 2024, and 2034



These transitions, driven by public policy shifts and market dynamics, are reshaping the SERC regional (and national) energy landscape. The reliance on natural gas, spurred by its lower emissions and replacement of coal, is set to increase as coal plants continue to retire. However, this creates a less diverse generation portfolio. Meanwhile, renewables such as photovoltaic solar and, to a lesser extent, wind are rapidly increasing in prominence, thanks to state-based

incentives and falling installation costs.¹⁴ Coal plant retirement drivers (see Figure 17),¹⁵ include state and corporate carbon reduction goals, federal emission reduction rules, and competition from other sources of generation.¹⁶ In Figure 18 shows a summary of renewable policies by state.¹⁷

Figure 16: U.S. Electricity Mix

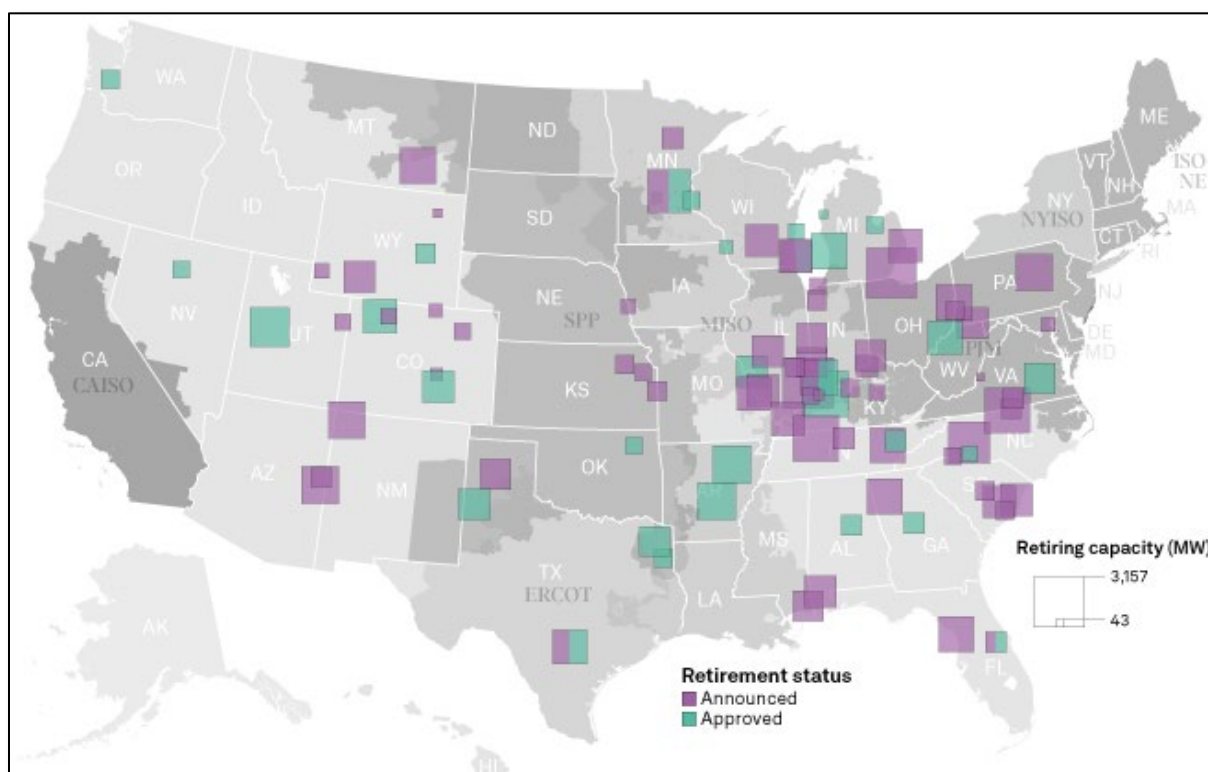
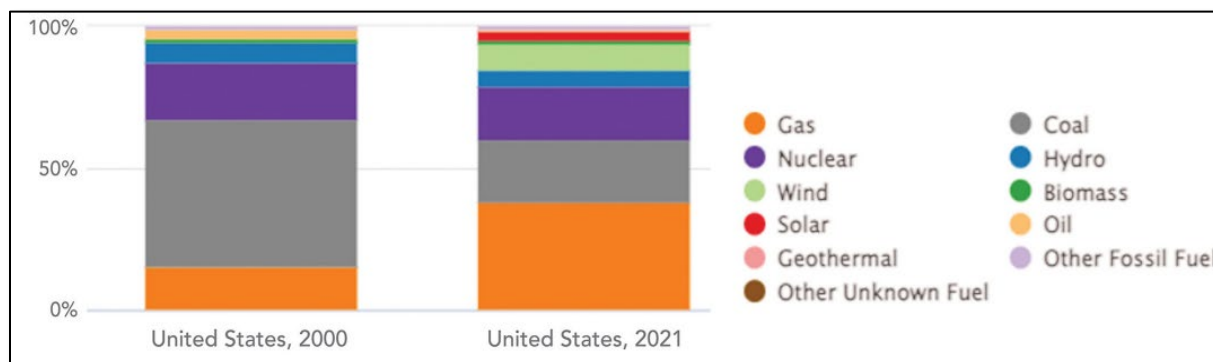


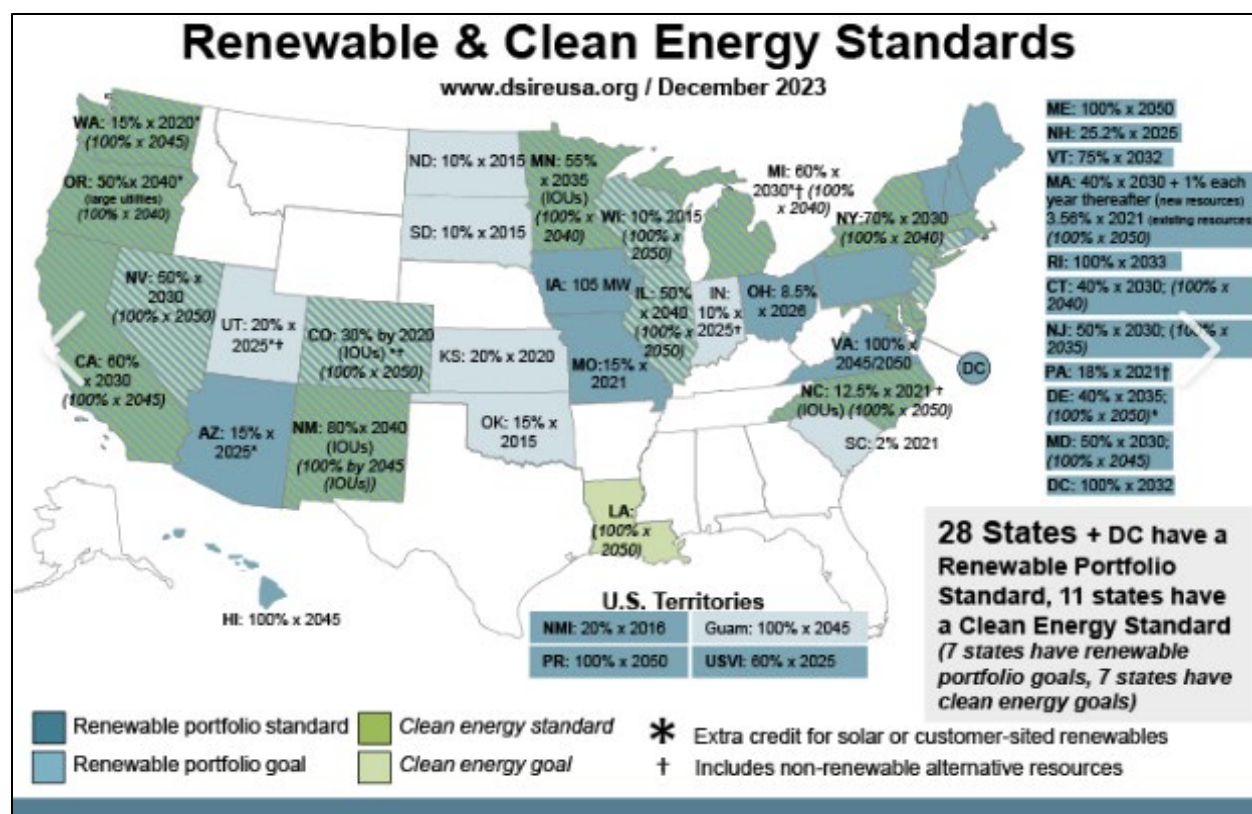
Figure 17: Future Coal Capacity Retirements (through 2025)¹⁶

¹⁴ Power Sector Evolution | US EPA

¹⁵ Mitigating Stranded Asset Risks to Utility Customers: an Exploration of Securitization and Retiring Coal Generation (naruc.org)

¹⁶ Inflation Reduction Act to accelerate US coal plant retirements | S&P Global Market Intelligence (spglobal.com)

¹⁷ Database of State Incentives for Renewables & Efficiency® - DSIRE (dsireusa.org)

Figure 18: DSIRE Renewable & Clean Energy Standard December 2023¹⁷

Although the risk is described as resource uncertainty, there is a need for continued improvements in planning approaches to support the reliability and security for the BPS through this transition and in the future. Historically, power generation was more predictable, with established resources that had well-understood dispatch capabilities and dynamic behaviors. These resources were connected to the grid in ways that allowed operators and planners to leverage years of documented knowledge and experience.

However, the new mix of generation of resources introduces greater complexity. Their availability fluctuates daily, and their dynamic behavior can be altered by simple firmware updates or software changes, making their operation less predictable. These newer resources are often smaller, quicker to deploy, and tend to cluster in areas with cheaper land, further complicating both real-time grid operations and short- and long-term planning efforts.

Inverter-based generators, in particular, require advanced Electromagnetic Transient (EMT) modeling to accurately predict and manage their behavior during transient events, which is critical for maintaining system stability. This complexity underscores the need for continuous improvements in planning approaches, ensuring that the BPS remains reliable and secure through this period of transition and beyond.

The 2018 PJM Grid Reliability and Resilience Study used the PJM system as a case study to illustrate the potential reliability consequences of two major risks: increased coal retirements and fuel insecurity. (See Figure 19.) The study showed that premature retirement of fossil fuel generation could adversely impact PJM's ability to meet reliability criteria.

PJM and other grid operators have issued warnings recently stating that the reliability of the grid in many regions of the country is at risk for electricity shortages due to:

- Premature retirement of dispatchable electricity resources driven by federal and state policies,
- Delays in new or replacement capacity coming on-line, and
- Increasing electricity demand (e.g., data centers and electrification).

Retirements have continued to increase since the 2018 study:

- More than one-third of the nation's coal fleet has retired.
- Electricity generators have announced plans to retire roughly one-third (more than 60,000 MW) of the remaining fleet within the next 5 years.
- In total, almost two-thirds of the nation's coal fleet will have retired by the end of 2028.¹⁸

Mitigation Status and Strategies: In 2023, MISO completed a foundational analysis of attributes, with a focus on three priority attributes where risk for the MISO system is most acute (see Figure 20):

- System adequacy is the ability to meet electric load requirements during periods of high risk. MISO focused on the near-term risk factors of availability, energy assurance, and fuel assurance.
- Flexibility is the extent to which a power system can adjust electric production or consumption in response to changing system conditions. MISO focused on the near-term risk factors of rapid start-up and ramp-up capability.

Figure 19: PJM's Increased Coal Retirements and Fuel Insecurity

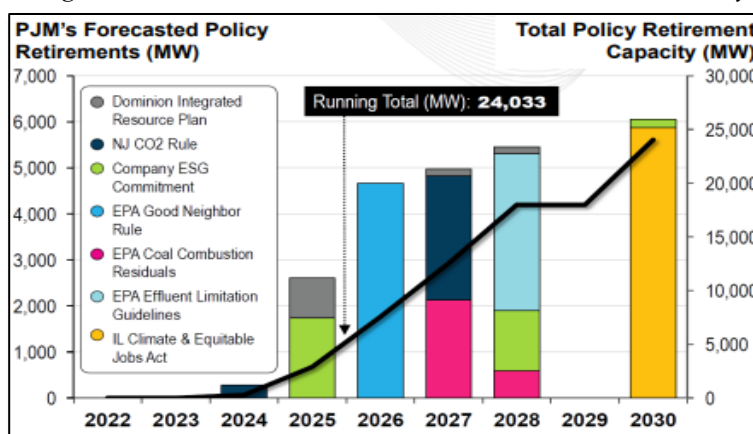
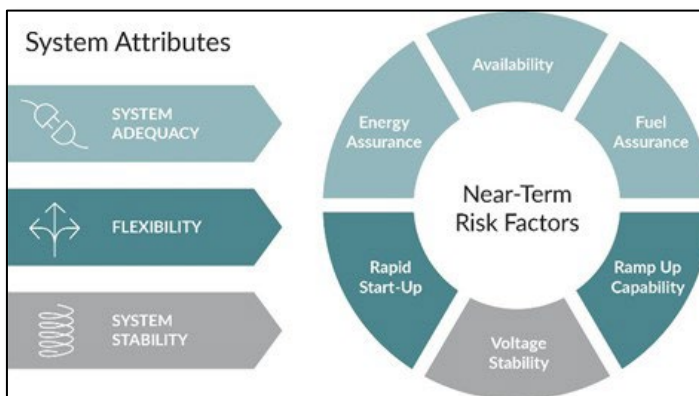


Figure 20: MISO System Attributes



¹⁸ <https://quanta-technology.com/wp-content/uploads/2020/05/Reliability-Challenges-Case-Study-for-PJM.pdf>

- System stability is the ability to remain in a state of operating equilibrium under normal operating conditions and to recover from disturbances. MISO focused on the nearest-term risk factor of voltage stability.¹⁹

Diverse Resource Needs:

- No single resource provides all necessary system attributes, necessitating a diverse resource fleet.²⁰

“The average time projects spend in queues before being built has increased. The typical project built in 2023 took almost five years between requesting an interconnection study to beginning commercial operations, compared to three years in 2015 and less than two years in 2008, the researchers said.”²¹ (See [Figure 21](#).)

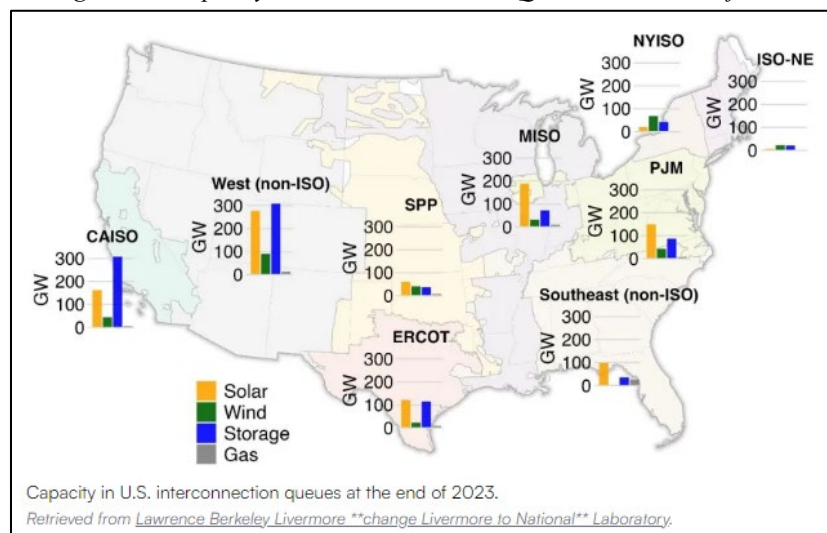
Figure 21: Berkeley Lab Active Projects in the Queues

Region	<i>n</i> (active)	Capacity (GW)
CAISO	995	523.3
ERCOT	1,090	269.2
ISO-NE	405	51.2
MISO	1,669	311.5
NYISO	492	131.6
PJM	3,309	286.7
SPP	703	144.9
Southeast (non-ISO)	1,134	173.3
West (non-ISO)	2,044	706.5

“Variable solar and wind resources contribute a smaller percentage of their nameplate capacity to resource adequacy and peak load compared to dispatchable generation like natural gas, the researchers noted.”

“Decarbonizing the electric sector requires higher levels of installed solar and wind capacity to achieve the same resource adequacy contributions.” (See [Figure 22](#).) Capacity in US Solar, battery and wind projects make up 95 percent of the capacity in the interconnection queues. More than 1 TW of solar and 1 TW of battery storage are in the queues, mostly in hybrid

Figure 22: Capacity in US Interconnection Queues at the end of 2023²¹



projects, followed by 360 GW of wind and 79 GW of gas-fired generation.²¹

To mitigate these risks, it is recommended that the SERC OC utilize insights from the Engineering Committee’s Variable Energy Resources Working Group (VERWG). The VERWG should present updates on their activities and best practices at future SERC OC technical committee meetings, allowing the OC to identify and address any pertinent issues.

¹⁹ Reliability Imperative (misoenergy.org)

²⁰ PowerPoint Presentation (lbl.gov)

²¹ Queued Up: 2024 Edition

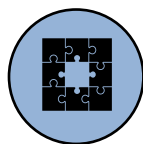
A gap exists in the modeling and monitoring of Distributed Energy Resources (DERs)²² within Energy Management Systems (EMS). To address this issue, it is recommended that SERC OC members with relevant expertise share best practices through presentations. Key topics presented so far include the registration and integration of DERs, as well as methods for modeling DERs within the EMS.

At its semi-annual meetings, the OC typically presents best practices, such as for solar and battery facilities. Additionally, the OC's newly established Generator Working Group discusses upcoming standard changes that address resource uncertainty, as well as share lessons learned.

The Planning Coordination Subcommittee (PLCS) will begin evaluating additional studies, including the impact of gas pipeline unavailability, the concentration of variable energy resources (like solar sites) in specific areas, and the need for resource diversity to enhance reliability. The Resource Adequacy Working Group (RAWG) will use its annual data collection to trend changes in the generation fleet and perform probabilistic sensitivity analyses to assess the impact of these uncertainties on the SERC footprint's resource adequacy.

²² Distributed Energy Resources (DERs) are small-scale energy systems that can be connected to the electricity grid or isolated to provide energy to nearby locations. DERs can include energy generation technologies and energy storage systems. DERs can be connected to the grid directly or on a customer's premises, behind the utility meter.

Risk 4: Shortage of Required Skillsets (Monitor)



Identified and evaluated by the Critical Infrastructure Protection, Engineering, and Operations Committees

Definition/Justification of risk: The electric sector is facing a significant

challenge in the form of a shortage of required skillsets. As the sector becomes increasingly digitized and reliant on advanced technologies, there is a growing need for skilled professionals who can effectively manage and protect critical infrastructure. However, the availability of individuals with the necessary expertise and experience is not keeping up with demand.

This shortage of skillsets poses various risks to the electric sector. It hampers organizations' ability to effectively respond to cyber threats and vulnerabilities. Without experienced professionals, identifying and mitigating potential risks becomes a daunting task, leaving critical systems exposed to attacks and disruptions. Additionally, the shortage of skillsets creates difficulties in implementing and maintaining robust cyber security measures, increasing the likelihood of successful cyber-attacks.

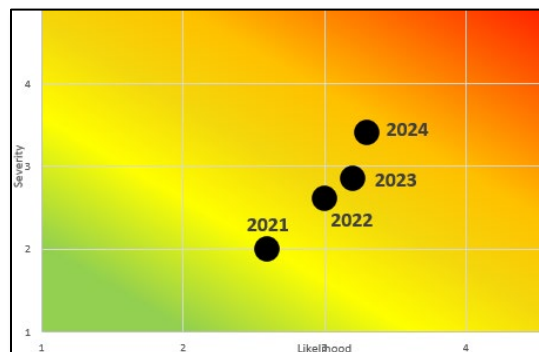


Figure 23: Skill Development



To address these risks, organizations in the electric sector have implemented several mitigation measures. One approach is investing in training and development programs to upskill existing staff and fill skill gaps. Offering specialized training courses, certifications, and workshops can help employees acquire the necessary knowledge and expertise in cyber security specifically tailored to the electric sector. By enhancing the capabilities of their teams, organizations can better respond to emerging threats and mitigate vulnerabilities effectively (see [Figure 23](#)).²³

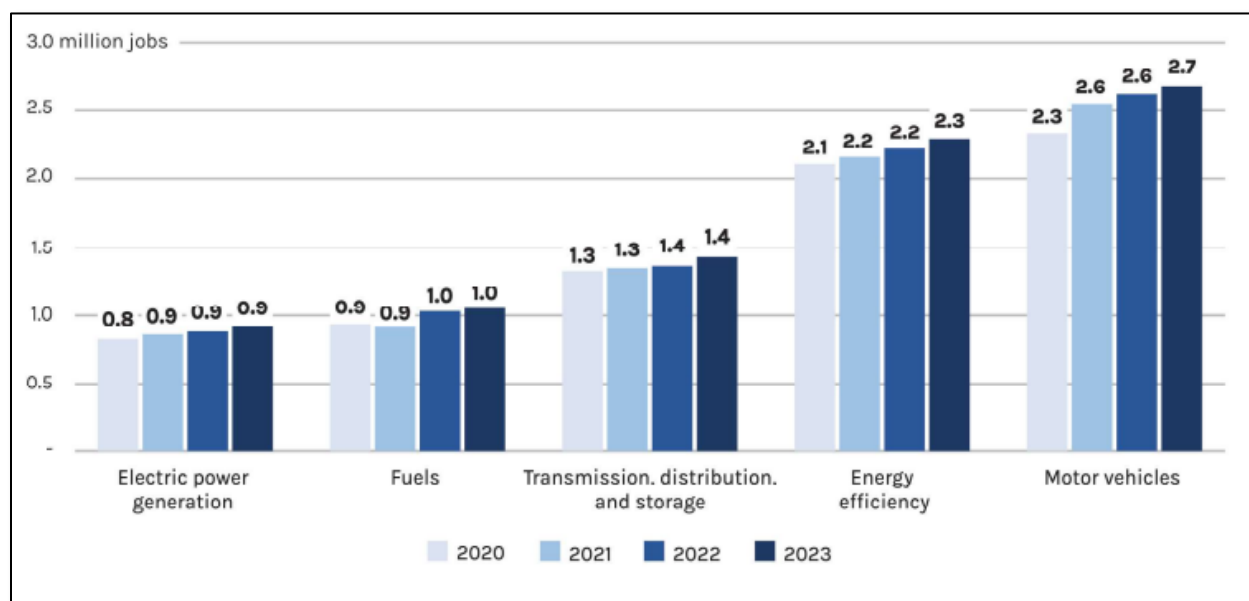
The US energy sector added 250,000 jobs in 2023 for a total of 8.35 million jobs. Of the energy jobs added, 146,000 (56 percent) were clean energy employment (see [Figure 24](#)).²⁴ Utilities are competing directly for employees with these clean energy companies in some cases. Additionally, there is a growing need for cyber security professionals across all industries. According to the Cyber security Workforce Study by ISC2 for 2024, an estimated 1.8 million cyber security jobs will go unfilled.²⁵

²³ https://www.researchgate.net/figure/Recommendations-for-skill-development-in-India_fig1_356566465

²⁴ 2024 USEER FINAL.pdf (energy.gov)

²⁵ ISC2-Cybersecurity-Workforce-Study.ashx

Figure 24: Energy Employment by Technology, 2020-2023 (Millions of Jobs)



The workforce has stabilized, suggesting focus may now turn to the composition of the workforce in terms of skills and diversity, as well as improving retention of existing energy workers. (See [Figure 25](#).)

Job Losses in the Energy Industry

“The United States Energy & Employment Report 2023 noted that from 2021 to 2022, the energy sector jobs grew 3.8 percent and clean energy jobs grew 3.9 percent, outpacing overall U.S. employment, which increased 3.1 percent in the same period.

Prior to the COVID-19 pandemic, the energy sector was one of the nation’s fastest-growing job markets. From 2015 to 2019, the annual growth rate for energy employment in the U.S. was 3.0 percent—double the 1.5 percent job growth in the U.S. economy as a whole. As of 2022, the energy sector has recovered 71 percent of the jobs lost in 2020. The energy sector has added back 596,000 of the 840,000 jobs lost during the first year of the pandemic.”

To address this trend, the industry will need to redirect some of its focus from recruiting to training, mentorship, and other programs to develop its younger, less experienced workforce.²⁶

The Center for Energy Worker Development’s (CEWD) 2023 Energy Workforce Survey Results highlight a critical shift in the energy sector’s workforce dynamics. While the workforce has stabilized, the focus must now shift toward enhancing workforce composition, particularly in terms of skills, diversity, and retention. The survey reveals a trend: 56 percent of the overall workforce have less than 10 years of service, with engineers and line workers exceeding 60 percent.²⁷

As shown in [Figure 26](#), SERC Audit staff have noticed an increasing number of System Operators with under five years of experience. Younger operators also tend to have shorter tenures, often transitioning into other roles due to career advancement opportunities. This underscores the urgency for the industry to pivot from a primary focus on recruitment to prioritizing training, mentorship, and development programs aimed at nurturing and retaining a younger, less experienced workforce. These efforts will be essential to building a skilled, diverse, and resilient energy workforce for the future.²⁷

²⁶ United States Energy & Employment Report 2023:

²⁷ CEWD-2023-Workforce-Report-Executive-Summary_2023-FINAL-1.pdf

Figure 26: CEWD Percent of All Employees by Age Group Figure 27: Percent of Employees by Years of Service

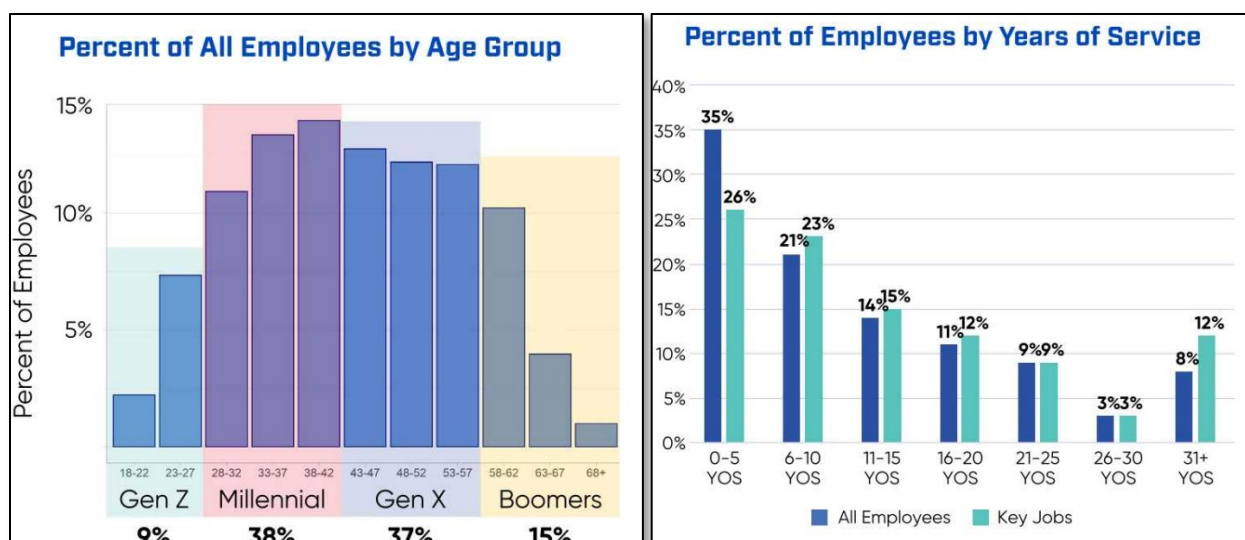


Figure 27 shows both retirement (in red) and non-retirement attrition (in blue), along with total hires (represented by the line) dating back to 2008. This data shows that both the non-retirement attrition and total attrition have increased significantly in the last two years and in 2022 hit their highest levels since the survey began. Hires to replace those workers are also at an all-time high. This follows a dip in these metrics in 2020 amid the pandemic and the uncertainty it brought.²⁸

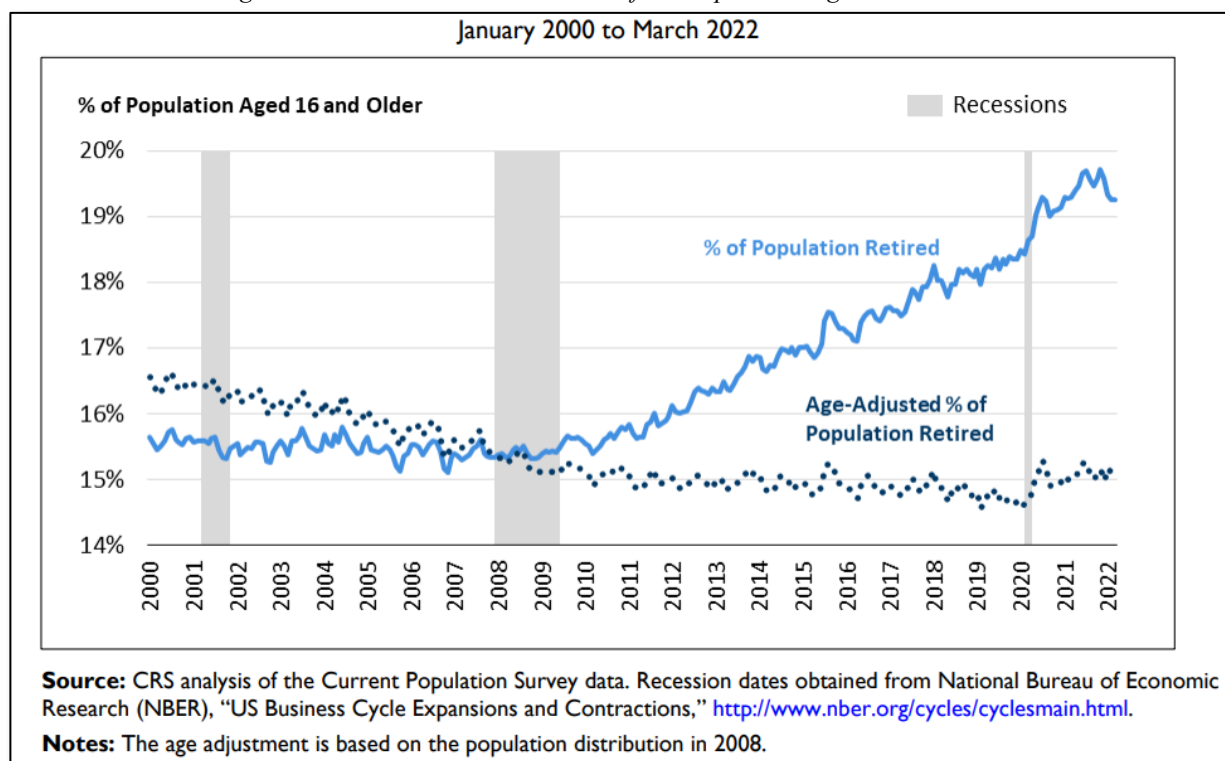
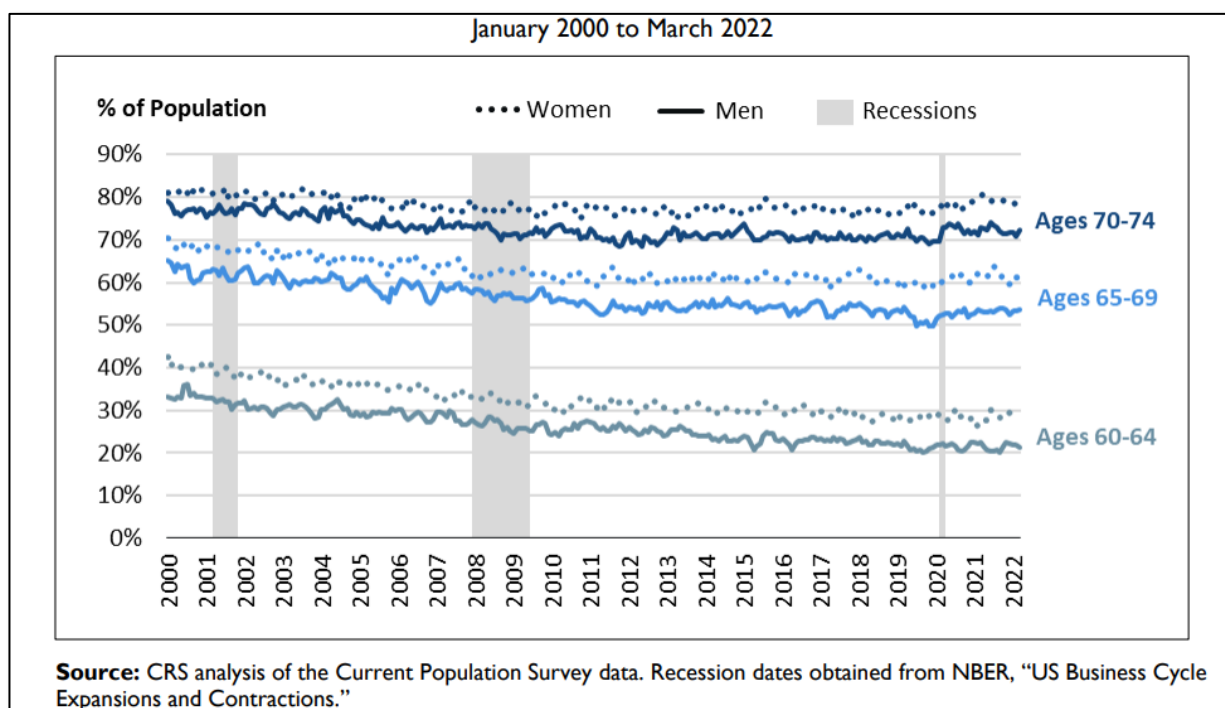
Figure 25: Total Attrition vs Total Hires – All Companies



The Congressional Research Service – Retirement Trends in the United States 2000-2022 indicates there is not an elevated rate of retirement age-adjusted percentage of the population retired. Figure 28 shows that the share of the population aged 16 and over who were retired stayed relatively stable at 15-16 percent from 2000 to 2011 before steadily increasing over the January 2012 (16.1 percent) to March 2022 (19.3 percent) period. Much of the increase since 2012 is attributed to the relatively large number of baby boomers (i.e., those born between 1946 and 1964) who are gradually attaining retirement age. (See Figure 29.)²⁹

²⁸ CEWD-2023-Workforce-Report-Executive-Summary_2023-FINAL-1.pdf

²⁹ Congressional Research Service: Retirement Trends in the US 2000-2022: IN11959 (congress.gov)

Figure 28: Retired Workers as Share of the Population Aged 16 and Older²⁹Figure 29: Retired Workers as a Shared Population by Select Age Groups and Gender²⁹

Mitigation Status and Strategies: To address skill gaps in the electric sector, organizations are implementing several key strategies:^{30,31}

- **Upskilling Through Targeted Training:** Organizations are investing in specialized training programs, certifications, and workshops to enhance the expertise of their existing staff, equipping them to effectively respond to emerging threats.
- **Fostering Collaborative Partnerships:** By collaborating with educational institutions, industry associations, and government agencies, organizations are developing a pipeline of skilled professionals through internships, apprenticeships, and scholarships specifically tailored to the electric sector's needs.
- **Expanding Recruitment Efforts:** To attract a diverse talent pool, organizations are broadening their recruitment strategies to include professionals from related industries, who possess transferable skills.
- **Cooperative Partnerships with Colleges and Universities:** SERC members are actively engaging in strategies to address workforce gaps in the electric sector by forming partnerships with academic institutions. These efforts include offering internships and cooperative education (co-op) programs at member companies, establishing mentorship programs to enhance industry knowledge, and serving as board members on Industrial Advisory Boards at colleges and universities. These initiatives aim to bridge the skills gap and strengthen the future workforce in the energy sector.
- **Leveraging Advanced Technologies:** Automation and advanced technologies, such as AI-driven monitoring tools and automated incident response systems, are being utilized to reduce reliance on scarce human resources, enabling existing professionals to focus on strategic tasks.

In the SERC footprint, the talent loss due to the return-to-office mandates has led utilities to consider remote work options, retention bonuses, and cost-of-living adjustments, although the effectiveness of these measures remains unclear. To strengthen their workforce, utilities are intensifying recruitment efforts, particularly at college job fairs, to showcase the advantages and challenges of working in the utility industry, especially in the context of the clean energy transition (e.g., wind, solar, DER). Emphasizing the industry's critical role in supporting communities, the economy, and key sectors is essential.³²

To remain competitive, utilities must continuously assess and adjust total compensation packages, ensuring they compare favorably with those in other industries. Additionally, expanding co-op and internship programs in key areas is crucial for building strong relationships between employers and educational institutions, as well as between employees and employers. Many utilities in the SERC footprint are already involved in consortiums, such as the Center for Energy Workforce Development, which helps build a pipeline of skilled talent for the energy sector.

The SERC System Operator Working Group (SOWG) continues to provide essential in-person and virtual training opportunities through its annual System Operator Conferences, benefiting system operators, support personnel, and operations management. Additionally, the

³⁰ Hedge, J. W., & Rineer, J. R. (2016). Workforce planning for talent management. *Journal of Applied Psychology*

³¹ International Energy Agency (IEA). (2020). *Skills development for energy sector innovation*

³² Recommendation from the Center for Energy Workforce Development. See Gaps-In-Energy-Careers-Report-2024_Final-pages.pdf (cewd.org)

Engineering Committee is providing an annual transmission planning and engineering training seminar, sponsored by members, to provide technical training focused on model building and engineering studies that are crucial to BPS reliability.

Key Recommendations for Building a Sustainable Energy Workforce Pipeline: ³³

- **Expand Career Awareness Initiatives:** Target youth, low-income students, women, military personnel, and transitioning workers to raise awareness of energy careers.
- **Promote Diversity, Equity, and Inclusion (DE&I):** Integrate DE&I into every step of workforce pathways to ensure the industry workforce reflects the communities it serves.
- **Implement Strategic Workforce Planning:** Prioritize workforce development strategies at the highest levels, with clear career pathways to reduce turnover within the first five years.
- **Pathways for Growth:** Companies must develop and set clear career pathways for growth within the company, to reduce turnover within the first five years.
- **Support State Energy Workforce Consortia:** Develop partnerships between energy employers, educational institutions, labor, and government entities to create programs tailored to skilled energy positions.
- **Focus on Renewable Energy:** Align training programs with the projected growth in the renewable energy sector to meet the demand for jobs in this area.

In conclusion, while the shortage of skills in the electric sector is a significant challenge, organizations are proactively addressing it through training, collaboration, strategic recruitment, and technology adoption. These efforts, supported by initiatives like those from SERC and SERC University, are crucial to develop, maintain, and enhance critical skillsets that contribute to maintaining a resilient and secure energy infrastructure.

³³ Recommendation from the Center for Energy Workforce Development. See Gaps-In-Energy-Careers-Report-2024_Final-pages.pdf (cewd.org)

Risk 5: Exploitation of Vulnerabilities (Manage)



Identified and evaluated by the Critical Infrastructure Protection Committee

Definition/Justification of risk: The electric sector has become increasingly digitalized and

interconnected, resulting in numerous benefits such as increased efficiency and improved services. However, this reliance on technology has also introduced various cyber security vulnerabilities that threat actors can exploit. Figure 30³⁴ is a visual example of the threat landscape. In this brief narrative, we will highlight some of the vulnerabilities and the corresponding mitigation measures implemented to safeguard the electric sector.

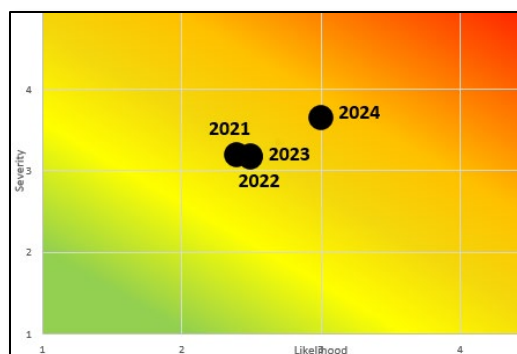
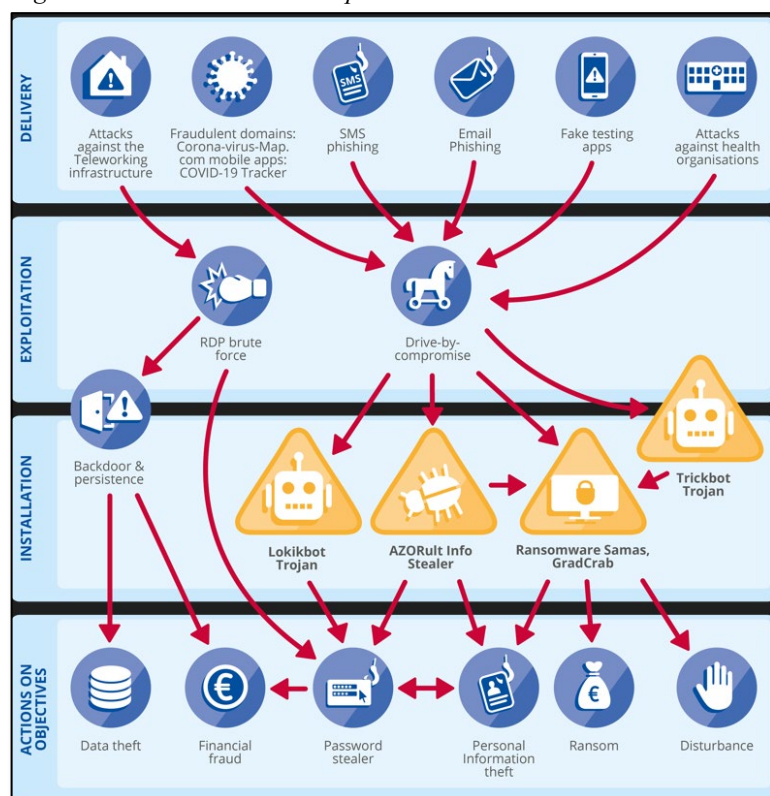


Figure 30: The Threat Landscape



The interconnected nature of the electric sector also creates vulnerabilities. Smart grids, which allow for the efficient flow of electricity by integrating various components, are susceptible to cyber-attacks. Threat actors can exploit weaknesses in the communication channels between different components, disrupting power distribution or even causing physical damage. To counter this, organizations have implemented measures like network segmentation and strong encryption protocols to protect the integrity and confidentiality of data transmitted between components. Cyber security vulnerabilities in the electric sector are a considerable concern due to the increasing reliance on digital technology. However, industry stakeholders have recognized the importance of safeguarding critical

infrastructure and have implemented various mitigation measures. By upgrading infrastructure, securing communication channels, protecting IoT devices, fostering employee awareness, and collaborating with relevant partners, the electric sector aims to enhance cyber security and ensure the reliable and secure delivery of electricity to consumers.³⁵

³⁴ What is the threat landscape, and why should you care? - Security Boulevard

³⁵ 3 Alarming Threats To The U.S. Energy Grid – Cyber, Physical, And Existential Events (forbes.com)

Key Elements:

- **Zero-Day Exploits:** Unknown vulnerabilities can be exploited before patches are available, posing severe risks.
- **Unpatched Software:** Running outdated software with known vulnerabilities on systems makes them easy targets for attacks.
- **Misconfigured Systems:** Incorrect settings or default credentials that expose vulnerabilities, can lead to potential breaches.
- **Inadequate Network Segmentation:** Poorly divided networks allow attackers to move laterally and access more sensitive areas.
- **Insufficient Security Controls:** Lack of proper security measures like intrusion detection systems (IDS) and firewalls increases the likelihood of successful exploitation.

Mitigation Status and Strategies: To effectively safeguard our industry, we must go beyond merely meeting compliance standards and focus on proactive risk management through comprehensive frameworks and internal controls programs. The rapid evolution of technology and the increasing sophistication of cyber threats demand continuous adaptation of our security measures to protect the BPS. Below are key initiatives and efforts that support this goal:

- **CIPC CIP Tools Working Group Initiatives:** The CIPC CIP Tools Working Group plays a vital role by introducing new technologies that strengthen BPS infrastructure resilience. This group fosters collaboration among members, committees, subcommittees, and working groups, allowing the industry to:
 - Leverage collective experience and expertise,
 - Share best practices, and
 - Address operational and security challenges more effectively.
- **Whitepaper Development:** Ongoing projects include the development of critical whitepapers to guide best practices:
 - Managing Transient Cyber Assets (TCAs) – Led by the CIP Tools Working Group
 - Securing Remote Sites (e.g., wind and solar installations) – Led by the CIPC Physical Security Subcommittee (PSS)
- **Educational Resources:** SERC CIPC sponsors a variety of courses throughout the year to educate members on essential cyber and physical security fundamentals and their practical applications.
- **Innovative Technology:** Researchers at Pacific Northwest National Laboratory (PNNL) have developed “*ChatGrid*,” an AI-powered tool that enhances grid operators' ability to monitor the power grid efficiently. Key benefits include:
 - Rapid access to relevant grid information through natural language queries.
 - Accelerated decision-making, particularly during disruptions (e.g., storms, equipment failures); and
 - Faster power restoration, boosting overall grid resilience.

By embracing new technologies, fostering collaboration, and prioritizing proactive risk management, our industry can build a more secure and resilient power grid, capable of withstanding the evolving landscape of threats.³⁶

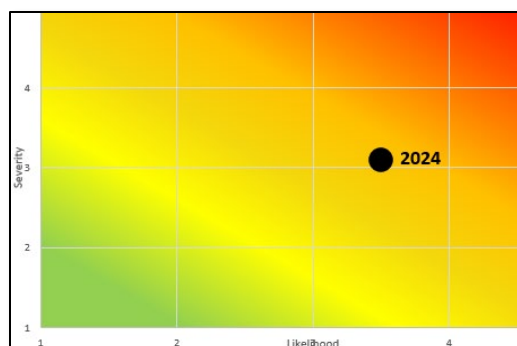
³⁶ New 'ChatGrid' technology could revolutionize the way we monitor our power grid: 'A new way to look at data' (thecooldown.com)

Risk 6: Rapid Nontraditional Load Growth (Manage)



Identified and evaluated by the Operations and the Engineering Committees

Definition/Justification of risk: The SERC footprint faces a major reliability risk due to rapid load growth driven by new data centers and crypto mining facilities, leading to potential transmission and distribution system stress. With some areas experiencing forecasted annual load increases of over 8 percent, planning and system enhancements are necessary to prevent transmission issues and potential resource inadequacies.



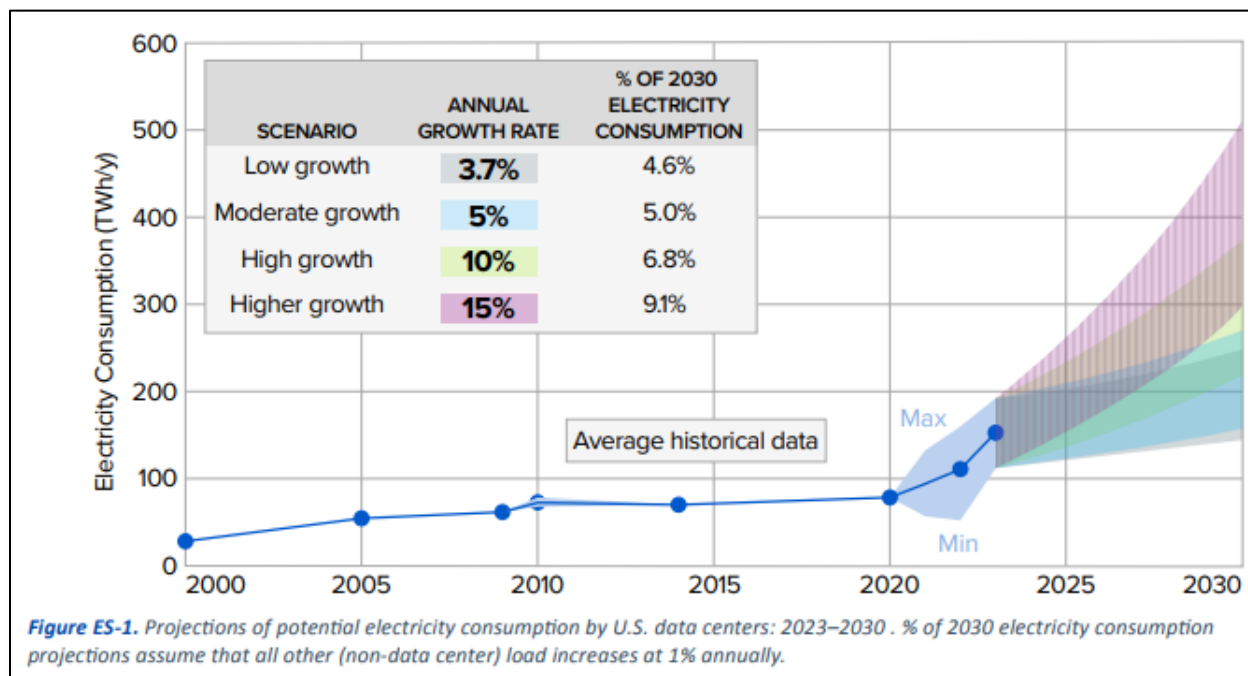
The southeastern US is facing significant challenges due to projected load growth in electricity demand.

The reliability of the BPS is critical, always requiring a balance between generation and demand. Increased load growth stresses aging infrastructure, leading to potential equipment failures and more frequent maintenance needs. When combined with the current risks of Extreme Weather and Supply Chain, this emerging risk is exacerbated.

Factors Driving Load Growth:

Data Centers and AI: The surge in data centers, driven by AI and machine learning, requires substantial power,³⁷ with some data centers consuming up to 1000 MW. [Figure 31](#)³⁸ and

Figure 31: Data Center Energy Consumption³⁸



³⁷ Moody's Report Reveals Surge in Data Center Demand Driven by AI Boom (datacenterknowledge.com)

³⁸ Powering Intelligence: Analyzing Artificial Intelligence and Data Center Energy Consumption (epri.com)

Figure 32³⁹ show this demand is significantly higher, most often due to the intensive energy consumption of Graphics Processing Units (GPUs) used in AI model training.

Figure 32: Data Centers by State³⁹

State	Data Centers
Virginia	473
California	279
Texas	279
Ohio	156
Illinois	154
New York	128
Florida	122
Oregon	109
Washington	90
Arizona	86
Georgia	84

Electric Vehicles (EVs): The Transportation Electrification in the Southeast 2023 report uses data from sources like the U.S. Census Bureau, Atlas EV Hub, and the U.S. Department of Energy, tracking key metrics such as EV sales, manufacturing jobs, utility investments, public funding, and charging infrastructure deployment.

EV adoption is rising, with projections showing EVs could account for 10 percent of peak power use by 2035. This transition offers multiple benefits, including reduced emissions, improved public health, job creation, and economic development. However, these gains depend on effective public policy and investment.

In Q2 2023, EV sales in the Southeast reached 6.2 percent of new light-duty vehicle sales, led by Georgia, Florida, and North Carolina, while Alabama trailed at 2.5 percent. The Southeast region also saw a 60 percent increase in DC fast charging stations, with Tennessee leading at 85 percent.

Manufacturing Investments: Federal incentives are driving a surge in U.S. manufacturing investments, especially in semiconductors, batteries, and solar panels, increasing electricity demand. The Bipartisan Infrastructure Law and Inflation Reduction Act (IRA) have unlocked significant funding for Electric Vehicle (EV) infrastructure through programs like National Electric Vehicle Infrastructure (NEVI) and the Charging and Fueling Infrastructure (CFI) Grant Program, supporting EV charger deployment across the Southeast. The region is becoming a major EV manufacturing hub, with over 60,000 announced jobs—more than a third of all U.S. EV manufacturing roles—led by Georgia in job creation. Figure 33.⁴⁰

Figure 33: Announced EV Manufacturing Jobs by State⁴⁰

State	Announced EV Jobs
Florida	314
Alabama	2,058
South Carolina	10,611
North Carolina	11,723
Tennessee	12,719
Georgia	27,817
Southeast Total	65,242

³⁹ USA Data Centers (datacentermap.com)

⁴⁰ Transportation Electrification in the Southeast, 2023 Report (cleanenergy.org)

Federal support from the Bipartisan Infrastructure Law and Inflation Reduction Act is accelerating EV infrastructure development, with \$5 billion allocated for public charging. As seen in [Figure 34](#) and [Figure 35](#).⁴¹

Figure 34: Southeast Electric Vehicles Indicators through June 2023⁴¹

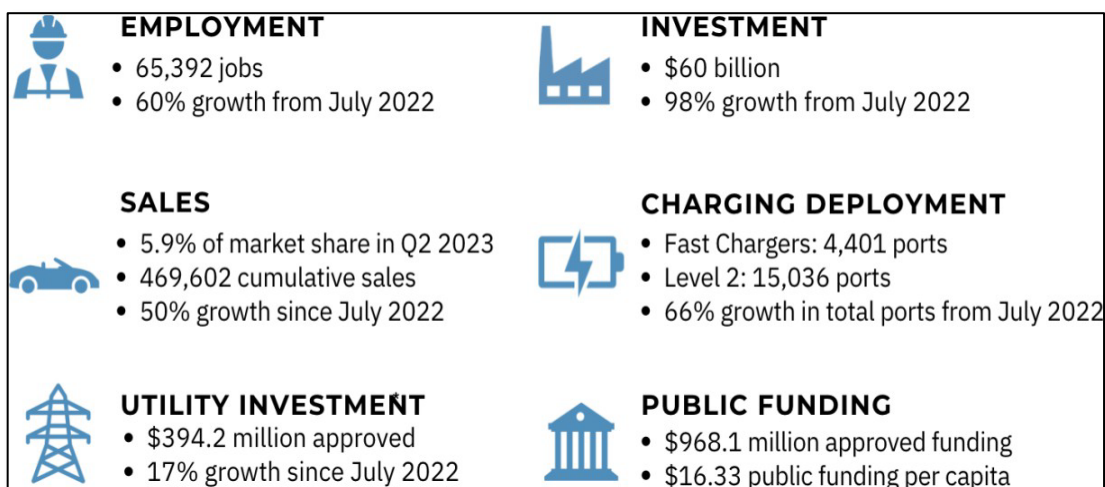
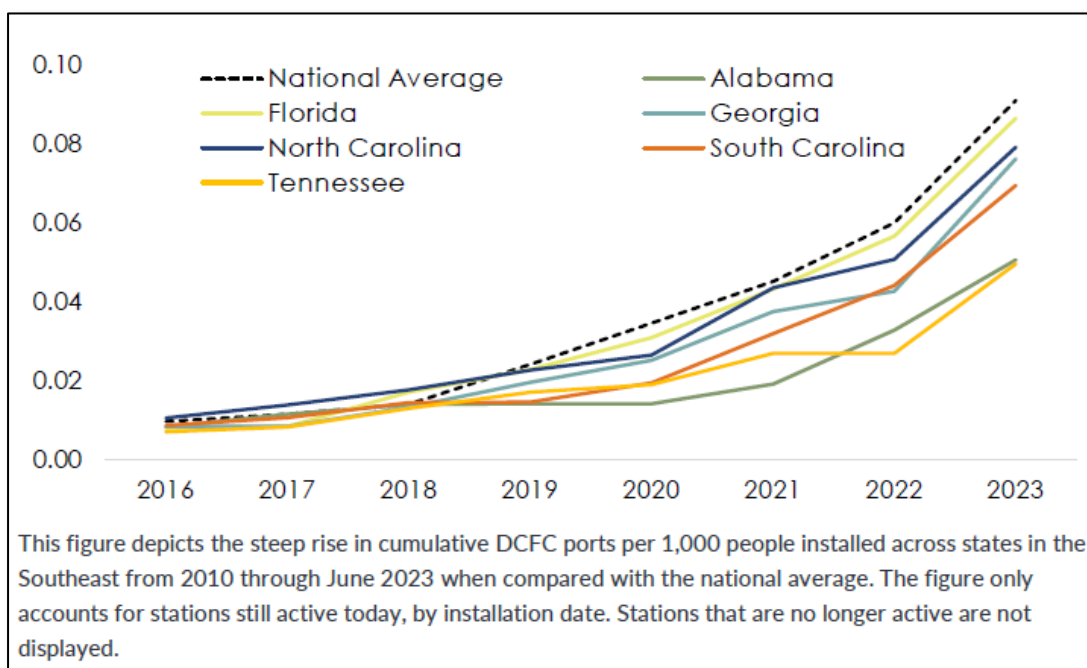


Figure 35: Cumulative Direct Current Fast Charging Ports per 1,000 People through June 2023⁴¹



Population Growth: The southeastern US is experiencing significant population growth, further driving up electricity demand (see [Figure 36](#)).⁴²

The projected load growth in the southeastern US presents significant risks and challenges that require immediate and strategic action. By focusing on the reliability of the BPS, addressing infrastructure strains, managing baseload generation, balancing generation-load-transmission timelines, and overcoming supply chain and labor constraints, the electric industry can work

⁴¹ Transportation Electrification in the Southeast, 2023 Report (cleanenergy.org)

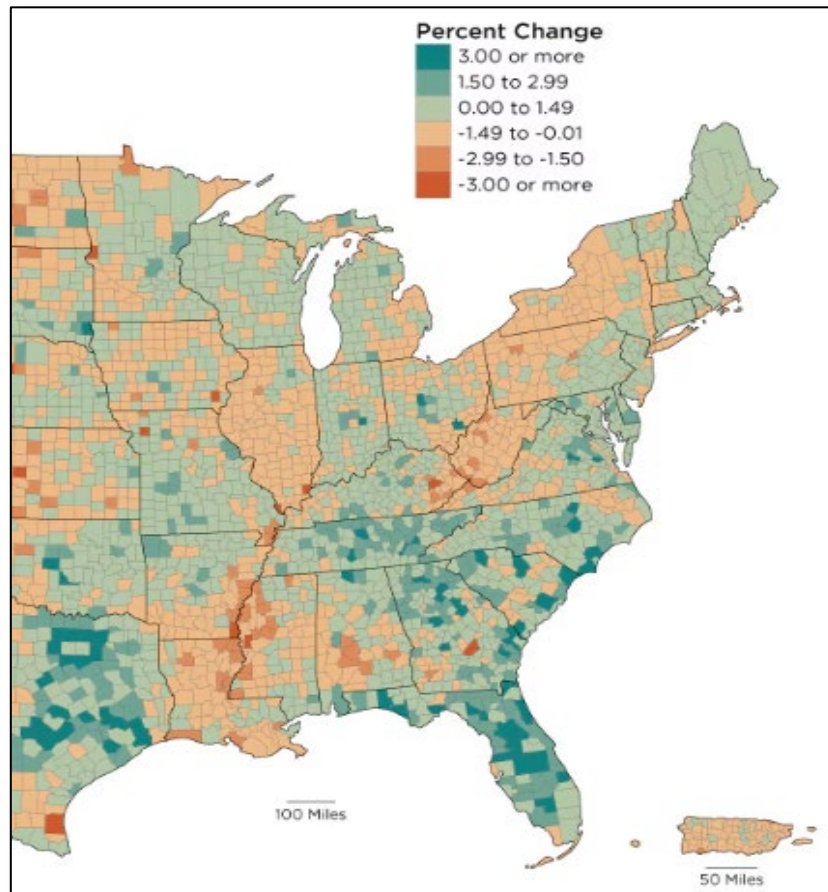
⁴² Percent Change in County Population: July 1, 2022, to July 1, 2023 (census.gov)

toward ensuring a stable and reliable electricity supply. The integration of new technologies and resources will be critical in meeting these demands while maintaining grid reliability.

Mitigation Status and Strategies: The Rapid Nontraditional Load Growth risk was nominated and accepted in 2024, and subsequently added to the 2024 Risk Registry and SERC Regional Risk Report. Given the pressing timeline, the SERC Technical Committees have had limited time to develop comprehensive mitigation strategies.

Here are some initial high-level suggestions for potential mitigation strategies and approaches as the SERC Technical Committees undertake a more detailed review and analysis of this risk and its impact on the BES:

Figure 36: Percent Change in County Population July 1, 2022, to July 1, 2023⁴²



- **Infrastructure Upgrades:** Enhance transmission and distribution systems to manage increased load from data centers, crypto mining, and population growth.
- **Leverage Federal Incentives:** Utilize funding from the Bipartisan Infrastructure Law and Inflation Reduction Act to support EV infrastructure and manufacturing investments.
- **Balance Generation and Load:** Ensure a stable power supply by expanding generation capacity and integrating flexible technologies.
- **Adopt New Technologies:** Implement energy storage and demand response systems to efficiently handle rising electricity demand.
- **Address Supply Chain and Labor Issues:** Overcome constraints to ensure timely infrastructure improvements and reliable grid operations.

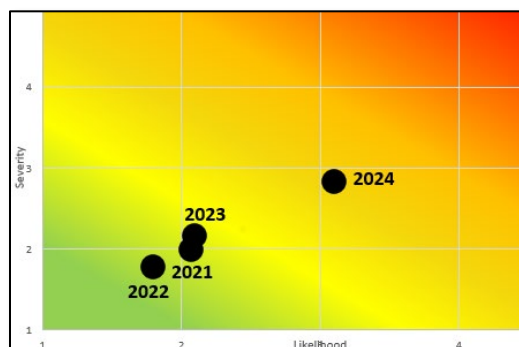
Risk 7: Lack of Fuel Diversity and Availability (Manage)



Identified and evaluated by the Operations and Engineering Committees

Definition/Justification of risk: Fuel sourcing, and its possible disruption because of weather or other extreme natural events, has the potential to adversely affect reliable operation of the BPS. Scenarios and case studies regarding the problems that can evolve from reliable fuel sourcing (or lack of it) show an ever-growing list of interdependencies that affect planning and operations.

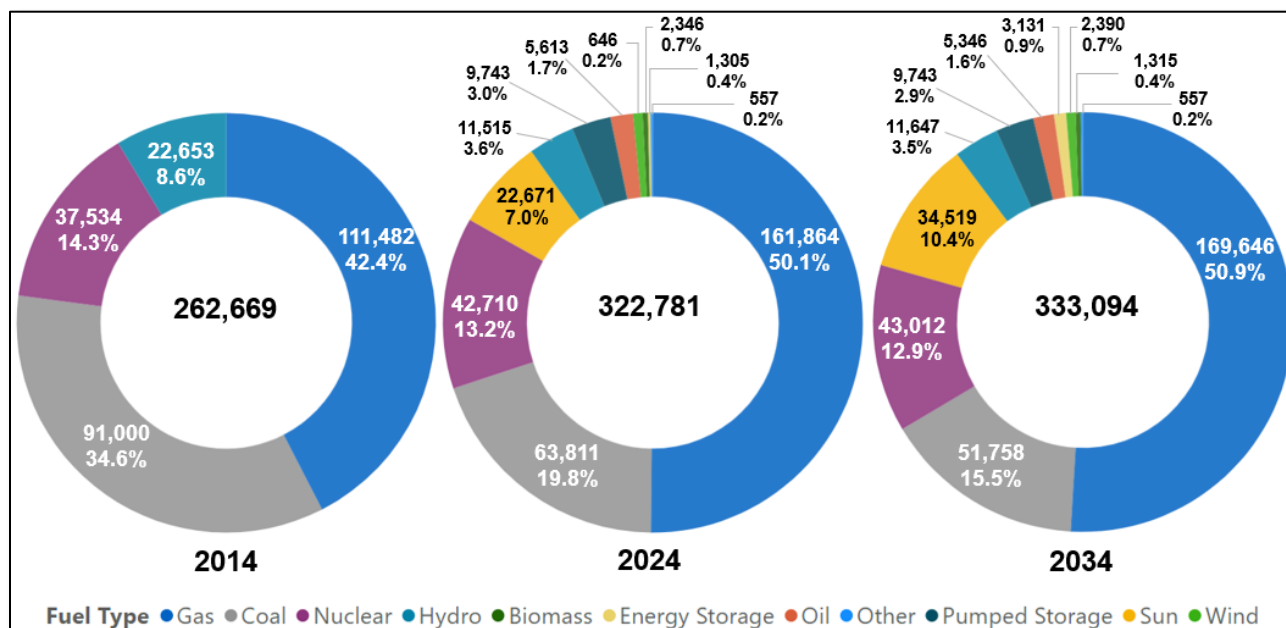
Natural gas currently accounts for more than 40 percent of generation within the SERC footprint, a reliance expected to grow as coal plants retire. While natural gas helps bridge the gap between retiring coal and the expansion of renewables, this shift poses challenges to fuel diversity, as the grid becomes more dependent on a single fuel type.



This Photo by Unknown Author is licensed under CC BY-SA

Ensuring the resilience of natural gas infrastructure and advancing renewable technologies is critical as increased reliance on natural gas reduces energy portfolio diversity and limits the system's ability to respond to fuel disruptions. Volatility in natural gas supply and aging infrastructure affect generation dispatch, while inadequate pipeline capacity for winter peaks heightens the need for fast-acting units. Additionally, environmental groups, activists, and public perception could significantly influence natural gas policy as dependence on this fuel source grows.

Figure 37: SERC Capacity Resources for 2014, 2024, and 2034 (SERC Database)



Reliability Assessment Portal Data

Year	Gas	Coal	Nuclear	Solar	Hydro	Pumped Storage	Oil	Energy Storage	Wind	Biomass	Other
2034	50.9%	15.5%	12.9%	10.4%	3.5%	2.9%	1.6%	0.9%	0.7%	0.4%	0.2%
2024	50.1%	19.8%	13.2%	7.0%	3.6%	3.0%	1.7%	0.2%	0.7%	0.4%	0.2%
2014	42.4%	34.6%	14.3%	0.0%	8.6%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%

Natural gas generation runs the risk of extended supply problems from natural gas infrastructure disruptions. Additionally, the availability and supply of natural gas can spike dramatically due to extreme cold weather, hurricanes, geopolitical incidents, and governmental policy. SERC has the delegated authority and a responsibility to annually collect data on Bulk Electric System generation facilities to conduct resource adequacy assessments.⁴³

Bulk Electric System generation is defined as a single unit greater than 20 MVA or a site (collection of generators) greater than 75 MVA.

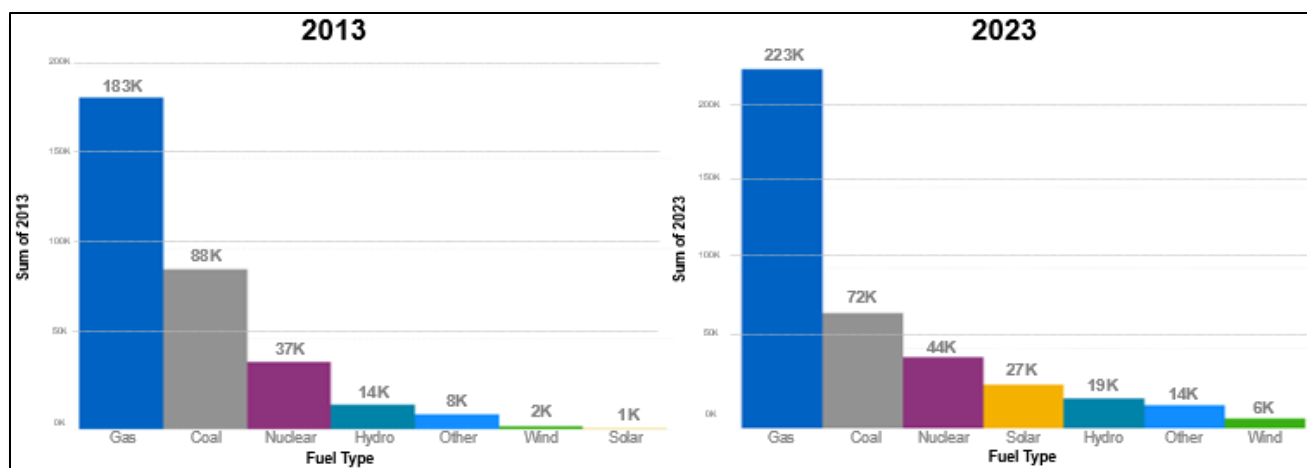
The NERC Glossary of Terms defines BES resources as "Dispersed power producing resources that aggregate to a total capacity greater than 75 MVA (gross nameplate rating), and that are connected through a system designed primarily for delivering such capacity to a common point of connection at a voltage of 100 kV or above."

The U.S. Energy Information Administration (EIA) under the Energy Policy Act of 1992 has jurisdiction/authority to annually collect data for all generators 1 MW or greater.

Figure 38 shows the EIA data change in generation from 2013 through 2023. The trends are comparable to those of the SERC Bulk Electric System; however, the total amount of generation captured in the EIA is much greater in magnitude. This demonstrates an additional area of risk. Of note, in 2023 the EIA reported 27,247 MW of solar generation (1 MW and above), whereas in 2024, the capacity reported to SERC for the Bulk Electric System data (75 MW and above) report shows a total of solar generation of 22,671 MW.

⁴³ On February 16, 2023, FERC issued an order Approving Extreme Cold Weather Reliability Standards, Docket RD23-1000, paragraph 11, directing NERC to collaborate with FERC staff to create a plan for collecting data on winterization.

Figure 38: Changing Capacity Resources for 2013 and 2023 (EIA Data)

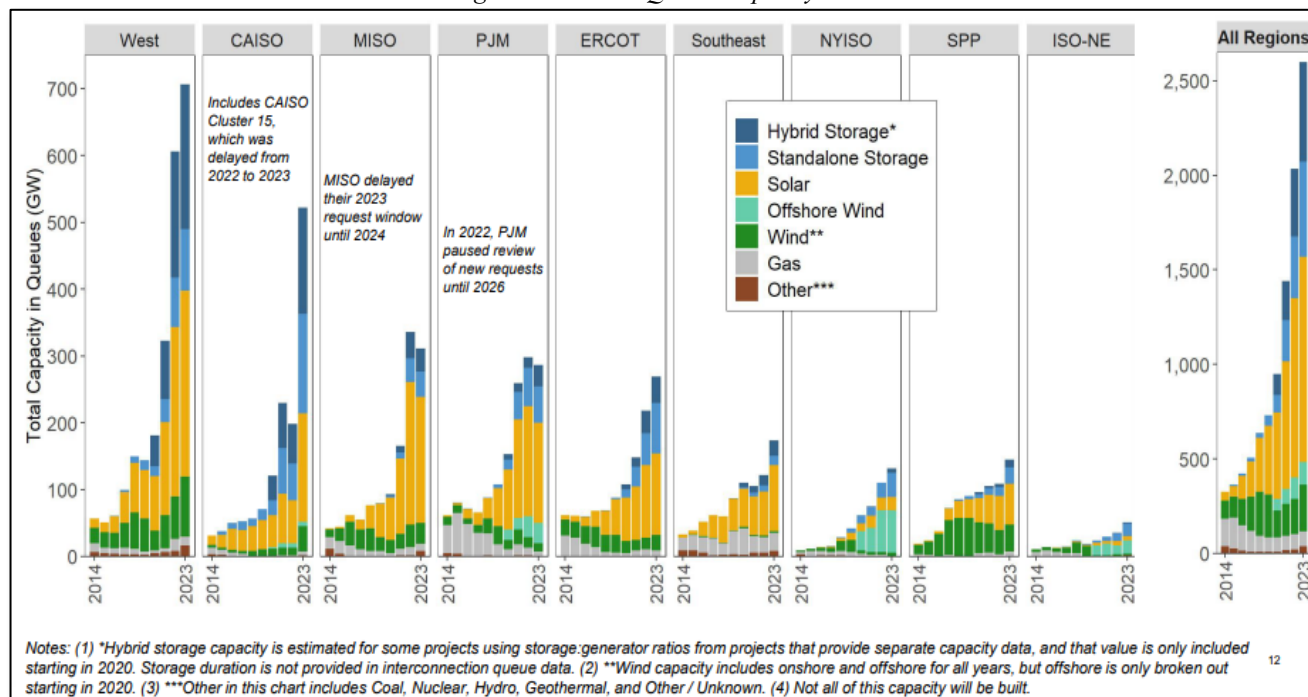


EIA Capacity Data

Year	Coal	Gas	Hydro	Nuclear	Solar	Wind	Other
2023	17.7%	55.1%	4.6%	10.9%	6.7%	1.5%	3.5%
2013	26.6%	55.1%	4.1%	11.1%	0.2%	0.5%	2.5%

Figure 39 shows the Active queue capacity is highest in the West (706 GW), followed by CAISO (523 GW). Several regions have delayed accepting or processing new requests due to backlogs.⁴⁴

Figure 39: Active Queue Capacity



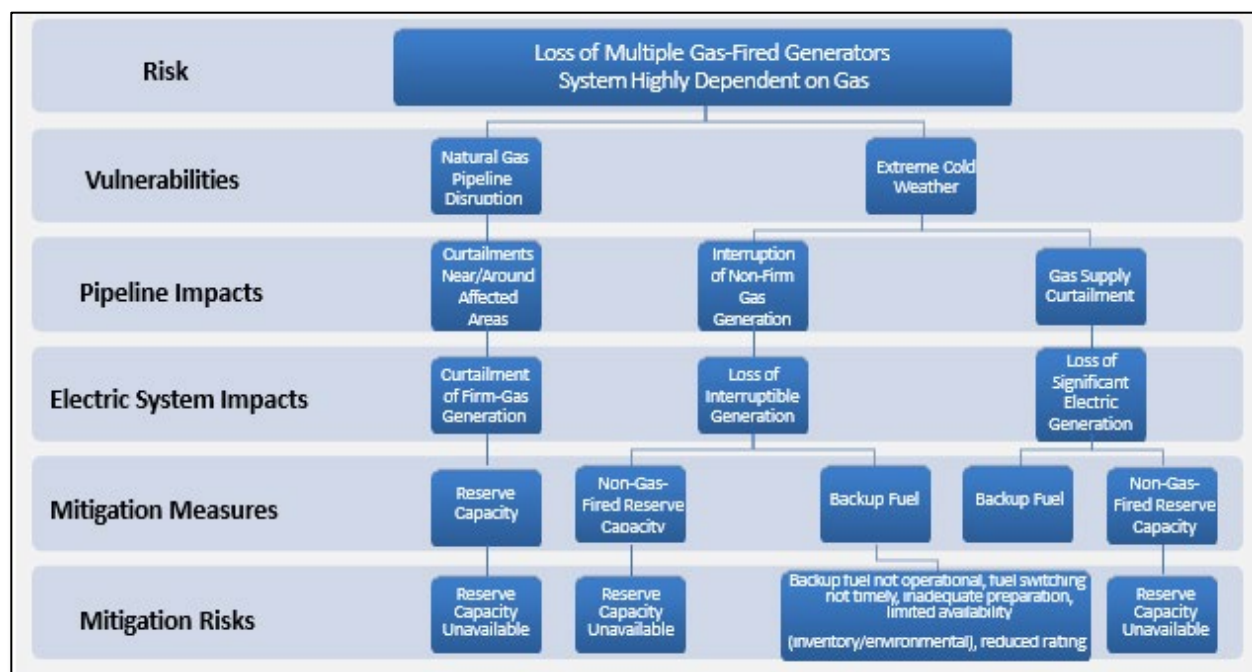
⁴⁴ Queued Up: 2024 Edition Characteristics of Power Plants Seeking Transmission Interconnection As of the End of 2023 (lbl.gov)

The retirement of coal and nuclear resources has positioned natural gas as the key fuel for U.S. power generation, making the gas and electric systems not merely interdependent but fully interconnected. This shift has created a new reliability paradigm, where the integration of gas and electric grids enhances both reliability and vulnerability. Cold weather events like Winter Storms Elliott and Uri have underscored the critical interdependence of these systems, revealing the need for more comprehensive risk management.

The Bulk Electric System, while bolstering reliability, also increases vulnerability due to its interconnected nature, as evidenced by historical blackouts in 1965 and 2003. These vulnerabilities led to the establishment of NERC, mandated by Congress to develop and enforce reliability standards. As the energy mix continues to evolve, these standards have adapted, yet the gas-electric interconnection remains a double-edged sword—supporting reliability and resilience while posing significant risks.

Moreover, the growing threats from cyber and physical attacks on both the gas and electric industries have sparked calls for the creation of a Natural Gas Reliability Organization (GRO), paralleling NERC's role in the electricity sector. This would ensure that the interconnected Bulk Energy System is robust enough to withstand and manage risks from any source until scalable, long-duration energy storage or clean energy production becomes viable.⁴⁵ [Figure 40](#)⁴⁶

Figure 40: Risk Assessments of Impacts and Mitigation



The output of inverter-based resources is, by nature, highly variable. This makes maintaining energy adequacy much more difficult. Traditional reserve margins based on traditional generation likely will not be sufficient for the generation resources supporting the grid in the future.

⁴⁵ Gas-Electric Systems Evolve Beyond Interdependency | Fortnightly

⁴⁶ 2013 Special Reliability Assessment: Accommodating an Increased Dependence on Natural Gas for Electric Power

Mitigation Status and Strategies: In response to fuel sensitivity risks, the OC has developed a multi-pronged strategy that includes ongoing monitoring and best practice sharing among members. The efforts of EC subcommittees and working groups focus on data collection, analysis, and strategic planning to address fuel type dependencies and resource adequacy challenges.

Below are the key components of these strategies and associated initiatives:

- **Tracking NERC Electric-Gas Working Group (EGWG) Efforts:**
 - The OC will track activities and insights from the NERC Electric-Gas Working Group (EGWG) to stay informed on electric-gas coordination and risk management.
- **Annual Fuel Type Trend Review:**
 - An annual review of generation resource trends based on fuel type will enhance awareness of fuel sensitivities and emerging risks.
- **Utilization of Existing Tools:**
 - The OC will continue leveraging existing tools, such as Conservative System Operations, to manage capacity constraints effectively.
 - Additionally, it is crucial to ensure that the operations perspective remains integrated into long-term planning scenarios.
- **Reliability Review Subcommittee (RRS) Data Collection:**
 - The RRS will continue to collect long-term data on the fuel types that comprise the generation fleet in the SERC footprint.
 - This data provides a foundation for guiding the RRS in identifying necessary assessments for the annual SERC Reliability Assessment.
- **Resource Adequacy Working Group (RAWG) Biennial Analysis:**
 - The RAWG will use the collected fuel type data and supplemental industry information, such as EIA 411 data, to conduct biennial probabilistic analyses.
 - These analyses will evaluate how shifts in fuel sources impact the resource adequacy of the SERC footprint.

The insights gained from these efforts will support members, the region, and policymakers in determining updated resource reserve margins, ensuring a reliable and resilient generation portfolio.

See these NERC documents for more information on this risk:

- [Reliability Guideline Fuel Assurance and Fuel-Related Reliability Risk Analysis for the Bulk Power System, March 2020](#)
- [Reliability Guideline Natural Gas and Electrical Operational Coordination Considerations](#)

Risk 8: Legacy Architecture Compatibility (Manage)

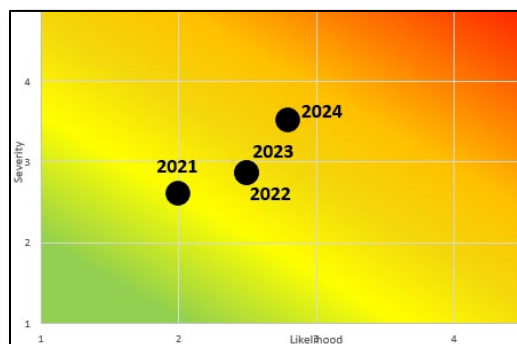


Identified and evaluated by the Critical Infrastructure Protection Committee

Definition/Justification of Risk:

Legacy architecture within the electric industry refers to outdated

infrastructure and systems that are still in operation. These legacy systems often lack modern security features, making them particularly vulnerable to cyber-attacks and other threats. The risks posed by these outdated systems are significant, as malicious actors can exploit them to disrupt critical operations and compromise the security of the electric grid.



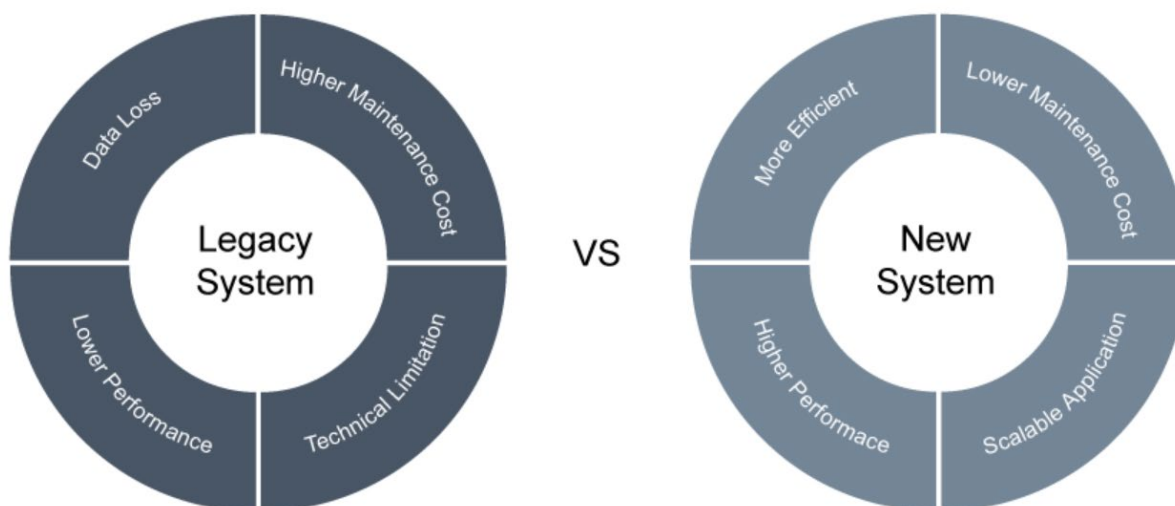
To address these risks, the electric industry has implemented a comprehensive mitigation strategy aimed at modernizing and securing its infrastructure. This strategy includes significant investments in upgrading infrastructure, replacing outdated equipment and software with more secure and resilient solutions.

Key components of this strategy include the implementation of strong access controls and network segmentation to minimize the potential impact of any attack. Regular vulnerability assessments and penetration testing are conducted to proactively identify and remediate security weaknesses before they can be exploited.

The industry also fosters collaboration with government agencies, intelligence organizations, and the private sector to share threat intelligence and adopt best practices. Additionally, robust training and awareness programs are in place to educate employees on safe computing practices and the latest security protocols.

Through these comprehensive mitigation measures, the electric industry is committed to enhancing cyber security, safeguarding critical infrastructure, and ensuring the continued reliability and security of the electric grid.

Figure 41: Legacy System vs New System



Mitigation Status and Strategies: Legacy architecture remains a significant vulnerability within the BPS as it struggles to keep pace with rapid technological advancements. The growing number of bad actors exploiting these vulnerabilities highlights the urgent need for robust mitigation strategies. Below are key efforts and resources aimed at addressing these risks:

- **Challenges of Legacy Architecture:**
 - Legacy systems are increasingly susceptible to cyber threats as they lack the capability to integrate modern security features.
 - Replacing or upgrading this equipment may be necessary in both the short and long term to mitigate these risks effectively.
- **Resources for SERC Members:** Fortunately, there are various resources available to help SERC members maintain and protect legacy systems:
 - The CIPC CIP Tools Working Group plays a critical role by introducing new technologies that strengthen BPS infrastructure.
 - This group creates a collaborative space for sharing experiences, best practices, and solutions to common challenges.
- **Industry Guidance and Knowledge Sharing:** Staying informed on risks associated with legacy equipment is crucial, and many industry outlets offer guidance and learning opportunities:
 - Events hosted by the ERO, Edison Electric Institute (EEI), and the North American Transmission Forum (NATF) provide valuable platforms for sharing knowledge.
 - Peer group exchanges allow participants to benefit from collective expertise and experiences.
- **Research and Innovation:** Ongoing research from National Labs and the Electric Power Research Institute (EPRI) is exploring innovative technologies to enhance BPS security.
 - These advancements not only improve security but also often bring enhanced functionality, helping drive the development of the “Grid of the Future.”

By fostering continued collaboration and innovation, SERC members can develop stronger, more secure systems. Working together ensures the integrity and resilience of the entire BPS, protecting it against both current and emerging threats.

Risk 9: Extreme Physical Events: Sabotage & Attacks (Manage)



Identified and evaluated by the Critical Infrastructure Protection Committee

Definition/Justification of risk:

Extreme physical events, specifically deliberate sabotage, and attacks, pose a significant threat to the electric sector. These man-made events can result in widespread disruptions to the power grid, leading to serious economic and societal consequences. It is important to define what Extreme Physical Events are and are not.

The Department of Justice defines intent or “mens rea”⁴⁷ as: “the resolve or determination with which a person acts to commit a crime.” The intent of the individual to disrupt the BPS whether they are successful or not defines it as an Event that meets the threshold of an Extreme Physical Event. Vandalism and copper theft are not Extreme Physical Events.

The largest risk considerations with Extreme Physical Events are considered interdependent with cyber security and the prospective impact of replacing long lead-time equipment damaged due to an Event. [Figure 42](#) shows the recent trend in physical attacks and vandalism.⁴⁸

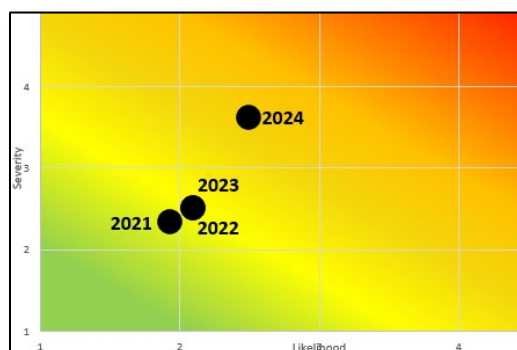
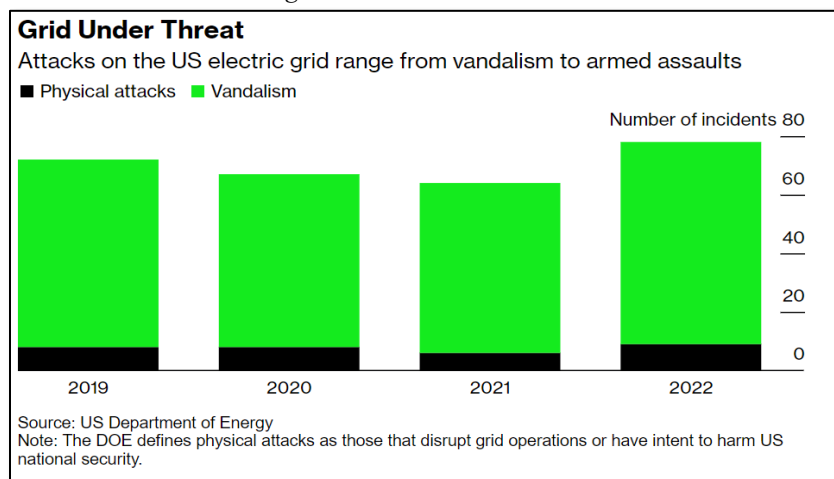


Figure 42: Grid Under Threat



Mitigation Status and Strategies:

To address this risk, the electric sector has implemented various mitigation steps. One crucial mitigation measure is enhancing the physical security of critical infrastructure. This includes implementing access controls, perimeter fencing, and surveillance systems to deter unauthorized access and monitor suspicious activities. Security personnel are

stationed at key facilities to monitor and respond to potential threats promptly.

To further mitigate the risk of sabotage and attacks, the electric sector emphasizes the importance of robust incident response and recovery plans. These plans outline the coordinated actions to be taken in the event of an attack, ensuring a swift and effective response. Regular drills and exercises are conducted to test the plans and identify areas for improvement.

Collaboration and information sharing play a vital role in mitigating man-made physical events. The electric sector partners with law enforcement agencies, intelligence organizations, and other stakeholders to share relevant threat intelligence and best practices. This collaboration ensures a collective effort to prevent and respond to potential threats.

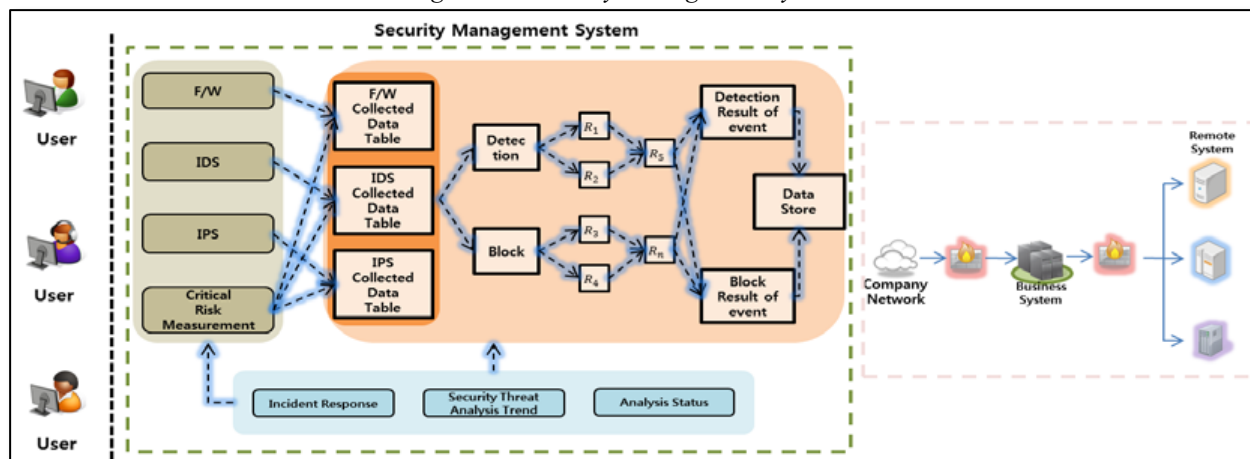
⁴⁷ “mens rea” refers to criminal intent. The literal translation from Latin is “guilty mind.”

⁴⁸ North Carolina Attack Shows US Power Grid Is ‘Extremely Vulnerable’ - Bloomberg

The CIPC serves as a vital platform for members to gain insight into best practices by engaging with industry experts who offer the latest information on safeguarding the most sensitive infrastructure. The Physical Security Subcommittee is dedicated to addressing physical security concerns, offering targeted guidance and strategies to enhance the protection of critical assets.

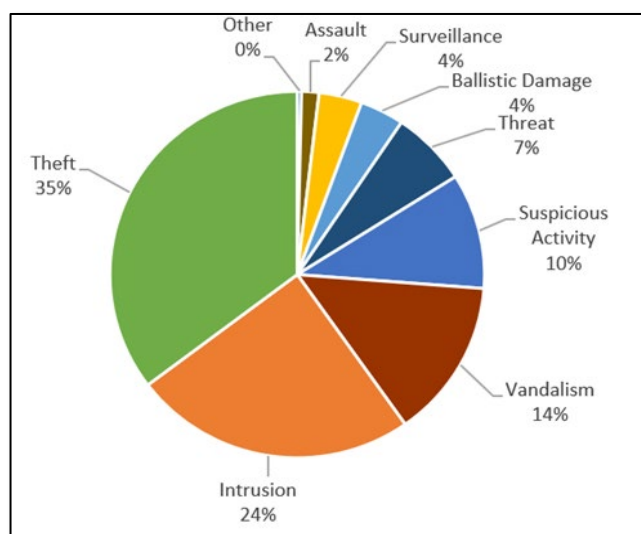
In addition to providing general guidance for both cyber and physical security professionals, the CIPC keeps members informed about new laws and potential legislation, particularly concerning Unmanned Aircraft Systems (UAS) and the legal frameworks that enable members to protect their infrastructure effectively. This continuous flow of up-to-date information ensures that members are well-prepared to navigate the evolving landscape of security challenges.

Figure 43: Security Management System



Organizations within the electric sector conduct vulnerability assessments and risk analyses to identify and address potential vulnerabilities in critical infrastructure. Many entities have participated in SERC's Physical Security Workshop, which teaches physical security concepts, vulnerability assessments and risk analysis and incorporates practical application of them. By identifying weak points and implementing necessary safeguards, the sector can strengthen its resilience against

Figure 44: E-ISAC July 2023 - August 2024 Electric Industry Incidents

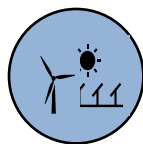


man-made physical events. A visual depiction of a security management system appears in Figure 43.⁴⁹

In summary, the electric sector recognizes the risks associated with sabotage and deliberate attacks on critical infrastructure and has implemented several mitigation steps. These include enhancing physical security measures, developing robust incident response plans, promoting collaboration and information sharing, and conducting vulnerability assessments. By taking these measures, the sector aims to minimize the potential impact of man-made physical events and ensure the reliability and security of the power grid.

⁴⁹https://www.researchgate.net/publication/270527040_Security_Management_Architecture_for_Secure_Smartwork_Center

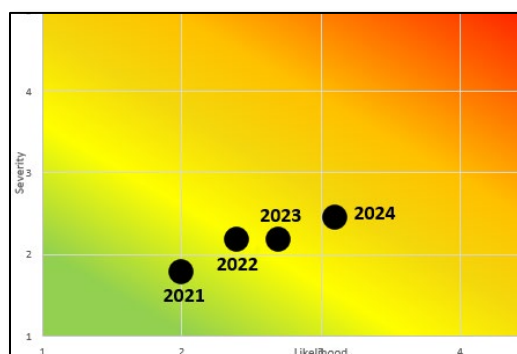
Risk 10: Integrating Variable Energy Resources (Monitor)



Identified and evaluated by the Engineering Committee

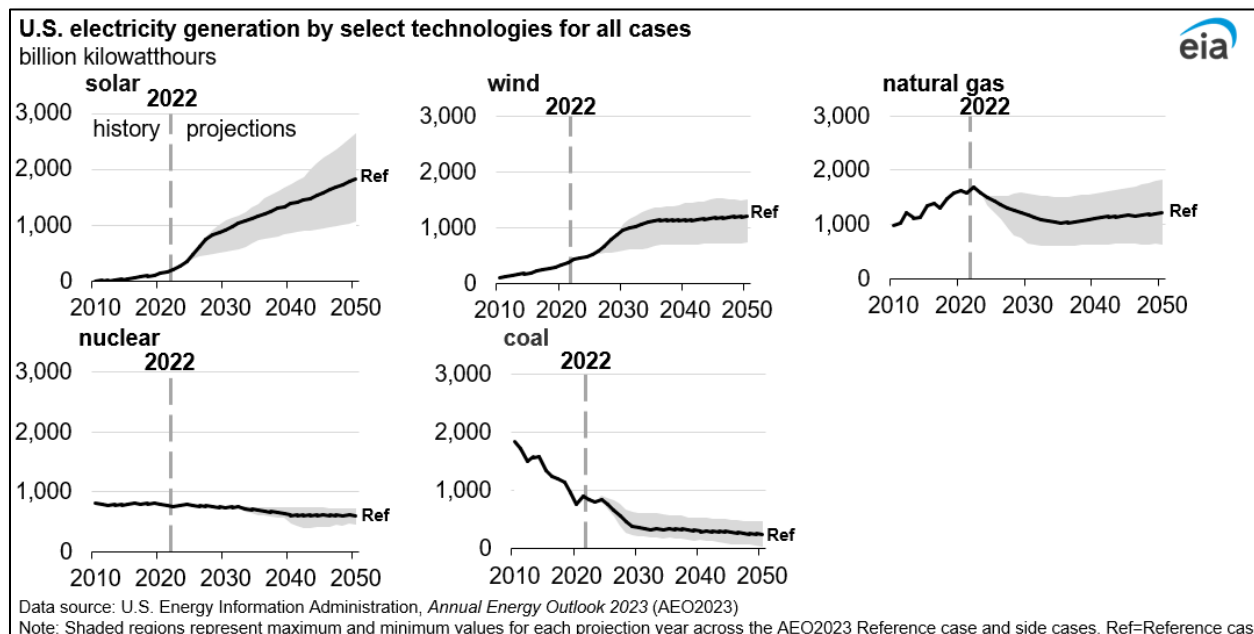
Definition/Justification of risk: The SERC footprint has seen a significant increase in renewable energy

resources, particularly wind in the west and solar in the south and southeast. Integrating Variable Energy Resources (VERs) presents challenges in areas such as voltage regulation, dynamic and transient performance, and sudden shifts in dispatch patterns.



Since the U.S. Energy Information Administration released the last Annual Energy Outlook (AEO), wind generation capacity has grown substantially, increasing by about 138 percent to 235 percent. This growth is accompanied by a rise in installed battery capacity across all scenarios to support renewable expansion. Additionally, natural gas generating capacity is projected to increase by 20 percent to 87 percent through 2050, depending on the scenario (see [Figure 45](#)).⁵⁰

Figure 45: US Electricity Generation – EIA



System operators and planners face significant challenges in integrating these resources, particularly in the following areas:

- Resource adequacy,
- Transmission adequacy,
- Frequency control, and
- Voltage control.

⁵⁰ Narrative 2023 - U.S. Energy Information Administration (EIA)

“Engineers from the NERC and the Energy Systems Integration Group (ESIG) stress the importance of deploying inverters with “grid-forming” (GFM) capabilities to support the increasing integration of inverter-based resources (IBRs) like wind and solar. They point out that today’s IBRs, which rely on “grid-following” inverters, are unable to correct faults that cause current fluctuations, making the deployment of GFM technology a pressing need. The establishment of market incentives and rewards for those who deploy GFM technology should be driven by FERC. NERC and system operators will need to focus on incorporating GRM technology and how it impacts the reliability and secure operation of the grid.

The National Renewable Energy Laboratory (NREL) reports that as traditional synchronous machines with rotating mass—such as those in natural gas, coal, and hydropower plants—are replaced by IBRs, the system’s inertia decreases, increasing the risk of frequency swings. Grid-following inverters in current IBRs do not significantly address these changes, but GFM inverters can detect and adjust power flow to limit them, offering a crucial advantage.”⁵¹

The integration of renewables has introduced new technologies such as energy storage, IBRs, and distributed energy resources (DERs). This influx of technology can complicate forecasting of anticipated net demand and impacts grid operation. Given that VERs are weather-dependent, planning for backup resources is crucial to maintaining system reliability during periods of low generation.

Mitigation Status and Strategies: The Generator Working Group (GWG) and the Variable Energy Resources Working Group (VERWG) are taking active steps to address the evolving challenges of integrating Inverter-Based Resources (IBRs) and Variable Energy Resources (VERs) into the Bulk Power System (BPS). Below are the key activities and initiatives of these groups:

- **Participation in NERC Standards Development:**
 - The GWG is actively involved in the NERC Standards Development process for projects related to IBRs, providing input on relevant guidelines to ensure industry alignment.
- **Monitoring and Developing Solutions for VER Integration:**
 - The VERWG continues to monitor the growth of Variable Energy Resources and is working on solutions to address the associated integration challenges.
 - The group is also updating the Inverter-Based Resource Interconnection Requirements Practices paper, which serves as a resource for SERC members. This paper:
 - Documents industry guidance and practices for IBR interconnections.
 - Facilitates discussions on best practices; and
 - Provides a foundation for entities to develop their own interconnection requirements.
- **Training and Knowledge Sharing:**
 - The Inverter-Based Resource Interconnection Requirements Practices paper will also be used as a training tool and a starting point for members to understand interconnection requirements and practices.
 - The VERWG will conduct seminars to share best practices and address challenges related to the increasing number of VERs.

⁵¹ As reliability concerns with renewables rise, upgrading inverters is urgent, analysts say | Utility Dive

- **Collaboration with Other Groups for Regional Assessments:**
 - The VERWG will work closely with the Planning Coordination Subcommittee (PLCS) and the Regional Reliability Subcommittee (RRS) to develop Regional Assessments that evaluate the reliability impacts of VER integration.

Through these collaborative efforts, the GWG and the VERWG are helping to ensure that SERC members are well-equipped to manage the growing role of inverter-based and variable energy resources, strengthening the overall resilience and reliability of the BPS.

NERC's Reliability and Security Guidelines offer key practices that users, owners, and operators of the BES can implement to ensure a highly reliable and secure power system. The following guidelines are particularly relevant to variable energy resource integration:

- [*Recommended Approaches for UFLS Program Design with Increasing Penetrations of DERs*](#)
- [*Performance, Modeling, and Simulations of BPS Connected Battery Energy Storage Systems and Hybrid Power Plants*](#)
- [*Reliability Guideline BPS-Connected Inverter-Based Resource Performance*](#)
- [*Recommendations for Simulation Improvement and Techniques Related to DER Planning White Paper*](#)
- [*Quick Reference Guide: Inverter Based Resource Activities \(nerc.com\)*](#)

5.0 New and Emerging Risks

This section introduces and acknowledges recently identified potential risks to the Bulk Power System (BPS). It provides an overview of the SERC Major Risk nominations received, highlighting ongoing collaboration with the SERC Technical Committees to address these concerns.

SERC has identified four emerging risks areas related to lack of generative artificial intelligence governance, coordination of critical generation or load concentration through a single substation, disruption of cloud services, and lack of insider threat governance programs.

5.1 Lack of Generative Artificial Intelligence Governance

Generative Artificial Intelligence (GenAI) is becoming integral to the electric industry for tasks like grid management and forecasting, but its adoption introduces risks, including data quality issues, cyber security threats, and ethical challenges. The National Institute of Standards and Technology (NIST) and the U.S. Department of Energy highlight the need for adaptive governance to mitigate these risks, especially as GenAI capabilities expand rapidly. Potential harm from GenAI, including impacts on reliability, security, and ethical considerations, make it essential for the SERC footprint to address this risk and develop effective risk mitigation strategies.

Generative AI vs Traditional AI: Generative AI refers to a type of AI that can create new content, such as text, images, music, or other data, by learning patterns from existing data. Unlike traditional AI, which typically performs tasks based on predefined rules or direct input, GenAI models can generate original outputs that mimic human-like creativity and problem-solving. Generative AI is trained on large datasets and learns to generate new content by identifying patterns. This enables generative AI to create original, creative outputs, such as text, images, and designs, based on the patterns it has learned.

GenAI is defined as a machine-based system that can, for a given set of human-defined objectives, make predictions, recommendations, or decisions influencing real or virtual environments.

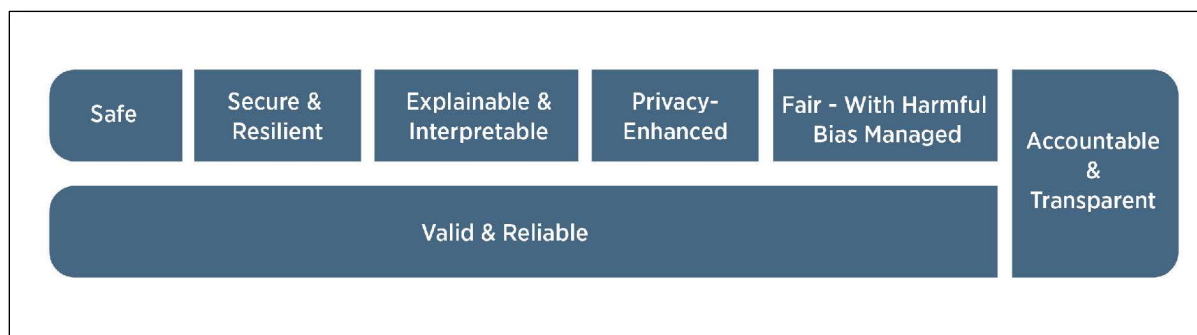
Reports of successful implementation of machine learning and artificial intelligence⁵² from across the North American BPS is prevalent as AI has been present in the electric utility industry for many years. Past and present applications of AI include forecasting, system planning, system protection, predictive maintenance, grid management, enhancing the output of wind and solar resources, storm recovery, wildfire risk assessment, customer analytics, etc.

A report released in April of 2024 from the U.S. Department of Energy highlights the rapid expansion of AI capabilities across the energy sector while cautioning that we need to, “navigate this emerging technology with a keen, technically grounded and risk-informed awareness of its potential and pitfalls.”⁵³

⁵² Machine learning (ML) is one among many other branches of AI. ML is the science of developing algorithms and statistical models that computer systems use to perform complex tasks without explicit instructions. The systems rely on patterns and inference instead. Computer systems use ML algorithms to process large quantities of historical data and identify data patterns. While machine learning is AI, not all AI activities are machine learning. Source <https://aws.amazon.com/compare/the-difference-between-artificial-intelligence-and-machine-learning/>

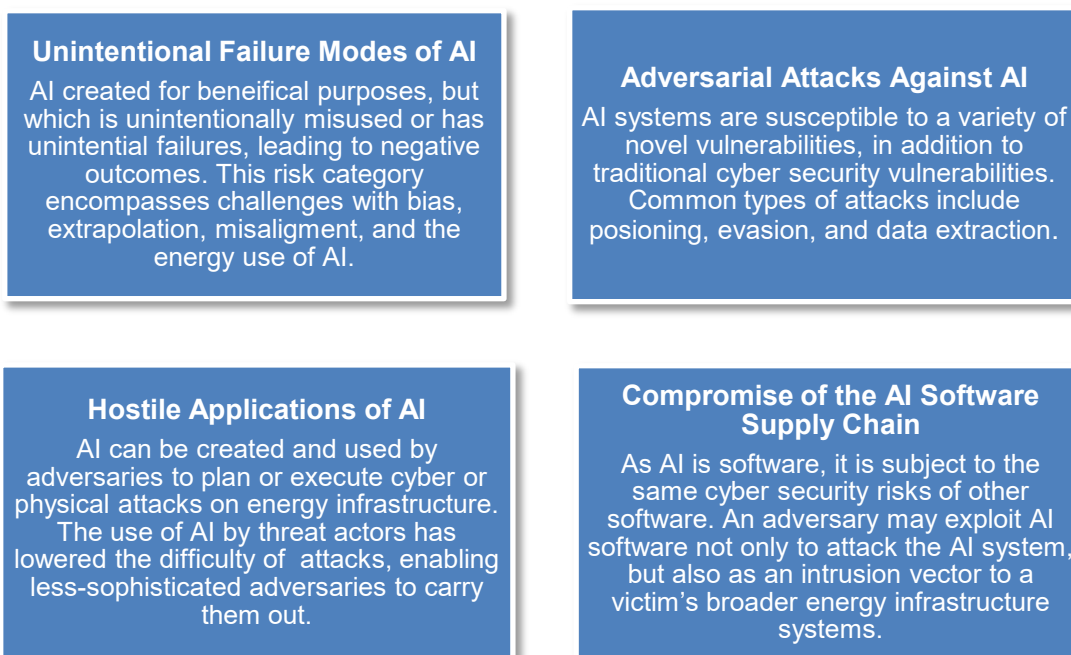
⁵³ Potential Benefits and Risks of Artificial Intelligence for Critical Energy Infrastructure, U.S. Department of Energy Office of Cybersecurity, Energy Security, and Emergency Response, April 2024.

Figure 47: Characteristics of trustworthy AI Systems



NIST released an AI Risk Management Framework in January 2023 that outlines seven characteristics of trustworthy AI (see [Figure 46](#)), which when addressed, can reduce negative AI risks,⁵⁴ while its 2024 companion document details twelve specific GenAI risks, ranging from harmful content creation to data privacy issues. These risks underscore the importance of a risk-informed, adaptive approach to AI governance. An initial assessment by the Office of Cyber security, Energy Security, and Emergency Response (CESER), with support from Lawrence Livermore National Laboratory (LLNL) researchers and energy sector experts, categorized AI and GenAI risks into four broad levels (see [Figure 47](#)).

Figure 46: Four Broad Levels of AI Systems



While AI offers significant benefits, it also introduces risks that must be managed through best practices and safeguards. Human-supervised AI can enhance accuracy and efficiency, reducing these risks. Effective GenAI governance—entailing continuous monitoring and updates—is vital for the safe, ethical, and efficient operation of AI within the Bulk Electric System. Without proper oversight, GenAI systems risk failing to adapt to changing grid conditions, leading to potential power outages, cyberattacks, ethical concerns, and regulatory breaches. Adopting a risk-informed approach to GenAI implementation, supported by robust generative governance frameworks, is crucial for maintaining a reliable and secure BPS while harnessing AI's potential.

⁵⁴ Artificial Intelligence Risk Management Framework (AI RMF 1.0), NIST, January 2023

5.2 Coordination of Critical Generation or Load Concentration Through a Single Substation

The coordination of critical generation or load concentration through a single substation poses a significant reliability risk to the BPS. If compromised, these substations could lead to substantial generation and load loss, potentially threatening the stability of the Eastern Interconnection. These high-impact, low-probability events require a focused approach to ensure BPS reliability and security.

Reliability Standard CIP-014-3 mandates Transmission Owners (TO) to identify and safeguard critical transmission stations, substations, and associated control centers that, if attacked, could cause instability or cascading failures. TOs must perform periodic risk assessments, using technical analyses to evaluate the potential impacts of substation loss on the Interconnection.

The CIP-014-1 standard was established following the 2013 Metcalf Substation attack in California and has undergone minor changes, resulting in the current CIP-014-3 standard. Due to an increase in physical attacks on substations, FERC mandated an evaluation of CIP-014-3 in 2023. NERC's subsequent report called for revisions to improve the assessment of instability and critical infrastructure identification. A standard drafting team is currently updating CIP-014-3 under project 2023-06 to clarify risk assessment methodologies and criteria for identifying critical stations. Including the risk of coordinating critical generation or load concentration through a single substation in the 2024–2026 SERC Regional Risk Report aims to raise awareness and emphasize the need for physical protection and resilience measures.

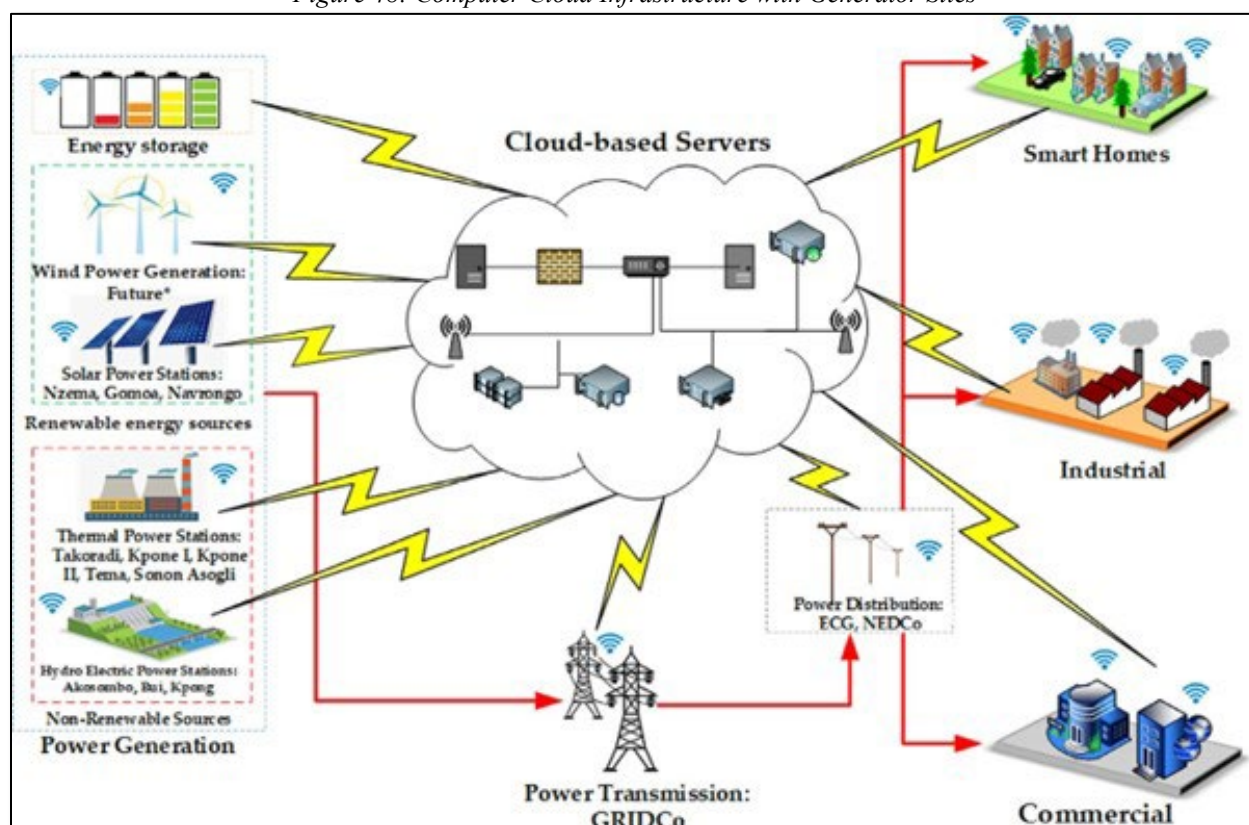
By identifying and addressing these risks through compliance with CIP-014-3 and ongoing updates to the standard, SERC is committed to enhancing the protection and resilience of critical substations. The inclusion of this risk in the 2024–2026 SERC Regional Risk Report underscores the importance of proactive measures to safeguard the BPS against potential disruptions.



5.3 Disruption of Cloud Services

From a security perspective, the electric industry landscape is facing an increase in the number and sophistication of cyberattacks, and security teams are seeking tools and capabilities to elevate their security programs. Security solutions with greater visibility, detection, correlation, analytics, and responsiveness are available using cloud services to help security teams to reduce potential impacts of security events and speed recovery. Cloud services play a critical role in providing greater capability across the security domains (see Figure 48)⁵⁵

Figure 48: Computer Cloud Infrastructure with Generator Sites



Concurrently, from an operational and reliability perspective, the modern power grid landscape is changing, driven by rapid grid modernization, digital transformation, decentralization of electric resources and decarbonization targets. These factors are increasing the data volumes required to continue operating a reliable and resilient grid and thus increasing the need for data analytics and resources such as computing, network, and storage.

Renewable capacity expansion is accelerating. The International Energy Agency updated its growth projections in 2022, to an estimate of 359.5 GW in renewable capacity growth in the US, 2022-2027. As renewable installations grow, site classifications may change from low to medium impact levels, putting operators at risk of having to revert to on-premises resources to meet compliance language rather than benefitting from the cloud services available to lower impact sites. Limiting the resources and capabilities available for operators to manage these use cases can result in impacts to reliability. The disruption of cloud services poses significant risks to the electric sector, as it increasingly relies on cloud-based technologies for various

⁵⁵ On cloud-based systems and distributed platforms for smart grid integration: Challenges and prospects for Ghana's Grid Network (sciencedirectassets.com)

critical operations, including data storage, real-time monitoring, and communication systems. If cloud services experience outages or breaches, it can lead to significant operational disruptions, data loss, and compromised security.

One major concern is the potential for downtime in energy management systems, which can affect grid reliability. According to the U.S. Department of Energy, the integration of cloud computing in smart grid technology enhances efficiency but also raises vulnerabilities.⁵⁶ For instance, if a cloud service goes down, electric utilities may struggle to gather and analyze real-time data crucial for grid management, leading to delayed responses to supply and demand changes or even outages.

Cyberattacks targeting cloud infrastructures can compromise sensitive data and operational capabilities. In its 2020 State of the Electric Industry Report, NERC emphasized the importance of robust cyber security measures, noting that “the electric sector is a prime target for malicious attacks aiming to disrupt services.”⁵⁷ The interconnectivity of cloud services also means that a failure in one area can have cascading effects, impacting other systems and services linked to the cloud.

Risks:

- 1. Operational Impact:** Disruptions in cloud services can hinder the functionality of critical grid management systems, which rely on cloud analytics for real-time monitoring and decision-making. The U.S. Department of Energy notes that while cloud technology enhances the efficiency of smart grid operations, it simultaneously exposes utilities to potential outages, which can severely affect grid reliability.
- 2. Cyber Security Vulnerabilities:** The rise in cyberattacks targeting cloud services further exacerbates risks. NERC highlights the electric sector's susceptibility to cyber threats, suggesting that unmitigated vulnerabilities in cloud infrastructures can lead to data breaches or operational disruptions that impact service delivery.
- 3. Cascading Failures:** The interconnectedness of cloud services means that a disruption in one cloud provider can trigger a series of failures across various electric sector operations, creating a domino effect that could compromise system integrity and public safety.

As the electric sector increasingly integrates cloud technologies for enhanced efficiency and data management, it simultaneously faces heightened risks associated with service disruptions and cyber security threats. Ensuring the resilience of cloud infrastructures through comprehensive risk mitigation strategies is essential for maintaining the integrity and reliability of the electric grid.⁵⁸

⁵⁶ 2020 Smart Grid System Report Download | Department of Energy

⁵⁷ NERC 2020 State of Reliability Report (nerc.com)

⁵⁸ Cyber Security Industry Updates: 2018 Edition (epri.com)

5.4 Lack of Insider Threat Governance Programs

Figure 4949: U.S. Department of Homeland Security (2019). *Insider Threat Program



The lack of insider threat governance programs poses significant risks to the electric sector, primarily by increasing the vulnerability of critical infrastructure to malicious or negligent actions by employees or contractors. Insider threats can manifest as intentional sabotage, data theft, or inadvertent errors that compromise systems, leading to potential service disruptions, financial losses, or safety hazards. In today's interconnected and rapidly evolving digital landscape, businesses face a deluge of cyber security risks.

While organizations often concentrate their efforts on defending against external threats, they must not overlook the formidable danger posed by

insiders. Employees, contractors, or trusted partners with access to an organization's sensitive data and systems have the potential to cause significant harm, intentionally or unknowingly. The absence of an effective insider threat program leaves businesses vulnerable and ill-prepared to tackle the potential risks arising from within their own ranks. By disregarding this imperative aspect of cyber and physical security, organizations expose themselves to a variety of detrimental consequences that can affect their reputation, financial stability, and overall business operations.

Insider threats occur when individuals with authorized access to an organization's resources misuse their privileges for personal gain, revenge, or ideological motives. These threats can take many forms, ranging from stealing intellectual property and siphoning off funds to damaging critical systems or spreading malware from within. What makes insider threats particularly insidious is the difficulty in detecting and preventing them. Unlike external hackers or malicious actors, insiders usually evade traditional security measures and go undetected for extended periods, increasing the potential damage they can cause.

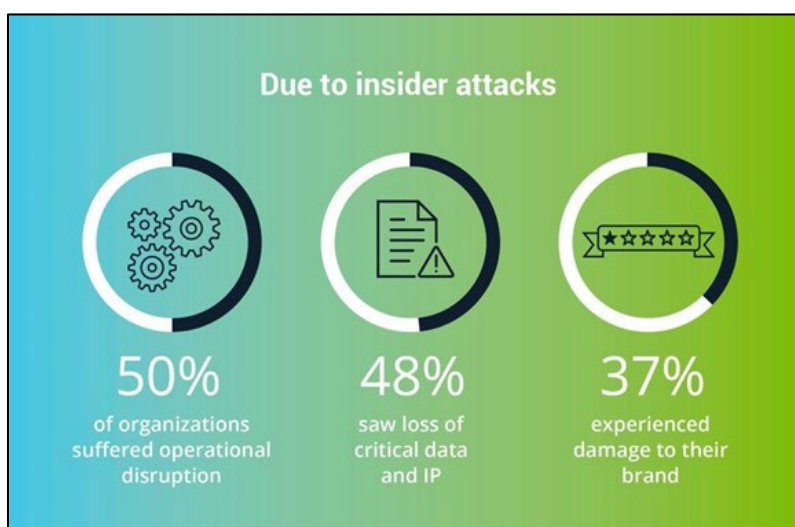
Operational Disruption: Insider threats can lead to severe disruptions in the operational integrity of electric utilities. According to the U.S. Department of Homeland Security (DHS), employees with access to sensitive information and critical systems can exploit their positions, potentially leading to unauthorized access or manipulation of crucial data.⁵⁹ A case study by the Electric Power Research Institute (EPRI) analyzed incidents where insider threats resulted in outages, highlighting the need for robust governance frameworks to mitigate such risks.⁶⁰

⁵⁹ U.S. Department of Homeland Security (2019). *Insider Threat Program. * Retrieved from www.dhs.gov <https://www.dhs.gov/>

⁶⁰ Electric Power Research Institute (2020). *Insider Threat: The Impact on the Electric Sector. * Retrieved from www.epri.com <https://www.epri.com/>

Figure 50: Insider Attacks⁶¹**Data Breaches and Security**

Risks: The absence of a structured insider threat program can leave utilities blind to potential risks from within. The Federal Bureau of Investigation (FBI) has reported that insider threats frequently lead to data breaches, creating vulnerabilities that can be pivotal for adversaries aiming to disrupt energy services.⁶² Additionally, a 2022 report from MIT Technology Review revealed that organizations without effective insider threat governance suffer higher incidents of data breaches and financial impacts due to both malevolent insiders and unintentional actions.⁶³



governance suffer higher incidents of data breaches and financial impacts due to both malevolent insiders and unintentional actions.⁶³

Increased Regulatory Scrutiny: Without established insider threat governance, electric utilities may also face regulatory scrutiny. NERC emphasizes the need for utilities to implement comprehensive risk management practices to satisfy regulatory requirements and protect against potential failures resulting from insider actions.⁶⁴ Failure to comply can result in fines or sanctions, further straining resources.

The lack of insider threat governance programs within the electric sector creates vulnerabilities that can lead to operational disruptions, heightened security risks, and increased regulatory pressures. Establishing comprehensive insider threat mitigation strategies is critical to safeguarding the integrity of energy infrastructure and ensuring reliable service delivery.

⁶¹ How to Prevent Insider Threats | Case Studies, Examples, Types (delinea.com)

⁶² The Insider Threat: An Introduction to Detecting and Deterring an Insider Spy — FBI

⁶³ MIT Technology Review (2022). *How Companies Can Combat Insider Threats*. * Retrieved from www.technologyreview.com <https://www.technologyreview.com/>

⁶⁴ North American Electric Reliability Corporation *2020 State of Reliability Report*. Retrieved from www.nerc.com <https://www.nerc.com/>

6.0 Mapping 2024 SERC Top 10 Reliability Risks to NERC Standard Families

Table 1 identifies families of NERC Standards that map to the top ten risks. This is not a complete list of possible and/or applicable standards to meet compliance and should not be utilized to meet compliance of the NERC Reliability Standards.

Table 1: SERC 2024 Top 10 Reliability Risks

Supply Chain Constraints	Family of Standards	Rationale/Impact Areas
	CIP, COM, PRC, EOP	• Prevention of unauthorized access into the Electronic Security Perimeter (ESP), Software authentication • Supply chain protection of critical components • Loss of electric generation/transmission • Loss of situational awareness monitoring • Loss of communication data/voice • Loss of protection systems • Market disruption
Extreme Weather Impacts	Family of Standards	Rationale/Impact Areas
	EOP, TOP, BAL, COM, INT, and/or TPL	• Planning to deal with an emergency event • Required reporting of extreme events • Ensuring that Network Modeling, State Estimation, and Real-Time Contingency Analysis (RTCA) are resilient in extreme weather and backup measures are in place if primary tools fail (e.g., State Estimator fails to solve) • Use of alternate tools - manual operation of system, loss of situational awareness • Loss of communication (data/voice) • Planning for extreme weather conditions
Resource Uncertainty	Family of Standards	Rationale/Impact Areas
	EOP, MOD, TPL, FAC, and/or BAL	• Emergency Operations Plans are in place to mitigate and respond to Capacity and Energy Emergencies • Increased coordinated studies required with SERC neighbors. • Increased penetration of distributed energy resources (DERs) and potential lack of visibility

		• Policy or legislation changes' impact on fuel demands due to environmental groups, activists, and public perception • Photovoltaic (PV) manufacturing defects impact a significant percentage of solar capacity • Potential transmission system expansion and lead time concerns • Increased penetration in variable energy resources (VERs) requires additional reserves and a need to plan/build a system to accommodate VERs unavailability • Additional scenario planning needed • Load forecasting, metered load • Lack of visibility of DERs and its effect on load forecasting, metered load, and state estimation as penetration levels increase • Greater commitment/dispatch challenges for operators, (ramping requirements, plus others) • Additional reserves required to address higher penetration of VERs; increase in need to plan and build the system when VERs are not available • Challenges to BES system strength (i.e., short circuit, inertia, etc.) due to replacing spinning rotation with more variable generation resources
Shortage of Required Skillsets	Family of Standards	Rationale/Impact Areas
	PER, COM, CIP, EOP, TPL	• Lack of personnel trained in real-time communications and emergency operations • Loss of staff and/or difficulty filling positions in the IT/OT and security areas • Knowledge and maintenance of ESPs, PSPs – EACMS, PACS, PCA • Reliable operation of BES • Training and adapting operators for reliable operations of a BES with significant integration of IBRs, coal retirements, and electrification • Loss of technical expertise in bulk planning due to strong recruiting efforts from non-utility entities • Increased complexity of system planning studies and modeling due to increased inverter-based generation resources

Exploitation of Vulnerabilities	Family of Standards	Rationale/Impact Areas
	CIP	• Unauthorized entry into the ESP • Required patching and updates • Loss of communication (data/voice) • Loss of coordination and situational awareness • Large block of power unexpectedly flowing through SERC
Rapid Nontraditional Load Growth	Family of Standards	Rationale/Impact Areas
	BAL, EOP, FAC, MOD, and TPL	• 2024 SERC Major Risk Nomination • Stress caused by rapid load growth from data centers, AI, and crypto mining on aging transmission systems • Resource availability to respond to sudden changes in loads • Increasing electricity demand caused by Federal incentives for growing industries like semiconductors and EVs • Rising EV use (estimated 10% of peak power by 2035) • Increasing electricity demand due to rapid population increases in the Southeast
Lack of Fuel Diversity and Availability	Family of Standards	Rationale/Impact Areas
	EOP, and/or BAL	• NERC Fuel Assurance Reliability Guideline • Resource adequacy • Sudden changes in dispatch and operating conditions • Forced operating conditions • Need for fast-acting capabilities of existing units • Significant event that would affect a certain fuel type • More reliance on natural gas

Legacy Architecture Compatibility	Family of Standards	Rationale/Impact Areas
	CIP, PER	• EMS support concerns • Patching and updating difficulties in: • Legacy Architecture – ESP interface • Integrating new complex architecture to replace legacy architecture – introducing vulnerabilities • Knowledge and maintenance of CIP aspects of legacy and replacement architecture • Tracking where all legacy systems are installed and having a comprehensive strategy for updating or maintaining systems
Extreme Physical Events: Sabotage & Attacks	Family of Standards	Rationale/Impact Areas
	CIP, EOP	• Insider Physical Threat: Physical attacks connected to a grievance of some kind, presenting the potential for moderate impact • Sabotage: deliberate, well planned, often with an insider component to it, and has the greatest potential for impact • E-ISAC Reports • E-ISAC Watch lists • FBI Bulletins and Flash Alerts • US-CERT • ICS-CERT
Integrating Variable Energy Resources	Family of Standards	Rationale/Impact Areas
	BAL, EOP, TOP, TPL, MOD, PRC, IRO, and/or FAC	• Operations planning tools and mitigations (e.g., operating procedures) in place for each entity to mitigate imbalances created by variable energy resources in a timely manner • Maintain sufficient reactive supply and voltage regulation • Additional reserves required, ramping requirements • Plan and build when VERs are not available • Additional scenario planning • Accurate Load forecasting • Harmonic impact on the system

7.0 2024 ERO Enterprise Risk Elements

Purpose

The strategic Compliance Monitoring and Enforcement Program (CMEP) Implementation Plan (IP)⁶⁵ highlights the focus of ERO Enterprise monitoring and enforcement efforts in 2024 on the risk elements identified within. The CMEP IP gives guidance to the employees of the ERO Enterprise⁶⁶ involved with monitoring and enforcement, and through public posting, informs the ongoing conversations with industry about the risks to mitigate. The risk elements described herein are all developed with the four risks designated “manage” in the 2023 RISC ERO Reliability Risk Priorities Report⁶⁷ as well as the four risk profiles unchanged from 2021. The risks designated “manage” are: 1) Changing Resource Mix, 2) Cyber Security Vulnerabilities, 3) Resource Adequacy and Performance, and 4) Critical Infrastructure Interdependencies. In addition, the report focuses on five risk profiles: 1) Grid Transformation, 2) Security Risks, 3) Extreme Events, and 4) Critical Infrastructure Interdependencies, and a new one added in 2023, 5) Policy, which will not be addressed in the 2024 CMEP IP. While compliance with Reliability Standards is evaluated as part of continuous monitoring, the focus of a mature CMEP is on how the ERO Enterprise and industry proactively identify and mitigate risks to the BPS.

The CMEP IP represents the ERO Enterprise’s high-level priorities for its CMEP activities. While the ERO Enterprise will decide how to monitor each registered entity based on its unique characteristics, registered entities should consider the risk elements and their associated areas of focus as they evaluate opportunities and priorities to enhance their internal controls and compliance operations to mitigate risks to reliability and security. There is no expectation that every risk element or every requirement mapped to a risk element should be contained within every possible engagement. Risk elements serve as an input in determining the appropriate monitoring of risks and related Reliability Standards and requirements in the Compliance Oversight Plan (COP) for each registered entity.

Impact of Risk Elements

The Compliance Enforcement Authorities (CEAs) evaluate the relevance of the risk elements to the registered entity’s facts and circumstances as they plan CMEP activities throughout the year. For a given registered entity, requirements other than those in the CMEP IP may be more relevant to mitigate the risk, or the risk may not apply to the entity at all. Thus, depending on regional distinctions or registered entity differences, focus will be tailored as needed. The 2024 risk elements included in Table 1 are mostly similar to the 2023 risk elements that reflect the maturation of the risk-based approach to compliance monitoring. The changes include a new physical security risk element, and the expansion of the cold weather risk element to extreme weather, which includes both hot weather and space weather⁶⁸ events. The discrete risks identified within the risk elements provide focus for measuring current state and validating registered entity progress. By tracking improvements, industry and the ERO Enterprise can justify focusing on different risks in the future.

Compliance monitoring is not the only tool available to address the risks identified. CMEP staff may assist in various forms of outreach with industry to understand how effective certain

⁶⁵ ERO CMEP Implementation Plan v1.0 - 2024 (nerc.com)

⁶⁶ The ERO Enterprise is comprised of NERC and the six Regional Entities, which collectively bring together their leadership, experience, judgment, skills, and supporting technologies to fulfill the ERO’s statutory obligations to assure the reliability of the North American BPS.

⁶⁷ 2023 ERO Reliability Risk Priorities Report

⁶⁸ Space weather events refer to disturbances in the Earth’s magnetosphere caused by solar activity,

obligations are being implemented and to encourage best practices to achieve the common goal of mitigating risk to the BPS. Enforcement may consider these risks when assessing risk from possible noncompliance, assisting with mitigation plans, or assessing penalties.

The resulting risk elements are shown in ERO Enterprise Risk Elements [Table 1](#) in [Figure 51](#). The table shown in [Figure 51](#) compares the 2023 and 2024 Risk Elements from the 2024 ERO Enterprise Compliance Monitoring and Enforcement Program Implementation Plan Version 1.0 October 2024.

Figure 51: Comparison of 2024 and 2023 Risk Elements⁶⁹

Table 1: 2023 and 2024 Risk Elements	
2023	2024
Remote Connectivity	Remote Connectivity
Supply Chain	Supply Chain
N/A	Physical Security
Incident Response	Incident Response
Stability Studies	Stability Studies
Inverter-Based Resources	Inverter-Based Resources
Facility Ratings	Facility Ratings
Cold Weather Response	Extreme Weather Response

[Figure 52](#) is a series of tables capturing the Areas of Focus for each of the 2024 Risk Elements in the 2024 ERO Enterprise Compliance Monitoring and Enforcement Program Implementation Plan Version 1.0 October 2024.

⁶⁹ NERC 2024 ERO Enterprise Compliance Monitoring and Enforcement Program and Implementation Plan V 1.0 October 2023 (nerc.com)

Areas of Focus

Figure 52: ERO 2024 Risk Elements⁷⁰

Table 2: Remote Connectivity			
Rationale	Standard	Req	Entities for Attention
Remote access to Critical Infrastructure Cyber Assets introducing increased attack surface, as well as possible increased exposure.	CIP-005-7	R2, R3	Balancing Authority Distribution Provider Generator Operator Generator Owner Reliability Coordinator Transmission Operator Transmission Owner
Malware detection and prevention tools deployed at multiple layers (e.g., Cyber Asset, intra-Electronic Security Perimeter, and at the Electronic Access Point) are critical in maintaining a secure infrastructure.	CIP-007-6	R3	Balancing Authority Distribution Provider Generator Operator Generator Owner Reliability Coordinator Transmission Operator Transmission Owner
Mitigation of the risks posed by unauthorized disclosure, unauthorized modification, and loss of availability of data used for Real-time Assessment and Real-time monitoring while such data is being transmitted between any applicable Control Centers.	CIP-012-1	R1	Balancing Authority Generator Operator Generator Owner Reliability Coordinator Transmission Operator Transmission Owner

⁷⁰NERC 2024 ERO Enterprise Compliance Monitoring and Enforcement Program and Implementation Plan V 1.0 October 2023 (nerc.com)

Table 3: Supply Chain			
Rationale	Standard	Req	Entities for Attention
Unverified software sources and the integrity of their software may introduce malware or counterfeit software.	CIP-010-4	R1	Balancing Authority Distribution Provider Generator Operator Generator Owner Reliability Coordinator Transmission Operator Transmission Owner
Mitigate risks to the reliable operation of the BES by implementing sound Supply Chain policies and procedures.	CIP-013-2	R1 R2	Balancing Authority Distribution Provider Generator Operator Generator Owner Reliability Coordinator Transmission Operator Transmission Owner

Table 4: Physical Security ²⁶			
Rationale	Standard	Req	Entities for Attention
Mitigate risks to the reliable operation of the BES as the result of a Physical Security Incident.	CIP-014-3	R4, R5	Transmission Operator Transmission Owner

Table 5: Incident Response			
Rationale	Standard	Req	Entities for Attention
Mitigate risks to the reliable operation of the BES as the result of a Cyber Security Incident.	CIP-008-6	R1, R2, R3	Balancing Authority Distribution Provider Generator Operator Generator Owner Reliability Coordinator Transmission Operator Transmission Owner

⁷¹ NERC Report on CIP-014-3.pdf

Table 6: Stability Studies

Rationale	Standard	Req	Entities for Attention
Planning studies are effective in identifying system performance issues following both minor and major system disturbances.	CIP-014-3	R1	Transmission Owner
	TPL-001-4, TPL-001-5.1	R4, R6	Planning Coordinator Transmission Planner

Table 7: Inverter-Based Resources

Rationale	Standard	Req	Entities for Attention
Clear and consistent interconnection requirements for IBRs	FAC-001-4	R1, R2	Generator Owner Transmission Owner
IBRs being adequately studied during the interconnection process	FAC-002-4	R1, R2	Generator Owner Planning Coordinator Transmission Planner
IBRs including in models provided from generator owners	MOD-026-1	R2	Generator Owner
IBRs staying online when needed	PRC-024-3	R1, R2	Generator Owner

Table 8: Facility Ratings

Rationale	Standard	Req	Entities for Attention
Ensuring entities maintain accurate Facility Ratings	FAC-008-5	R6	Generator Owner Transmission Owner

Table 9: Extreme Weather Response			
Rationale	Standard	Req	Entities for Attention
Ensure plans are developed and implemented to mitigate operating Emergencies	EOP-011-2	R1, R2, R3, R6, R7, R8	Balancing Authority Generator Owner Reliability Coordinator Transmission Operator
Ensure plans are developed and implemented to mitigate extreme cold weather	EOP-012-1 (comes into effect October 2024)	R1, R2, R3, R4, R5, R6, R7	Generator Owner Generator Operator
Planned performance during geomagnetic disturbance (GMD) events.	TPL-007-4	R1, R2, R4, R5, R7	Planning Coordinator Transmission Planner

Appendix A SERC Consolidated Risk Registry

Major Risk	Justification	Risk Ranking	Associated NERC 2023 RISC Risk Profile
Supply Chain Constraints	2023 NERC RISC Report. The Executive Orders to secure the BPS prohibit any acquisition, importation, transfer, or installation of BPS electric equipment which has a nexus with any foreign adversary and poses an undue risk to national security, the economy, or the safety and security of Americans. While no equipment is actually prohibited at this time, since the Secretary of Energy has not issued regulations to implement the Order, there is a possibility that prohibited equipment may be identified within SERC's footprint at a later point in time. Technologies and Services and Interdependencies from the critical infrastructure sectors, such as Communications, Financial Services, Oil and Natural Gas Subsector, and Water, where sector-specific vulnerabilities can impact BPS reliability. 1. Increased reliance on third-party service providers and cloud-based services for BPS operations and support. 2. Cyber security risks in the supply chain: software integrity and authenticity; vendor remote access; information system planning; and vendor risk management and procurement controls.	1	Energy Policy A. Federal Grid Transformation A. Bulk Power System Planning B. Resource Adequacy and Performance F. Changing Resource Mix Security Risks B. Cyber

Major Risk	Justification	Risk Ranking	Associated NERC 2023 RISC Risk Profile
Extreme Weather Impacts	SERC has 69,000 miles of coastline and is therefore susceptible to hurricane impacts: Category 5 hurricanes with historical impacts (Hugo, Katrina, Rita, Matthew, Irma, Michael), most recent major Hurricanes with significant impacts to the SERC Region (Ida, Ian, Idalia, Helene, and Milton).⁷² Thunderstorm activity in the Southeast often produces tornado activity. 2022 was a record year for tornado activity, and trend continues with 165 tornadoes in the southeast in 2023.⁷³ Drought (2012, 2022) and flooding (2015, 2021) conditions are also a concern.⁷⁴ Extreme cold weather (2018 South Central Event, 2021 Winter Storm Uri, Winter Storm Elliott 2022, Winter Storm Heather 2024) events occur.	2	Resilience / Extreme Events A. Extreme Natural Events, Widespread Impact B. Other Extreme Natural Events
Resource Uncertainty	A transition is taking place in the resource mix within SERC (and nationally) propelling natural gas dominance as a fuel source (replacing coal), renewables (photo voltaic and to a lesser extent, wind) are making a surge.	3	Energy Policy A. Federal B. State Grid Transformation A. Bulk Power System Planning B. Resource Adequacy and Performance F. Changing Resource Mix

⁷² 2022 Atlantic Hurricane Season (noaa.gov)

⁷³ List of North American tornadoes and tornado outbreaks - Wikipedia

⁷⁴ Record drought gripped much of the U.S. in 2022 | National Oceanic and Atmospheric Administration (noaa.gov)

Major Risk	Justification	Risk Ranking	Associated NERC 2023 RISC Risk Profile
Shortage of Required Skillsets	2023 NERC RISC Report Lack of experience leads to poor decision making/delayed actions to resolve emergencies. Knowledge transfer from the Aging Workforce can mitigate this lack of experience. Factors affecting Work Force Capacity and Capability: 1. Retirements 2. Active movement between employers 3. Staff having to accommodate process changes in addition to routine work 4. Documenting knowledge to inform the next generation of professionals 5. Incentivizing technical development of staff 6. Increasing complexity and fast pace of technological change within the power grid, for example IBR calling for a new layer of knowledge and modeling. 7. Increasing complexity of power grid procedurally, for example, Order 881, changing standards, etc. 8. Training and development of a new generation of professionals (GOLD Graduates of last decade – IEEE Term) 9. Expanding marketing to and development of the workforce of the future. Some workforce needs are quite specific as well as highly technical; and therefore, they require specific attention from SERC. An example is the Electromagnetic Transient Modeling Task Force at NERC and the ERO.	4	Grid Transformation E. Human Performance and Skilled Workforce Security Risks B. Cyber

Major Risk	Justification	Risk Ranking	Associated NERC 2023 RISC Risk Profile
Exploitation of Vulnerabilities	2023 NERC RISC Report Consolidation of two risks from the 2020 report: the rapid growth in sophistication and widespread availability of tools and processes designed to exploit vulnerabilities and weaknesses in BPS technologies and in connected IT networks and systems. Cyber security threats result from exploitation of both external and internal vulnerabilities. 1. Exploitation of employee and insider access. 2. Weak security practices of host utilities, third-party service providers and vendors, and other organizations. 3. Unknown, undisclosed, or unaddressed vulnerabilities in cyber systems. 4. Growing sophistication of bad actors, nation states, and collaboration between these groups, including the use of Artificial Intelligence.	5	Security Risks B. Cyber
Rapid Nontraditional Load Growth	2024 SERC Major Risk Nomination Rapid load growth from data centers, AI, and crypto mining is stressing aging transmission systems, requiring urgent upgrades to prevent failures. AI-driven data centers, consuming up to 1000 MW, are significantly increasing energy use. Federal incentives for industries like semiconductors and EVs are raising electricity demand. Rising EV use could account for 10% of peak power by 2035, stressing the system further with increased demand and charging infrastructure needs. Rapid population increases in the southeast are driving up electricity demand. The combination of rapid load growth, extreme weather, and supply chain disruptions heightens the risk of resource shortages and reliability issues.	6	Grid Transformation A. Bulk Power System Planning B. Resource Adequacy and Performance D. Situational Awareness Challenges

Major Risk	Justification	Risk Ranking	Associated NERC 2023 RISC Risk Profile
Lack of Fuel Diversity and Availability	Currently, natural gas accounts for more than 40% of the generation in SERC footprint. There is a greater reliance on natural gas generation with baseload coal plant retirement. This will give rise to a less diverse portfolio, and the ability to respond to loss of fuel type. 1. Lack of coordination between electric and natural gas industry 2. Natural Gas infrastructure affecting dispatch to generation 3. Adequacy of natural gas pipeline infrastructure to serve extreme winter peak load along with traditional firm gas needs 4. Policy or legislation changes could impact fuel availability	7	Energy Policy A. Federal B. State Grid Transformation A. Bulk Power System Planning B. Resource Adequacy and Performance F. Changing Resource Mix Resilience / Extreme Events A. Extreme Natural Events, Widespread Impact B. Other Extreme Natural Events Critical Infrastructure Interdependencies D. Natural Gas
Legacy Architecture Compatibility	2023 NERC RISC Report Legacy architecture coupled with the increased connectivity of the grid expands the attack surface of BPS protection and control systems. 1. Increased automation of the BPS through control systems implementation 2. The trend toward increased integration of IT operating systems may increase the attack surface and associated attack risk 3. IT/operational technology (OT) control system infrastructure management, out-of-date operating systems, and the lack of patching capability/discipline	8	Grid Transformation C. Increased Complexity in Protection and Control Systems Security Risks A. Physical B. Cyber

Major Risk	Justification	Risk Ranking	Associated NERC 2023 RISC Risk Profile
Extreme Physical Events: Sabotage & Attacks	2023 NERC RISC Report - E-ISAC Watch list & Reports - FBI Bulletins & Flash Alerts - US-CERT - ICS-CERT Note: Physical attacks often are connected to a grievance of some kind and present the potential for moderate impact. Insider Physical Threat. Note: Sabotage is defined as deliberate, well planned, often with an insider component, and has the greatest potential for impact.	9	Grid Transformation D. Situational Awareness Challenges Resilience / Extreme Events A. Extreme Natural Events, Widespread Impact B. Other Extreme Natural Events Security Risks A. Physical Critical Infrastructure Interdependencies A. Communication
Integrating Variable Energy Resources	There has been an increase in the SERC renewable energy resource portfolio, wind energy in the west and solar in the south and southeast. Integration of VERS raises the risk of voltage regulation, dynamic response, and sudden change in dispatch patterns. The changing characteristic of the grid with the growth of VERs will affect how the grid is operated in the future. Since VERs are weather dependent, planning for backup resources in the absence of generation becomes essential to maintaining the reliability of the system.	10	Energy Policy A. Federal B. State Grid Transformation A. Bulk Power System Planning B. Resource Adequacy and Performance F. Changing Resource Mix
Parallel /Loop Flow Issues	Parallel/loop flows put firm transactions at risk. These firm transactions are usually being utilized as distribution network reconfigurations (DNRs) for SERC entity load.	11	Grid Transformation D. Situational Awareness Challenges F. Changing Resource Mix
Resource deficiency during outage periods	Identified in 2011 Southwest blackout report	12	Grid Transformation A. Bulk Power System Planning B. Resource Adequacy and Performance

Major Risk	Justification	Risk Ranking	Associated NERC 2023 RISC Risk Profile
Lack of Generative Artificial Intelligence Governance	Artificial Intelligence has been present in the industry for many years in the form of forecasting, predictive maintenance, grid management, enhancing the output of wind and solar resources, storm recovery, wildfire risk assessment, customer analytics, etc. However, the introduction and widespread availability of generative AI brings to the forefront the need to assess and address the risks posed by AI systems.	13	Grid Transformation D. Situational Awareness Challenges E. Human Performance and Skilled Workforce Security Risks A. Physical B. Cyber
Loss of Critical Application(s) (EMS/SCADA)	Event Reports indicate more loss of EMS events have occurred in the last two years for SERC entities.	14	Grid Transformation D. Situational Awareness Challenges E. Human Performance and Skilled Workforce
Generation and Transmission Modeling Inaccuracies	Accurate representation of transmission and generation modeling is the key to planning the transmission adequacy and to maintaining system reliability in the future. The rise of renewable generation models and use of inaccurate and black box models introduces errors and significantly affects planning studies. Above normal demand in energy Accurate governor deadband and droop settings are important to reliably operate the bulk energy system.	15	Grid Transformation A. Bulk Power System Planning B. Resource Adequacy and Performance D. Situational Awareness Challenges E. Human Performance and Skilled Workforce F. Changing Resource Mix
Facility Ratings Inaccuracies	SERC has noted a pattern of ratings that does not follow the entities' Facility Rating Methodology. In particular, a trend has been noted where the Most Limiting Element (MLE) was not accounted for in the rating calculation, so the Entity was operating to a higher limit than was accurate. FAC-008	16	Energy Policy A. Federal Grid Transformation A. Bulk Power System Planning F. Changing Resource Mix
Equipment Failure compounded by relay mis operation	Voltage depression events point to the consequences associated with this compound risk.	17	Grid Transformation C. Increased Complexity in Protection and Control Systems D. Situational Awareness Challenges E. Human Performance and Skilled Workforce

Major Risk	Justification	Risk Ranking	Associated NERC 2023 RISC Risk Profile
Coordination of Critical generation or load concentration through a single substation	As a result of audits related to the CIP-014 standard requirements, SERC has identified several substations that under certain contingencies could result in the loss of substantial load and/or generation.	18	Grid Transformation A. Bulk Power System Planning Resilience / Extreme Events A. Extreme Natural Events, Widespread Impact B. Other Extreme Natural Events
Extreme Physical Events: Vandalism	2023 NERC RISC Report Note: Vandalism is defined as a random and opportunity-based, and although it occurs most frequently, results in low impact.	19	Security Risks A. Physical Critical Infrastructure Interdependencies A. Communications
Vegetation causing cascade event	Vegetation mismanagement could cause cascading outages.	20	Grid Transformation A. Bulk Power System Planning
Pandemic - Operational Impacts	The next potential pandemic could impact staffing levels; cause load reductions; impact to load profile; and/or impact maintenance and capital project plans.	21	Resilience / Extreme Events B. Other Extreme Natural Events
Inaccurate Modeling of interchange in study models	Failing to correctly model firm transfers raise reliability risks associated with improper assessments of near-term operating conditions, as well as development of incorrect expansion plans. The firm transfers associated with many utilities have not been accurately modeled in LTWG/MMWG models and their derivative models used by TOs, TSPs, TPs, and PCs for several years. These models are used for a variety of reliability related studies throughout the Eastern Interconnection. These models are also widely used for evaluation of transfer capability and transmission service requests.	22	Grid Transformation A. Bulk Power System Planning B. Resource Adequacy and Performance
Solar Eclipse (2024) - Operational Impacts	Loss of solar generation during the event could lead to under frequency or loop flows.	23	Resilience / Extreme Events B. Other Extreme Natural Events
Lack of Organizational Coordination - Non-Industry Stakeholders	2023 NERC RISC Report Ineffective teamwork and collaboration among the federal, provincial, state, local government, private sector, and critical infrastructure owners can exacerbate cyber events.	24	Security Risks B. Cyber Critical Infrastructure Interdependencies A. Communications

Major Risk	Justification	Risk Ranking	Associated NERC 2023 RISC Risk Profile
Pandemic - Planning Impacts	The specific concern is the uncertainty regarding the impact the COVID-19 pandemic will have on long-term planning.	25	Grid Transformation A. Bulk Power System Planning E. Human Performance and Skilled Workforce
Protection system and single points of failure	The rate of protection mis-operation in the SERC footprint is around 7.5% of the total operation, which is considered high.	26	Grid Transformation D. Situational Awareness Challenges E. Human Performance and Skilled Workforce
Changing RC/BA ownership/boundary changes within the SERC footprint	The acquisition of a company may result in reestablishing BA boundaries, changing Reliability Coordinators, and adjusting the SERC sub-regional footprints. This is for an atypical situation and not for common boundary changes	27	Grid Transformation A. Bulk Power System Planning B. Resource Adequacy and Performance D. Situational Awareness Challenges F. Changing Resource Mix
Load Pockets/Voltage Concerns	The concern is about specific load pockets or areas that have potential for voltage collapse (i.e., Savannah, Atlanta, Washington DC, and Alabama South Region). A voltage collapse or blackout in one of these “known” areas could have major political impacts and implications to the SERC Region. This is a result of Generation retirements.	28	Grid Transformation A. Bulk Power System Planning B. Resource Adequacy and Performance
Disruption of Cloud Services	New to risk registry in 2024 – under evaluation by the RRWG	New	Grid Transformation C. Increased Complexity in Protection and Control Systems Security Risks B. Cyber
Lack of Insider Threat Governance Programs	New to risk registry in 2024 – under evaluation by the RRWG	New	Grid Transformation E. Human Performance and Skilled Workforce Security Risks A. Physical B. Cyber

Appendix B Definitions

Term or Acronym	Definition
AEO	Annual Energy Outlook
AES	Advanced Encryption Standard
AI	Artificial Intelligence
BA	Balancing Authority
BES	Bulk Electric System - as all transmission elements operated at 100 kV or higher and real power and reactive power resources connected at 100 kV or higher. Facilities used in the local distribution of electric energy are not included in this term.
BESS	Battery Energy Storage Systems
BPS	Bulk Power System - A) facilities and control systems necessary for operating an interconnected electric energy transmission network (or any portion thereof); and (B) electric energy from generating facilities needed to maintain transmission system reliability.
BPSA	Bulk-Power Situational Awareness
CEAs	Compliance Enforcement Authorities
CEII	Critical Energy Infrastructure Information
CESER	Office of Cyber security, Energy Security, and Emergency Response
CEWD	Center for Energy Worker Development's
CFI	Charging and Fueling Infrastructure
CIPC	Critical Infrastructure Protection Committee
CMEP	Compliance Monitoring and Enforcement Program
CMEP IP	Compliance Monitoring and Enforcement Program Implementation Plan
COP	Compliance Oversight Plan
CPP	Clean Power Plan
CRISP	Cyber Risk Information Sharing Program
CRPA	Cyber security Risk Assessment Program
CS	Control System
DEI	Diversity, Equity, and Inclusion
DER	Distributed Energy Resource
DCFC	Direct Current Fast Charging
DHS	Department of Homeland Security
DNR	Designated Network Resource
DP	Distribution Provider
EACMS	Electronic Access Control or Monitoring Systems
E-ISAC	Electricity Information Sharing and Analysis Center
EC	Engineering Committee
EEl	Edison Electric Institute
EGWG	Electric-Gas Working Group (NERC)
EIA	Energy Information Administration / Electronic Industries Alliance
EIPC	Eastern Interconnection Planning Collaborative
EMS	Energy Management System

Term or Acronym	Definition
EPA	Environmental Protection Agency
EPRI	Electric Power Research Institute
ERAG	Eastern Interconnection Reliability Assessment Group
ERM	Enterprise Risk Management
ERO	Electric Reliability Organization
ESIG	Energy Systems Integration Group
ESP	Electronic Security Perimeter
EV	Electric Vehicle
FBI	Federal Bureau of Investigation
FERC	Federal Energy Regulatory Commission
FFR	Faster Frequency Response
FRCC	Florida Reliability Coordinating Council, Inc
FTP	File Transfer Protocol
GenAI	Generative Artificial Intelligence
GFM	Grid-Forming (GFM) inverters are advanced inverters designed to operate independently, creating a stable voltage and frequency reference for the power grid.
GO	Generator Owner
GOP	Generator Operator
GPU	Graphics Processing Units
GRO	Natural Gas Reliability Organization
GW	Gigawatt 1GW = 1,000 MW or 1,000,000,000 Watts
IBR	Inverter-based Resources
ICS	Industrial Control Systems
IoT	Internet of Things
IOU	Investor-Owned Utilities
IPP	Independent Power Producer
IRA	Inflation Reduction Act
IROL	Interconnection Reliability Operating Limit
IS	Information System
ISO	Independent System Operator
IT	Information Technology
JIT	Just-in-time
LFSTR	Loop Flow Study Task Force
LLNL	Lawrence Livermore National Laboratory
LTSG	Long-Term Study Group
LTWG	Long-Term Working Group
MaaS	Malware as a Service
MISO	Midcontinent Independent System Operator
MLE	Most Limiting Element
MMWG	Multi-regional Modeling Working Group
MW	Megawatt – 1MW = 1,000 Watts
NATF	North America Transmission Forum

Term or Acronym	Definition
NERC	North American Electric Reliability Corporation
NEVI	National Electric Vehicle Infrastructure (NEVI)
NG	Natural Gas
NIST	National Institute of Standards and Technology
NOAA	National Oceanic and Atmospheric Administration
NPP	Nuclear Power Plant
NREL	National Renewable Energy Laboratory (NREL)
NTI	Non-public Transmission Function Information
OATI	Open Access Technology International
OC	Operating Committee
OEM	Open Electric Market / Original Equipment Manufacturer
OFO	Operational Flow Order
OPSEC	Operations, Planning, and Security Executive Committee
OT	Operational Technology
PC	Planning Coordinator
PACS	Physical Access Control System
PCA	Protected Cyber Asset
PCS	Protection & Control Systems
PFV	Parallel Flow Visualization
PJM	PJM Interconnection, LLC
PLCS	Planning Coordination Subcommittee
PNNL	Pacific Northwest National Laboratory
PSP	Physical Security Perimeter
PSS	CIPC Physical Security Subcommittee
PV	Photovoltaic
RaaS	Ransomware as a Service
RAWG	Resource Adequacy Working Group
RC	Reliability Coordinator
RE	Regional Entities
RISC	Reliability Issues Steering Committee (NERC)
RISTF	Renewable Impact Study Task Force
RRS	Reliability Review Subcommittee
RRWG	Reliability Risk Working Group
RTCA	Real-Time Contingency Analysis
RTO	Regional Transmission Organization
SA	Situational Awareness
SCADA	Supervisory Control and Data Acquisition
SCS	Southern Company Services
SeRC	Reliability Coordinator for the Southeast
SERC	SERC Reliability Corporation
SIR	System Inertial Response
SME	Subject Matter Expert

Term or Acronym	Definition
SOL	System Operating Limit
SOWG	System Operator Working Group
SPOD	Single Point of Disruption
TCA	Transient Cyber Assets
TLR	Transmission Loading Relief
TO	Transmission Owner
TOP	Transmission Operator
TP	Transmission Planner
TSP	Transmission Service Provider
TVA	Tennessee Valley Authority
TW	Terawatt – 1TW = 1,000 GW or 1 Trillion Watts
UAS	Unmanned Aircraft System
UEA	Unauthorized Electronic Access
USB	Universal Serial Bus
VACS	VACAR South (Virginia-Carolinas sub region)
VER	Variable Energy Resource
VERWG	Variable Energy Resources Working Group
VPN	Virtual Private Network

Appendix C Enhancements

The Reliability Risk Working Group (RRWG) started 3 key initiatives in 2021 to enhance the Regional Risk Report with a focus on improving scoring, developing a consistent group of impact areas, and further defining the Major Risk Nomination Process.

Key Initiatives:

- Review the current scoring and develop a methodology to improve consistency in scoring by enhancing the Probability of Occurrence and Severity of Impact ratings and developing the overall total risk score.
- Review the current Impact Areas to improve the number of Impact Areas and provide consistency across all the Technical Committees in the development of their risk registry.
- Review the current Nomination Process for Potential Major Risk and incorporate changes to ease the use of the Nominations for Potential Major Risks form. In addition, develop a process and timeline for submitting and vetting nominations for consideration in the current year.

Outcome:

- Scoring for the Probability of Occurrence: The task force provided further clarification and definition to each score in the range (1 to 5) for consistent application.
- Scoring for the Severity of Impact: The task force implemented a weighted scoring consisting of four different areas.
 1. Materiality (M) – impact of given event finances, load, and/or industry reputation) weight 40%.
 2. Controllability (C) – the amount of influence we have over mitigating the event/risk) weight 40%.
 3. Span (S) – the length of time the event/risk could potential last once occurrence happens) weight 10%.
 4. Velocity (V) – how quickly the event could materialize) weight 10%.
- Scoring Formulas:
 - Probability of Occurrence (PO) for each Impact Area – Score of 1 to 5 = PO
 - Severity of Impact for each Impact Area (SOI) – Score of 1 to 5 for each
 - Multiplied by weighting: $(M \cdot .4) + (C \cdot .4) + (S \cdot .1) + (V \cdot .1) = \text{SOI}$
 - Impact Area Risk Score = $\text{PO} \cdot \text{SOI}$
 - Total Risk Score is the average of all the Impact Area Risk Scores
- Scoring Philosophy Change on the use of zero (0):
 - For consistent scoring, the RRWG has elected to promote the use of zero (0) for Impact Areas that may not apply to each Major Risk being evaluated and scored on the same set of 15 Impact Areas.
- Impact Area:
 - Developed a set of 15 high-level Impact Areas that may apply to the Major Risks Identified across the EC, OC, and CIPC:
 - System Control, Loss of Communication, Loss of Generation, Loss of Transmission, Loss of Load, Infrastructure Damage (Transmission, Generation, Communication), System Projection, Modeling & Forecasting, Long-Term Planning, Technology Integration, Fuel Supply, Transmission & Generation Adequacy, Personnel / Human Resources, Market Disruption, and Loss of CEII / NTI Data.

- The set of 15 impact areas were implemented and assessed, as applicable, for the 2021 Major Risks identified.
- All 15 impact areas will be scored using the enhanced scoring methodology defined above.
- Nomination Process
 - Developed a timeline to accept, review, and add Major Risk Nomination Forms to the Risk Registry.
 - RRWG Leadership team reviews nomination forms for completeness and accuracy in December.
 - The RRWG reviews submitted nomination forms with submitter(s) to accept and move forward in January.
 - The RRWG reviews submitted nomination forms to determine the Monitor or Manage classification and identify the technical committee(s) (EC, OC, CIPC) to which the Nominated Risk belongs in February.
 - The Nomination Form will be presented to the applicable Technical Committee(s) at the Spring Technical Committee Meetings in March.
 - If accepted by the Technical Committee(s), the nominated Major Risk will be added to the Risk Registry in May.
 - Developed a tracking log of submitted accepted and rejected Major Risk Nomination Forms.
 - Enhanced the Nomination for Potential Major Risk Form and developed a completed example form.

Zero (0) Rationale:

The RRWG believes the consistent scoring of risks using the methodology described previously will help ensure the scores are standardized and comparable across each committee (CIPC, EC, and OC). However, it is important to note that not all impact areas will apply to every risk. To maintain consistent and equitable scoring for every risk, the RRWG found it necessary to establish parameters for this situation.

Before beginning the 2021 risk scoring and prioritization process, the RRWG discussed the appropriateness of utilizing a score of zero (0) for the Probability of Occurrence. In the past, the use of a zero (0) for a Probability of Occurrence was not contemplated; each of the Impact Areas used in developing the individual risk score were uniquely identified during the scoring process, suggesting that Impact Areas with no impact were simply not considered (or included) in the process. However, when using a consistent set of 15 Impact Areas, it is appropriate to consider that one or more Impact Areas may have negligible or no impact on a proposed SERC major risk. In this case, a score of zero (0) would be appropriate for the Probability of Occurrence.

As a result, the RRWG agreed to allow the use of a zero (0) for a Probability of Occurrence for an individual Impact Area.

Appendix D Regional Risk Framework

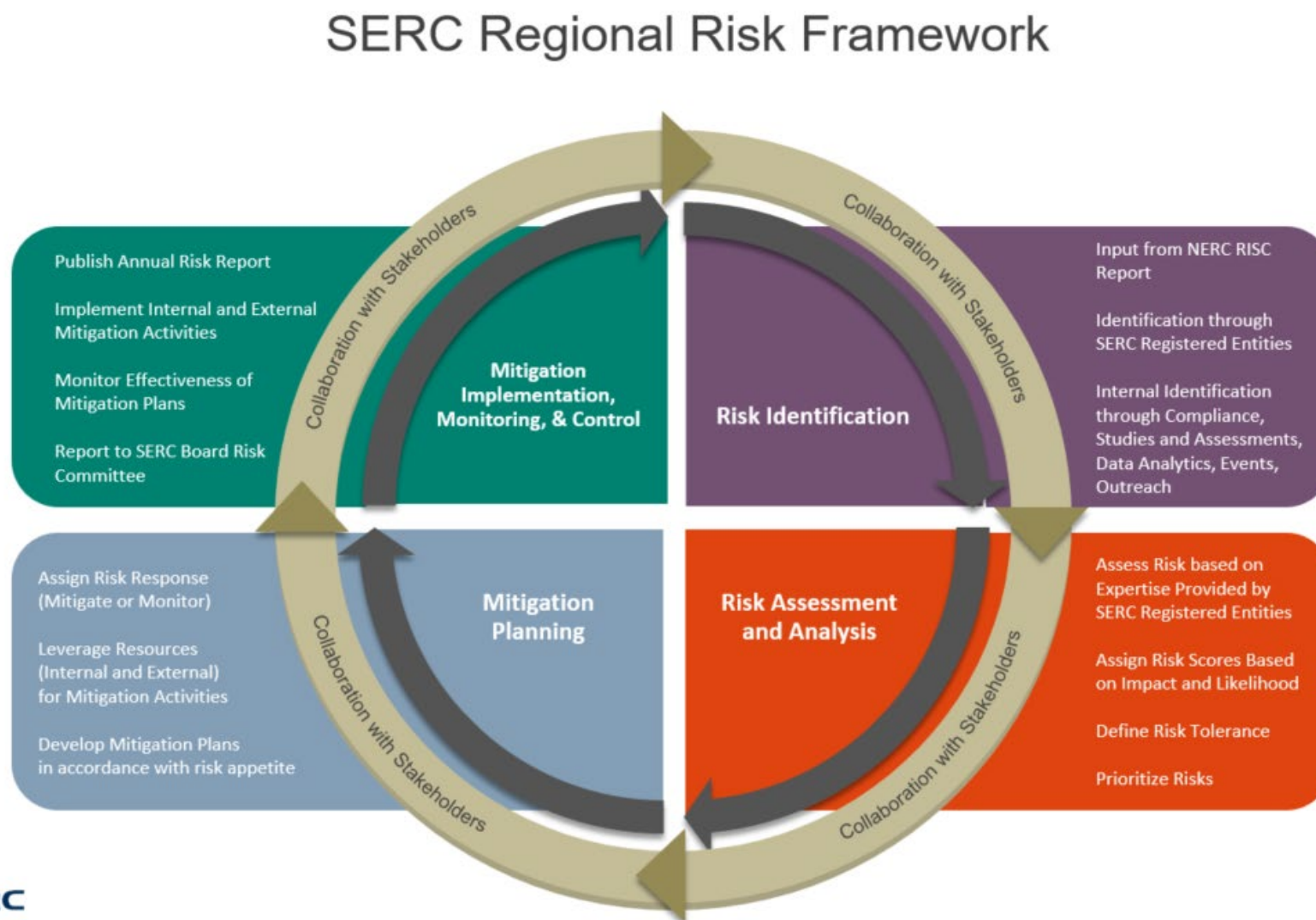
SERC's Mission is to assure effective and efficient reduction of risks to the reliability and security of the bulk power system. To achieve this objective, the SERC RRWG develops the annual regional risk registry. It is important to have a clear understanding of risk, mitigation process, stakeholder participation, outreach, and measures of effectiveness to manage the risk to reliability and security of the BPS.

A defined risk framework provides the structure required for efficient resource allocation and prioritization. The framework also helps SERC to evaluate the effectiveness of the deployed mitigation, which in turn informs the future risk prioritization. The structure helps to establish responsibility and accountability.

SERC, in collaboration with its stakeholders, developed a four-step regional risk framework. Each phase has its own unique set of processes, tools, and engagement with internal and external stakeholders.

1. **Risk Identification:** SERC and its stakeholder subject matter experts are actively engaged in identifying the current and emerging risks through RRWG engagement. The risk identification also takes into consideration the NERC Reliability Issues Steering Committee (RISC) report, and SERC internal identification through compliance monitoring, studies and assessments, events, and outreach.
2. **Risk Assessment and Analysis:** The SERC data analytics group assesses the risk based on the available data on the subject and analysis of the data. The stakeholders also help provide valuable input in determining the prioritization of the quantified risk by defining the impact areas and probability of occurrence. The framework provides provision to set the risk tolerance for a particular risk.
3. **Mitigation Planning:** The planning starts with the determination of whether the risk needs to be managed or monitored. If managed, the risk mitigation planning/development is the responsibility of SERC and its stakeholders through the SERC technical committees, subcommittees, and the working groups. The technical committees also take advantage of all the work carried out by ERO Enterprise, trade associations, vendors, and other industry initiatives.
4. **Mitigation Implementation, Monitoring, and Control:** The SERC technical committees oversee the mitigation implementation, monitoring, and control through their annual work plan development. SERC and the stakeholders also deploy measures to monitor the effectiveness of the mitigation implementation.

Figure 53: SERC Regional Risk Framework



In Figure 53, the outer circle is included to emphasize that SERC is actively engaging and collaborating with stakeholders throughout the entire process.